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Enterprise Data Preparation User Guide

Informatica Enterprise Data Preparation User Guide
10.5.10
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Preface

Use the *Enterprise Data Preparation User Guide* to learn how to use the Enterprise Data Preparation application. You can search for, discover, and prepare data for analysis. Learn how to manage projects and create visualizations for data assessment.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Network

The Informatica Network is the gateway to many resources, including the Informatica Knowledge Base and Informatica Global Customer Support. To enter the Informatica Network, visit https://network.informatica.com.

As an Informatica Network member, you have the following options:

- Search the Knowledge Base for product resources.
- View product availability information.
- Create and review your support cases.
- Find your local Informatica User Group Network and collaborate with your peers.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

Informatica Documentation

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Informatica Product Availability Matrices

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Informatica Velocity

Informatica Velocity is a collection of tips and best practices developed by Informatica Professional Services and based on real-world experiences from hundreds of data management projects. Informatica Velocity represents the collective knowledge of Informatica consultants who work with organizations around the world to plan, develop, deploy, and maintain successful data management solutions.

You can find Informatica Velocity resources at http://velocity.informatica.com. If you have questions, comments, or ideas about Informatica Velocity, contact Informatica Professional Services at ips@informatica.com.

Informatica Marketplace

The Informatica Marketplace is a forum where you can find solutions that extend and enhance your Informatica implementations. Leverage any of the hundreds of solutions from Informatica developers and partners on the Marketplace to improve your productivity and speed up time to implementation on your projects. You can find the Informatica Marketplace at https://marketplace.informatica.com.

Informatica Global Customer Support

You can contact a Global Support Center through the Informatica Network or by telephone.

To find online support resources on the Informatica Network, click **Contact Support** in the Informatica Intelligent Cloud Services Help menu to go to the **Cloud Support** page. The **Cloud Support** page includes system status information and community discussions. Log in to Informatica Network and click **Need Help** to find additional resources and to contact Informatica Global Customer Support through email.

The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Introduction to Enterprise Data Preparation

This chapter includes the following topics:

- Enterprise Data Preparation Overview , 9
- Enterprise Data Preparation Concepts, 10
- Data Discovery and Analysis Process, 12

Enterprise Data Preparation Overview

With the advent of data engineering technologies, many organizations are adopting a new information storage model called data lake to solve data management challenges. The data lake model is being adopted for diverse use cases, such as business intelligence, analytics, regulatory compliance, and fraud detection.

A data lake is a shared repository of raw and enterprise data from a variety of sources. It is possible to store large volumes of structured and unstructured data from various enterprise systems within and outside the organization. Data in the lake can include raw and refined data, master data and transactional data, log files, and machine data.

Enterprise Data Preparation helps you derive more value from the data sources such as Hive, HDFS, Microsoft Azure data warehouse using a native connector, and Microsoft SQL Server using a JDBC V2 connector, and make data available to all users in the organization.

Organizations are looking to provide ways for different kinds of users to access and work with all of the data in the enterprise, within the Hadoop data lake as well data outside the data lake. They want data analysts and data scientists to be able to use the data lake for ad-hoc self-service analytics to drive business innovation, without exposing the complexity of underlying technologies or the need for coding skills. IT and data governance staff want to monitor data related user activities in the enterprise. Without strong data management and governance foundation enabled by intelligence, data lakes can turn into data swamps.

Enterprise Data Preparation is a collaborative self-service data discovery and preparation solution for data analysts and data scientists. It enables analysts to rapidly discover and turn raw data into insight and allows IT to ensure quality, visibility, and governance. With Enterprise Data Preparation, analysts to spend more time on analysis and less time on finding and preparing data.

Enterprise Data Preparation provides the following benefits:

- Data analysts can quickly and easily find and explore trusted data assets within the data lake and outside the data lake using semantic search and smart recommendations.

- Data analysts can transform, cleanse, and enrich data in the data lake using an Excel-like spreadsheet interface in a self-service manner without the need for coding skills.
- Data analysts can publish data and share knowledge with the rest of the community and analyze the data using their choice of business intelligence and data analysis tools.
- IT and governance staff can monitor user activity related to data usage in the lake.
- IT can track data lineage to verify that data is coming from the right sources and going to the right targets.
- IT can enforce appropriate security and governance on the data lake
- IT can operationalize the work done by data analysts into a data delivery process that can be repeated and scheduled.

Enterprise Data Preparation Concepts

To successfully use Enterprise Data Preparation, you must understand the concepts that are used in the application.

Data Lake

The data lake used by Enterprise Data Preparation is a centralized repository of large volumes of structured and unstructured data. A data lake can contain different types of data, including raw data, refined data, master data, transactional data, log file data, and machine data.

The data lake can be deployed on-premise or in the cloud. The data lake utilizes Hive, the Spark engine, and a Hadoop-compatible file system. The data lake must be collocated with the Informatica application services associated with Enterprise Data Preparation. For example, if the data lake is deployed in Amazon EMR, you must also deploy Enterprise Data Preparation, Enterprise Data Catalog, and the Informatica services in Amazon EMR.

Enterprise Data Preparation works with the security mechanism used by the cluster, including Kerberos, Apache Ranger, and Apache Sentry, to securely access data in the data lake.

You can upload assets such as comma-separated value files as Hive tables in the data lake using Enterprise Data Preparation.

You can also ingest data from external sources outside the data lake using Hadoop tools or Informatica Mass Ingestion.

You can prepare a Hive or file-based asset that exists in Enterprise Data Catalog. When you prepare a file-based asset, Enterprise Data Preparation creates an external temporary table in the Hive schema specified in the Enterprise Data Preparation Service.

When you publish prepared data, Enterprise Data Preparation writes the transformed input source to a Hive table or to a file in the data lake.

Data Asset

A data asset is source data that you work with as a unit. A data asset can include items such as a flat file, a table, or a table view. A data asset can include data stored in or outside the data lake, such as in external databases.

You use Enterprise Data Preparation to search for assets described in the Enterprise Data Catalog, which serves as a centralized repository that stores the metadata for assets extracted from various external sources. Enterprise Data Catalog contains metadata such as profile statistics, asset ratings, data domains, and data relationships for each asset. When you search for assets in Enterprise Data Preparation, you actually search the metadata in Enterprise Data Catalog.

After you find the data asset you are interested in, you can add the data asset to a project and begin preparing the data for analysis.

Projects, Worksheets and Recipes

A project is a container that stores data assets and worksheets.

When you add a data asset to a project, Enterprise Data Preparation creates a corresponding worksheet that contains a sample of the data in a spreadsheet-like presentation. You use the worksheet to preview the data and to prepare it for publication. You do not directly change the data in the input source.

When you publish the prepared data, Enterprise Data Preparation applies the recipe to the data in the input source and creates a new data asset.

Enterprise Data Preparation captures each step you complete in the worksheet in the recipe for the worksheet. When you publish the prepared data, Enterprise Data Preparation applies the recipe to transform the data.

Data Publication

Data publication is the process of making data that you prepared in a worksheet available in the data lake.

When you publish prepared data, Enterprise Data Preparation writes the transformed input source to a Hive table in the data lake. Other analysts can add the published data to their projects and create new data assets, or use a third-party business intelligence or advanced analytic tool to run reports to further analyze the published data.

Data Visualization

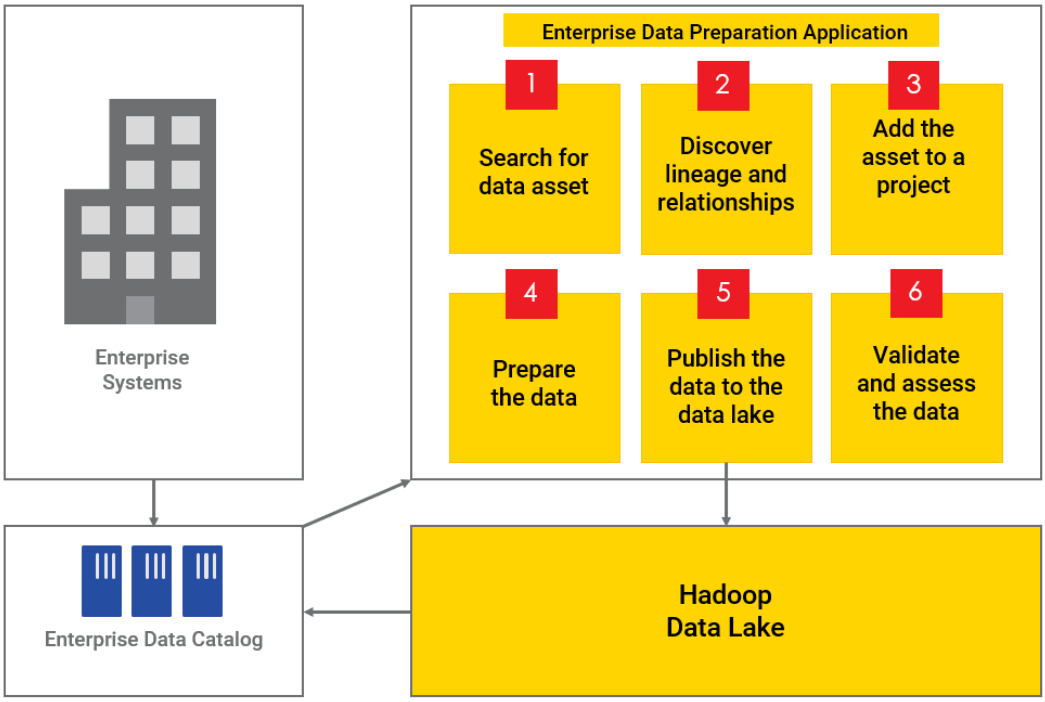
Data visualization is the process of assessing and validating published data using ad-hoc queries on the data to generate charts, tables, and other visual formats.

You can run queries and use dynamic forms to view the data in the form of graphs, charts, and other visual formats to assess its relevance for your business purposes. You can only run visualization on worksheets that are prepared and published. Collaborators on the project can access the notebook of the worksheet. You can share the notebook so that all users with the required permissions can view the visualization of the worksheet.

Data Discovery and Analysis Process

Use Enterprise Data Preparation to discover and prepare data.

The following image shows the high-level tasks that you complete to prepare data for analysis using the application:



Complete the following high-level activities in Enterprise Data Preparation to prepare data.

Search for data assets.

Search Enterprise Data Catalog and the data lake for data assets you can use. Data assets can include items such as delimited files, tables and views, Avro files, JSON Lines files, Parquet files, and more.

For more information, see Chapter 2, “Search for Data Assets” on page 14.

You can also move data from outside sources such as external databases into the data lake. If you find data outside the data lake that you want to work with, ask your administrator to add the data to the data lake.

For more information, see “Importing a Data Asset into the Data Lake” on page 31.

Examine each asset and understand how it relates to other assets.

After you discover an asset in Enterprise Data Catalog, view details about the data within the asset, the end-to-end data flow for the asset, and relationships between the asset and other data assets in the catalog.

For more information, see Chapter 3, “Discover Data” on page 17.

Add the data asset to a project.

Add the discovered data assets to a project, which is a container for organizing the worksheets and input sources that you use to prepare the data.

For more information, see Chapter 6, “Create and Manage Projects” on page 33.

Prepare the data for analysis.

Prepare the data by combining, cleansing, transforming, and structuring the data so that it is ready for analysis. Enterprise Data Preparation provides you with numerous options for preparing data.

For more information, see Chapter 7, “Prepare Data” on page 40.

Publish the prepared data to the data lake.

After you finish preparing the data in the asset, you publish the prepared data to the data lake, where analysts and other users can access it.

For more information, see Chapter 8, “Publish Data” on page 96.

You can schedule publication of the prepared data to regularly update the asset in the data lake on a recurring basis.

For more information, see Chapter 10, “Schedule Export, Import and Publish Activities” on page 105.

Assess the quality of the published data.

After you publish the data to the data lake, you can visualize the data to ensure the content and quality are sufficient for use by analysts and other users. If you determine that the data is not ready for use, you can modify the data preparation steps and republish the data.

For more information, see Chapter 9, “Visualize and Assess Published Data” on page 102.

Note: Activities and operations you perform in Enterprise Data Preparation do not support high precision mode.

CHAPTER 2

Search for Data Assets

This chapter includes the following topics:

- Overview, 14
- Search Strings, 14
- Search Results, 15

Overview

You can search for and discover any data asset described in Enterprise Data Catalog. Data assets can include items such as a flat file, table, or view.

You can search for a data asset by name, use a wildcard character in your search, and Enterprise Data Preparation displays probable matches as you type the name of a required data asset. You can also search for assets such as users and data domains.

You can sort search results, filter search results, and see more information about data assets. After finding the data asset that you need, you can annotate and enrich it with custom attributes.

After you find the data asset that you're looking for, you can view the profiling details and the lineage of the data asset. You can see the relationship between data assets. You can also add the data asset to a project.

Search Strings

You can use search strings to search for data assets. You can use wildcard characters in search strings. Enterprise Data Preparation also lists search suggestions when you enter a search string.

Wildcard Search

You can use the asterisk character (*) and the question mark character (?) to perform a wildcard search.

- Asterisk character (*). If you enter `*` in the **Search** box and click the **Search** icon, Enterprise Data Preparation lists all the data assets in the catalog. You can also use the asterisk with parts of a data asset name. For example, to search for all the data assets that end with the string "_name," enter *_name in the **Search** box. Enterprise Data Preparation lists all of the data assets that match the search criteria, for example, `Associate_Name`, `party_name`, and `cust_name`.

- Question mark character (?). Use the `?` character to substitute individual characters in the name of an asset. For example, if the data asset that you want to search for begins with the letters HR, is followed by two numbers that denote the year, and ends with REPORT, enter the following string in the **Search** box: `HR??REPORT`. Enterprise Data Preparation lists all of the data assets that match the search criteria, for example, `HR12REPORT`, `HR13REPORT`, and `HR14REPORT`.

If you know the name of a column in a table, you can enter the column name and search for the table. The search results display all tables that have data or columns with the name as your search string.

Search Suggestions

When you type the first few letters of an asset name into the Search box, Enterprise Data Preparation suggests matching data asset names.

If you type the name of a data asset incorrectly, Enterprise Data Preparation compares the typed letters with names of existing data assets in the catalog. The probable matches then appear as search suggestions. For example, if you typed `sela` to search for the data asset named `salary`, Enterprise Data Preparation suggests `salary` as a probable match.

Note: The suggestions include all custom attributes that are of type string. Data assets or custom attribute names that include special characters do not get listed.

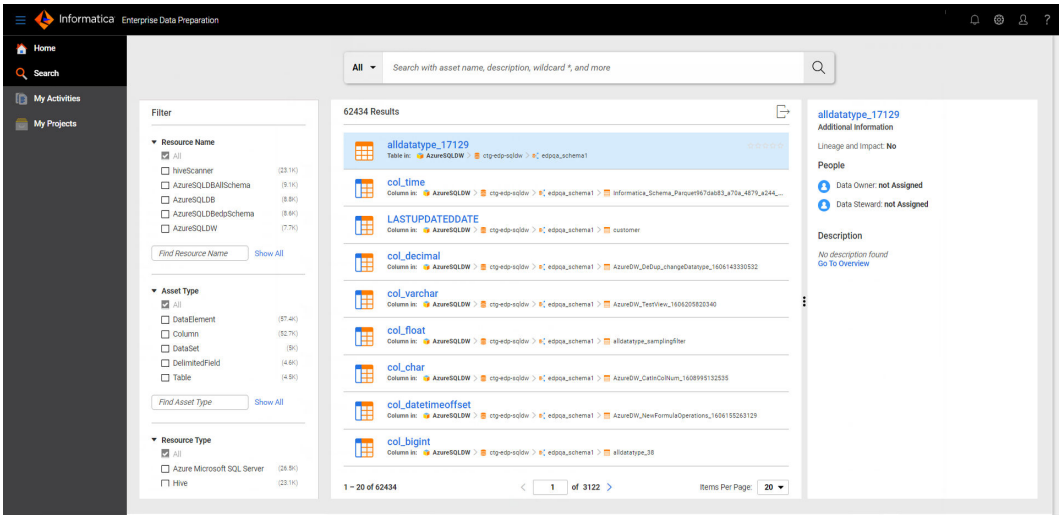
Search Results

Search results show assets that are in the data lakehouse as well as assets that are not yet in the data lake. Data assets that are in the data lakehouse are indicated with a blue dot in the icon.

To further refine the search results, you can use the available search filters in the **Filter By** section. The data lake containing assets is indicated by a blue dot in the Location filter, and resources in the data lake are denoted by a blue dot in the Resource Name filter.

You can also search and view data assets with filters under the **Project** > **Add Worksheet** > **Search** page.

The following image shows the Search Results page:



The Search Results page includes the following tabs:

Data Objects

Displays the data objects that are available in the catalog. A data object refers to a collection of tabular and mapping assets in a resource. Examples of data objects are tables, views, synonyms, and public synonyms.

Resources

Displays the resources that are available in the catalog. By default, the **Resource** filter option is selected in the **Asset Type** filter. The Resources tab displays only resources and not any other asset type.

Data Elements

Displays the data elements that are available in the catalog. A data element refers to any asset within a data set. For example, a column or view column within a table is a data element. Examples of data assets are columns, view columns, fields, and attributes.

Business Terms

Displays the business terms that are available in the catalog. Business terms include Axon glossaries and Business Glossary assets.

Reports

Displays the reports that are available in the catalog. Reports might be based on resources, such as Qlikview, Cognos, and SAP Business Objects.

CHAPTER 3

Discover Data

This chapter includes the following topics:

- Data Discovery Overview, 17
- Data Asset Views, 17
- Deleting a Data Asset, 23
- Access to Data, 24

Data Discovery Overview

After you search for data assets in the catalog, click a data asset to discover the data. During data discovery, you can view details about the data asset, lineage for the data asset, and relationships between the selected data asset and other data assets in the catalog. You can also import data assets into the lake and export data assets from the lake.

You can add an asset that is in the data lake to a project. For more information on adding a data asset to a project, see "Adding a Data Asset to a Project" on page 34.

To prepare a data asset that is not stored in the data lake, you must import the asset into the data lake. For information about importing data assets into the data lake, see "Importing a Data Asset into the Data Lake" on page 31.

Data Asset Views

Use the data asset views to get more information about the data asset by viewing the data asset details, previewing the data, and looking at the data relationship and lineage diagrams.

Enterprise Data Preparation displays additional details about the selected data asset in the following views:

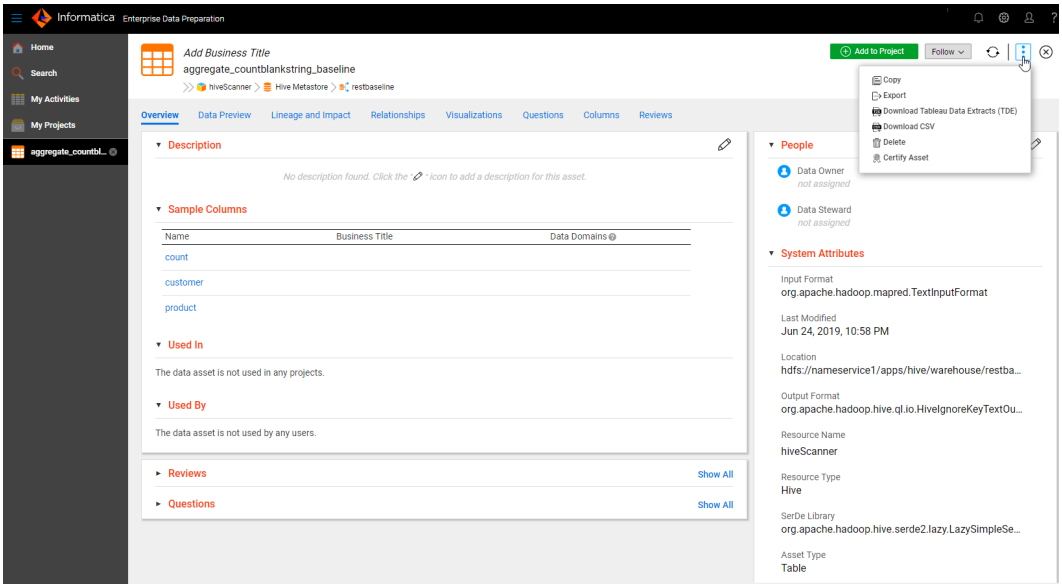
- Overview. Displays the details about the data asset. Details include the type of the data asset, the last time the data asset was updated, and the data source, along with other details.
- Data Preview. Displays a sample of the data in the data source.
- Lineage and Impact. Displays the lineage for the data asset selected. The lineage diagram shows the end-to-end data flow for the selected data asset.
- Relationships. Displays the relationships between the selected data asset and other data assets in the catalog.

- Columns or Fields. Displays details about the columns in the data asset, including the column data types and the percentage on distinct and non-distinct values. The tab displayed depends on the data asset type.

Asset View

When you select a data asset from the **My Activities** or **Projects** view, each data asset opens in an asset view labeled with the data asset name.

The following image shows the **Overview** view:



Let's explore what you can do in the **Overview** view:

- Click **Add to Project** to add the data asset to a project.
- Click the **Menu** icon to edit data asset properties, copy the data asset, export the data asset, download the data asset as a CSV file or TDE file, or delete the data asset.
- Select an option from the **Follow** menu to receive notifications when changes are made to the asset.
- Click the **Notifications** icon in the upper right corner of the page to view notifications about assets you follow.
- If you are a data owner, data steward, or subject matter expert for the asset, you can certify the asset in Enterprise Data Catalog. You can also decertify a certified asset if the asset contains obsolete information.
- Preview data on the **Data Preview** tab.
- Discover the lineage for the flow of data for the data asset on the **Lineage and Impact** tab.
- Discover the relationships between the open data asset with other data assets on the **Relationships** tab.
- Add questions you have about the asset on the **Questions** tab.
- View details on the data in the asset on the **Columns** or **Fields** tab. The tab displayed depends on the data asset type.
- Read reviews of the asset from other users or add your own review on the **Reviews** tab.
- For assets that are published to the data lake, you can click the **Visualize** tab to visualize the data before you operationalize the recipes created during publishing.

Certifying a Data Asset

If you are a data owner, data steward, or subject matter expert for the asset, you can certify the asset in Enterprise Data Catalog. You can also decertify a certified asset if the asset contains obsolete information.

You can certify an asset to endorse the asset so that other users can use the asset as a trustworthy one over the assets that are not certified. Certifying the asset indicates the reliability of the asset in Enterprise Data Catalog.

You can add a description and a maximum of 10 tags to the asset when you certify an asset. You add tags to make the asset easily searchable in the catalog. You can also filter certified assets in search results.

1. Select the menu icon, and the select **Certify Asset**.
2. Add a comment and associate tags. You can add a maximum of 10 tags in the Add Tags field.
3. Click **OK**.

Associating a Business Term with an Asset

You can associate a business term with any asset in the catalog that is not a business glossary asset. You can associate one business term with a data asset.

When you associate a business term with a data asset, Enterprise Data Preparation displays the business term with the asset in the following places:

- In the search results next to the asset name.
- In the Asset Details view next to the asset name.
- In the Lineage and Impact diagram above the asset name when you show business terms in the diagram.

Complete the following steps to associate a business term with a data asset.

1. In the Asset Details view for the asset, click **Associate a Business Term**.

   Alternatively, you can click the Edit Properties icon for the asset in the search results.
2. In the Edit Properties dialog box, click **Business Terms**.
3. The options to associate a term vary based on whether the asset has inferred terms:
   - If one or more data domains with an associated business term are assigned to the asset, the inferred terms appear in the **Recommended Business Terms** section. To accept an inferred term, click the check mark icon next to the term.
   - If the asset has no inferred terms or you want to associate a different term with the asset, select the business glossary from the **Glossary** list. Then, select the term that you want to associate from the list of terms.

     If the list of terms is long, you can search for a term in the list by entering the first few characters of the term name in the search field.
4. Click **OK**.

Data Preview View

The **Data Preview** view displays the first 500 rows of a data asset. The **Data Preview** is also available for data assets that are not in the data lake.

You can view data from tables and views on the Data Preview view. Each page shows 20 rows of data. You can use pagination options to navigate if a table or view contains more than 20 rows of data. Consider the following guidelines:

- To preview data assets that are not in the lake, you must connect to the data source. Contact your Administrator for the connection details to the required source.
- If the data asset preview does not show any data, click **Select Connection**. The Connections dialog box appears with a list of the valid connections. Select the connection from the list. Click **OK**.
- If Enterprise Data Preparation does not support the data type of a column of the data source, the column will appear as a blank column.
- For data sources such as Microsoft SQL Server and Microsoft Azure Data Warehouse, the blank values are displayed as NULL values. If you apply filters on the blank values, there will be no operation on the Data Preview screen.

Filter Data Assets

You can also filter the data during data preview to better assess data assets. You can add filters for multiple fields and apply combinations of such filters. If data preview contains profiling results, you can view column value frequencies for string values. In the absence of value frequencies, you can use the free form text filtering options.

1. Click the filter icon.
2. To add a filter to a column, click **Add Field**.
3. Use one of the following comparison operators to assign a value to filter:
   - Equals
   - Less than
   - Less than or equal to
   - Greater than
   - Greater than or equal to
   - Between
   - Is NULL
   - Is not NULL

   Filter conditions depend on the data types. Additionally, you can use one of the following filters:
   - Not Blank
   - Is Blank
   - Any time
   - Past 30 days
   - Past 365 days
   - Custom Range
4. Click **Apply** to view the filtered results.

Import Data Asset to Lake

You can import a data asset to lake from the Data Preview view.

1. In Data Preview view, click **Import to Lake**.

   A window with connection name and type appears.
2. (Optional) Click **Apply Row Filters for this import** to import the filtered data assets.

   You can use this option if you have applied filter in the Data Preview view.
3. Proceed to select the resource and specify the target details.
4. Proceed to specify the type of import.
   - **Import Now**: Available for all scenarios.
   - **Schedule Import**: Available only if you have not applied filter on the imported data assets.
5. Click **OK** to start the import.

You can view the status on the **My Activities** view.

Lineage View

The **Lineage and Impact** view shows the start point of the data, describes the path, and shows how the data asset arrives at the end point. The lineage diagram shows the end-to-end data flow for the selected data asset.

You can use the following icons to view the lineage diagram based on your requirements:

- Use the **Zoom in** icon to increase the magnification and the **Zoom out** icon to decrease the magnification.
- Click **View Adjacent Assets** to toggle between displaying and hiding the immediate neighboring data assets.
- Click the **Reset** icon to reset the lineage diagram back to the initial view size.

Expanded Lineage View

The **Expand Lineage Path** icon in the lineage diagram indicates that there are more objects in the data flow. You can place your pointer on any data asset toward the origin or end point in the data flow to view the expanded path.

Click the number with a plus sign (+) in the origin or the end point to view all the data assets included.

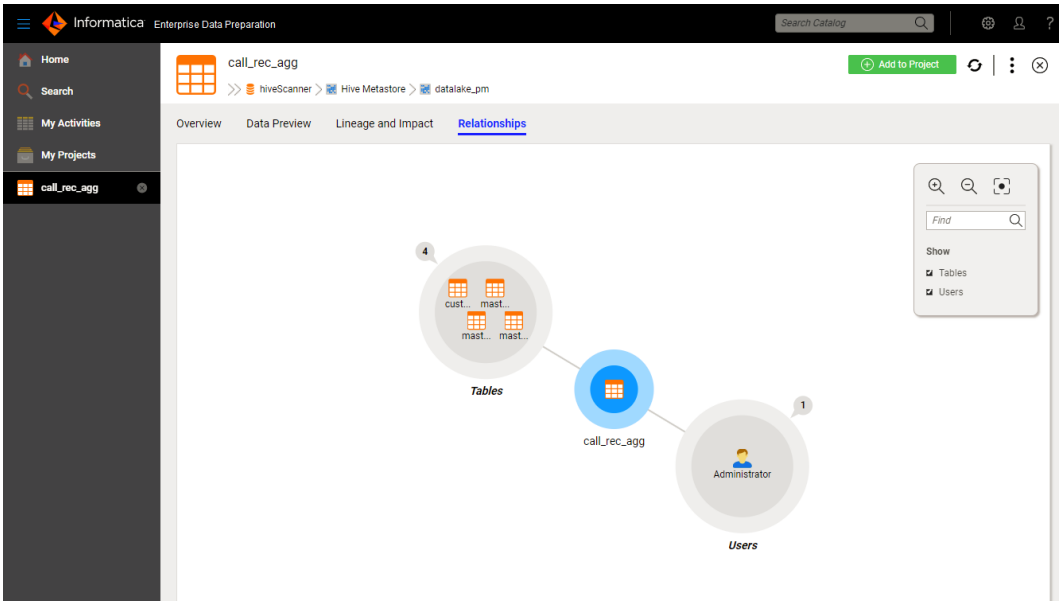
Click any of the data assets listed on the lineage diagram to view a summary of the details for that data asset.

You can also view lineage of individual columns in a table. Activities including copy, import, export, publication, and upload will add column-level lineage information in the Catalog. You can also view column-level lineage details for data preparation operations that you perform such as transformations, join, lookup, union, and aggregate. Expand the data asset and select a column to view its lineage details.

Relationship View

The **Relationship** view displays the relationship in a diagram that shows how the selected data asset is related to the other assets. You can view the relationships between the selected data asset and other data assets in the catalog.

The following image shows the **Relationship** view:

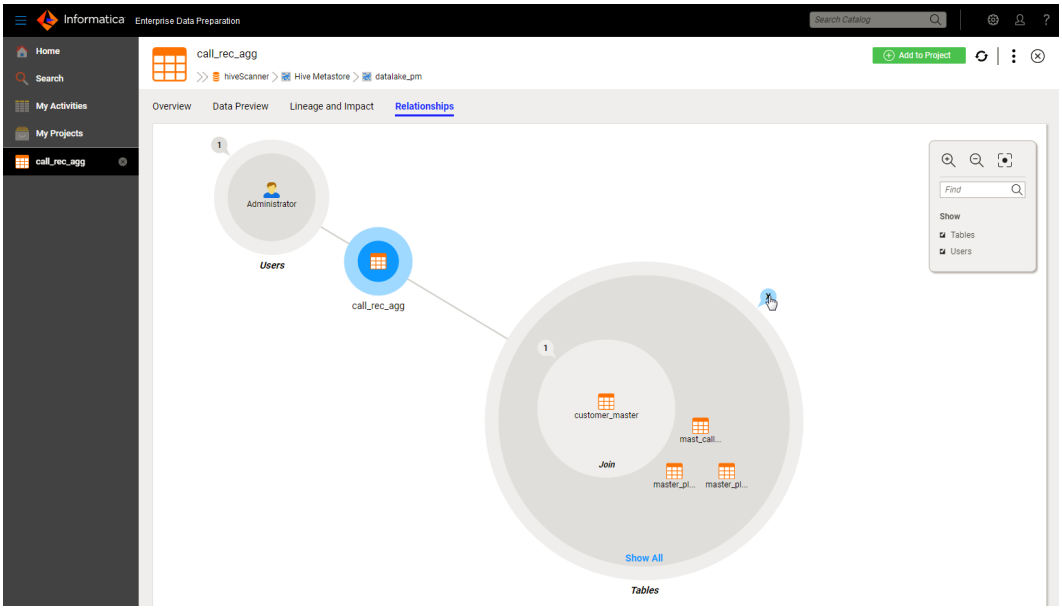


Based on the type of data asset, you can use the following icons to view the relationship between the selected data asset and other data assets:

- Use the **Zoom in** icon to increase the magnification and the **Zoom out** icon to decrease the magnification.
- Click the **Show** option to toggle between displaying and hiding the related data assets. This option changes based on the data asset that you selected.
  For example, if you selected a view that has related tables, you get the option **Tables** to show the tables related to that view. If you selected a table that has related views, you get the option **Views** to view all the views associated with the selected table. Or, if you selected a table that has related users, you get the option **Users** to view all the users for the selected table.
- Click the Reset icon to reset the relationship diagram back to the initial view size.

Click a related data asset to view a summary of the details for that data asset.

The number at the top of a related asset in the diagram represents the number of related asset types included in the selected data asset. Click this number to display all the related asset types included in the data asset. The following image shows the impact of clicking the number 4 above Tables:



Click **Show All** to view all the related data assets included in the selected data asset.

Deleting a Data Asset

You can delete a data asset stored in the lake. You need to check before deleting data assets as they can be used by other users, projects, and publications.

1. On any data asset view, click the **Manage Data Assets** icon and then click **Delete**.

   The **Delete** dialog box appears.
2. If the data asset is not part of any project, you can click **OK** to delete the project. If the data asset is in a project, the following message appears:

   This data asset is being used in 1 projects. It is recommended to go to Object Overview page to see details of which project it is used in. Click **OK** to delete anyway.
3. Click **Cancel** if you want to view the details of the project. If the project needs to be deleted, go to My Projects page and delete the project.

   Note: You need to check before deleting data assets and projects. Data assets can be shared by different users, projects, and publications. Check with the collaborators before deleting a data asset in a shared project.
4. Click **OK** to delete the project. Check the progress of the delete activity on My Activities page. You can see if the deletion is in progress, completed, or if it has failed.

Access to Data

Based on your organization's security policies, you might not have access to a data asset stored in the data lake.

If you find a data asset in the search results but find that you cannot view the data asset details, you might not have the permission to view the data asset. You must contact your administrator to request access to the data asset

CHAPTER 4

Upload Data

This chapter includes the following topic:

- Data Upload Overview, 25

Data Upload Overview

Data upload is the process of adding your own data to the data lake from a local drive. After you upload a file, you use Enterprise Data Preparation to prepare the data and publish it to the data lake.

You can upload files in the following formats to the data lake:

- Microsoft Excel (.xls, .xlsx)
- comma delimited files (.csv, .dat, .txt)
- JavaScript Object Notation (.json)
- Apache Avro (.avro)
- Apache Parquet (.parquet)

Depending on the upload option you choose, you can upload data to a Hive table in the data lake, or to an HDFS location in the data lake. To publish to an HDFS location, your catalog administrator must create and configure an HDFS resource that represents the location in Enterprise Data Catalog.

You can choose from the following upload options:

Upload a file directly to the data lake.

You can upload a comma delimited file, an Avro file, a JSON file, or a Parquet file in UTF-8 format directly from your local drive to the data lake, without previewing the data. You might choose this option if you want to upload a file without previewing the data.

Allow CLAIRE to determine the file structure, and then upload the file to the data lake.

You can upload the data in a comma delimited file or a Microsoft Excel spreadsheet to the data lake. When you upload the file, Enterprise Data Preparation uses the CLAIRE embedded discovery engine to determine the structure of the file and display a preview of the data.

When you use this option to upload an Excel spreadsheet, the CLAIRE engine discovers the sheets and tables in the spreadsheet. You can select the sheet and table you want to preview.

Define the file structure, and then upload the file to the data lake.

You can upload the data in a comma delimited file from your local drive to the data lake. When you upload the file, you can preview the data, specify the structure of the file, and configure column

attributes to meet your requirements. You might choose this option if you need to modify column attributes before you upload the file.

Note: All files that you upload to the data lake must use the UTF-8 character encoding format.

Uploading a File Directly to the Data Lake

You can upload a comma delimited file, an Avro file, a JSON file, or a Parquet file in UTF-8 format directly from your local drive to the data lake, without previewing the data.

1. In the **Projects** view, click **Upload Data**.

   The **Upload Data** dialog box appears.
2. Drag the file into the dialog box, or click **Choose File** browse to the file.

   If you browse the file, select the file in the Resource Explorer.
3. Select **Upload file as is**.
4. Select the resource that represents the location in the data lake to which you want to upload the file.

   The application displays the folders within the location.
5. Select the folder to which you want to upload the file.
6. Verify the file name and the file extension.
7. Select the action to take:
   - Select **Create** to create a new asset in the data lake.
   - Select **Overwrite** to overwrite an existing file with the same file name in the data lake.
8. Click **Upload**.

Uploading a File with CLAIRE® Discovered Structure

You can upload the data in a comma delimited file or a Microsoft Excel spreadsheet to the data lake. When you upload the file, Enterprise Data Preparation uses the CLAIRE® embedded discovery engine to determine the structure of the file and display a preview of the data.

When you upload a file to an HDFS location, you can choose whether to save the file as a comma separated value file, an Avro file, or a Parquet file.

If you upload an Excel spreadsheet that contains multiple sheets, you can select the sheet to preview. If a sheet contains multiple tables, you can select the table to preview. The preview displays the first sheet and the first table in the sheet by default.

Note: Uploading Excel spreadsheets is a preview feature.

1. In the **Projects** view, click **Upload Data**.

   The **Upload Data** dialog box appears.
2. Drag the file into the dialog box, or click **Choose File** browse to the file.
3. Select the **Upload with CLAIRE discovered structure** option.
4. Click **Next**.
5. When you upload an Excel spreadsheet, the preview displays the first sheet and first table in the worksheet by default. Select a different sheet or table to preview the data within the sheet or table.
6. Click **Next**.
7. Column types are inferred from the file by default. Click **Strings** to convert all columns to the string type.

8. To modify a column name, click the column name in the Column Configuration panel. The column name cannot exceed 120 characters.

   The column name updates in the preview panel below.
9. Click **Next**.
10. Select the resource that represents the location in the data lake to which you want to upload the file.
    - To upload the file to Hive, select the Hive resource that represents the Hive schema in the data lake.
      - Select the Hive schema to which to upload the file.
      - Verify the file name and type.
      - Select **Create** to create a new table in the data lake.
      - Select **Overwrite** to overwrite an existing table with the same name in the data lake.
      - Select **Append** to append the data to an existing Hive table. The data and the Hive table must have the same structure. You can only select this option if you upload a file to Hive.
    - To upload the file to HDFS, select the HDFS resource that represents the HDFS location in the data lake.
      - Select the folder to which you want to upload the file.
      - Verify the file name.
      - Select whether to save the file to the folder as a comma separated value file, an Avro file, or a Parquet file.
      - Select **Create** to create a new file in the data lake.
      - Select **Overwrite** to overwrite an existing file with the same file name in the data lake.
11. Click **Upload**.

Uploading a File with User Defined Structure

You can upload the data in a comma delimited file from your local drive to the data lake. When you upload the file, you can preview the data, specify the structure of the file, and configure column attributes to meet your requirements.

Note: You cannot upload numeric data with a precision of more than 15 digits.

1. In the **Projects** view, click **Upload Data**.

   The **Upload Data** dialog box appears.
2. Browse for a file or drag a file into the dialog box.

   The name of the selected file appears on the window.
3. Select **Upload with user provided structure**.
4. Click **Next**.
5. Specify the code page and format of the file.

The following table describes the file properties that you specify:

Property	Description
Code Page	Character encoding used in the file. The default code page is UTF-8.
Delimiters	Character used to separate columns of data in the file. You can use multiple delimiters in the file. The default delimiter is a comma.
Text Qualifier	Character used to specify the start and end of a text string. If you select single quotes, the Enterprise Data Preparation Service ignores delimiters within pairs of single quotes. If you select double quotes, the service ignores delimiters within pairs of double quotes.
Start import at row	Row number at which Enterprise Data Preparation Service starts the import. The default starting import row is 1.
Import field names from first row	Select this option to use the values in the first row as names of the imported columns.

The page also displays a preview of the rows and columns to be created from the file.

6. Click **Next**.
7. Optionally click a column in the Column Configuration panel to modify the column attributes.

   The following table describes the column attributes that you specify for the columns:

Attribute	Description
Column Name	Name of the column. The column name cannot exceed 120 characters.
Column Type	Data type of the column.
Precision	If the column has a numeric data type, set the precision. The value of the precision must be larger than the value of the scale.
Scale	If the column has a numeric data type, set the scale. The value of the scale must be smaller than the value of the precision.
Format	If the column has a data type of DateTime, set the format.

8. Click **Next**.
9. Specify Hive schema and table properties.

The following table describes the Hive schema and table properties that you specify:

Property	Description
Hive Schema	Name of the hive schema in which to upload the file.
Hive Table	Name of the table in the Hive schema in which to upload the file. The name of the table must be in lower case. The name can include letters, numbers, and the underscore character. The first character of the name cannot be an underscore or a number.
Action on Table	Action that the Enterprise Data Preparation Service can perform during the import process. - To create a Hive table, select **Create**. - To append to an existing Hive table with the same structure, select **Append**. You cannot append to a partitioned table. - To overwrite an existing Hive table, select **Overwrite**. The Enterprise Data Preparation Service drops the existing table and creates a new table for the uploaded data. You cannot overwrite a partitioned table.

10. Click **Upload**.

CHAPTER 5

Import and Export Data

This chapter includes the following topics:

- Overview, 30
- Importing a Data Asset into the Data Lake, 31
- Exporting a Data Asset from the Data Lake, 32

Overview

You can import data assets from external sources to the data lake. After you import an asset, you use Enterprise Data Preparation to prepare the data. You can also export a published asset from the data lake to an external source.

Import data from external sources to the data lake.

You can import a data asset from an external source into the data lake. You can import the data with multibyte characters except special characters [\s!@#$%^&*()\-+={}[]|\\;:"'<>,\.\/\?] and multibyte characters that have known Hive limitations.

The metadata for an asset you want to import must exist in Enterprise Data Catalog. Your Enterprise Data Catalog administrator must create a resource to scan the external source.

Your Enterprise Data Catalog administrator must also create an HDFS resource for each HDFS location into which you want to import data.

You can import data from the following sources:

- Amazon Redshift
- Microsoft Azure SQL Database
- Microsoft SQL Server
- Oracle
- Teradata

Export data to external targets from the data lake.

You can export a data asset or a publication from a Hive table in the data lake to an external target.

You can choose to schedule import and export activities. Scheduling an activity enables you to import or export updated data assets on a recurring basis. For more information about scheduling activities, see Chapter 10, “Schedule Export, Import and Publish Activities” on page 105.

Importing a Data Asset into the Data Lake

You can import a data asset from an external source into the data lake.

The metadata for an asset you want to import must exist in Enterprise Data Catalog.

You can choose to import an asset immediately, or you can schedule the import activity. Scheduling an import activity enables you to import updated data assets on a recurring basis.

For more information about scheduling an import activity by creating a schedule or using an existing schedule, see Chapter 10, “Schedule Export, Import and Publish Activities” on page 105.

1. Click **Search**, and then search for an asset you want to import into the data lake.
2. Select the asset in the search results.
3. On the Overview page for the selected asset, click **Import to Lake**.

   The **Import** dialog box appears.
4. Select the connection to use to connect to the external source, and then click **Next**.

   You must have **Read** and **Execute** permissions for the database connection you use to import a data asset.
5. Select the resource that represents the location in the data lake into which you want to import the data.
   - To import the data to Hive, select the Hive resource that represents the Hive schema in the data lake.
     - Select the Hive schema to which to import the data.
     - Verify the file name and type.
     - Select **Create** to create a new table in the data lake.
     - Select **Overwrite** to overwrite the existing table in the data lake.
     - Select **Append** to append the data to an existing table. The data and the table must have the same structure.
   - To import the data to HDFS, select the HDFS resource that represents the HDFS location in the data lake.
     - Select the folder to which to import the data.
     - Verify the file name.
     - Select whether to save the file to the folder as a comma separated value file, an Avro file, or a Parquet file.
     - Select **Create** to create a new file in the data lake.
     - Select **Overwrite** to overwrite an existing file in the data lake.
6. Choose whether to import the asset now, or to schedule the import activity.
   - Select **Import Now** to start the import activity.
   - Select **Schedule Import** to schedule the import activity.
     If you decide to schedule the activity, specify what to do if the asset already exists in the data lake.
     - Select **Append** to append the data to an existing table.
     - Select **Overwrite** to overwrite the existing table.
7. Click **OK**.

   You can monitor the progress of the import activity on the **My Activities** page in Enterprise Data Preparation.

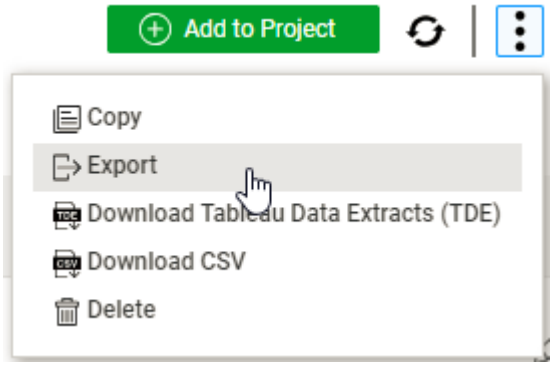
Exporting a Data Asset from the Data Lake

You can export a data asset or a publication from a Hive table in the data lake to external targets.

You can choose to export the asset immediately, or you can schedule the export activity. Scheduling an export activity enables you to export updated data assets on a recurring basis. For more information about scheduling activities, see Chapter 10, “Schedule Export, Import and Publish Activities” on page 105.

1. Click **Search**, and then search for the asset you want to export.
2. Click the asset in the search results.
3. On the **Overview** page for the data asset, click the **Manage Data Assets** icon, and then click **Export**.

   The following image shows the **Export** option selected from the **Manage Data Assets** icon:



   The **Export** dialog box appears.
4. Select the connection to use to connect to the target, and then click **Next**.

   You must have **Read** and **Execute** permissions for the connection you use to export a data asset.
5. Enter the name of the schema to which to export the asset.
6. Enter the name of the table to create in the target. The name can include letters, numbers, and the underscore character. The first character of the name cannot be an underscore or a number.
7. If a table with the same name already exists in the target, choose whether to append the data to the table, or to overwrite the table.
   - Select **Append** to append the data to the existing table.
   - Select **Overwrite** to overwrite the existing table.

     Choose to overwrite the table if the source table has a different schema structure than the exported table. Enterprise Data Preparation drops the table and re-creates the table with the data from the exported table.
8. Choose whether to export the asset now, or to schedule the export activity.
   - Select **Export Now** to start the export activity.
   - Select **Schedule Export** to schedule the export activity.
     If you decide to schedule the activity, specify what to do if the asset already exists in the target.
     - Select **Append** to append the data to an existing table.
     - Select **Overwrite** to overwrite the existing table.
9. Click **OK**.

   You can monitor the progress of the export activity on the **My Activities** page in Enterprise Data Preparation.

CHAPTER 6

Create and Manage Projects

This chapter includes the following topics:

- Projects Overview, 33
- Creating a Project, 34
- Adding a Data Asset to a Project, 34
- Editing a Project, 34
- Copying a Project, 35
- Viewing Project Flows, 35
- Viewing Project History, 36
- Sharing a Project, 36
- Changing the Project Owner, 37
- Locked Projects, 37
- Adding a Worksheet, 38
- Deleting a Worksheet, 38
- Deleting a Project, 39

Projects Overview

A project is a container for organizing the worksheets and input sources that you use to prepare data. You can add data assets that are stored in the data lake as Hive tables to a project.

You can import a data asset from external sources into the data lake such as Oracle and Teradata databases. Once the import activity is completed successfully, a Hive table is created in the data lake that you can add to a project. After you publish a data asset to the data lake, the publication appears in the project.

Projects are private and can only be viewed by the user who creates the project. However, you can share projects with other analysts to collaborate on activities related to the project. If you only share a project with view permissions, the collaborators can only view the project.

You can also edit a project to change the project properties or delete a redundant project. When you edit a shared project, the project is locked for editing by the user who edits the project. In a project, you can also get recommendations for data assets based on the data assets added to the project.

You can access projects from the **Home** view or the **My Projects** view. The projects in the **Projects Overview** section appear as cards by default.

Creating a Project

Create a project to organize the worksheets and input sources that you use to prepare data. You can add data assets that are stored in the data lake to any project.

1. You can create a project in the following ways:
   - On the **Home** page, click **New Project**.
   - On the **My Projects** page, click **New Project**.
   - To create a project from the **Search Results** or **Open Assets** view, select a data asset and click **Add to Project**. Then click **New Project**.

   The **Add to Project** dialog box appears.
2. Enter a name and an optional description.
3. Click **Finish**.

Adding a Data Asset to a Project

Add a data asset to a project to prepare the data asset. Enterprise Data Preparation adds the data asset as a worksheet to the project to prepare the data asset.

When you view the details of a data asset, you can add the data asset to a project. On the **Search Results** view or the **Open Assets** view, click the name of a data asset. You can also add a data set when you add a worksheet. For more information, see "Adding a Worksheet" on page 38.

1. On the data asset view, click the **Add to Project** icon. You can also add a data asset by clicking **Search for asset** on the project page.

   The **Add to Project** dialog box appears.
2. Add the data asset to an existing project or to a new project.
   - To add to an existing project, select a project under **Recent Projects** or **All Projects**.
   - To add to a new project, click **Create New** and enter a name and optional description for the project.
3. Click **OK**.

   The data asset appears as a worksheet in the selected project.

Editing a Project

Edit a project to change the name and description of the project.

To edit a project, you must have edit permissions. You can also hover over the project name or description, and then click  to edit a project.

1. In the **Projects** view, select a project.
2. Click the **Manage Projects** icon and select **Edit Properties**.

   The **Edit Project** dialog box appears.
3. Edit the name or description.

4. Click **Save**.

Copying a Project

When you copy an existing project, Enterprise Data Preparation also copies the recipes and ingredients in the project.

The following rules apply when you copy a project:

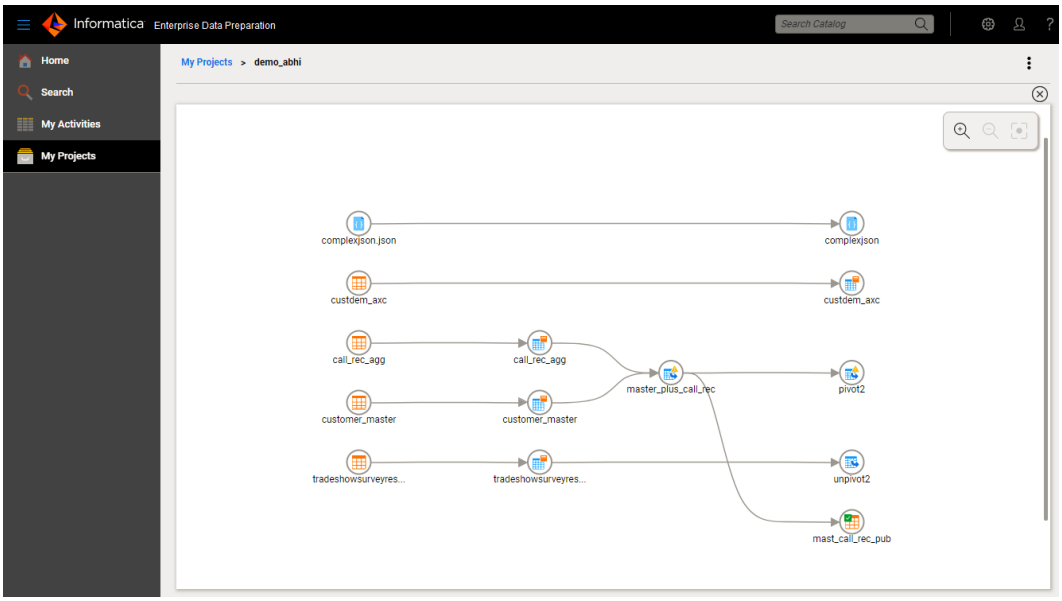
- The published datasets and collaborators of the project do no get copied.
- After you copy a project and then open it to prepare, Enterprise Data Preparation triggers the sampling of the copied worksheets.
- You can prepare and initialize only the existing worksheets in the copied project in one go and perform data preparation and publishing.

1. In the Projects view, select and open a project.
2. Navigate to **Manage Project** > **Copy Project**.
3. Enter a new name, and add a description to create a copy.

Viewing Project Flows

You can view a flow diagram that shows you how worksheets in a project are related and how they are derived. The diagram is especially useful when you work on a complex project that contains numerous worksheets and includes numerous assets.

The following image shows the flow diagram for a project:



1. In the **Projects** view, select a project.

2. Click the **View Project Flow** icon.

   The project flow view displays the assets in the project and the relationships between assets.
3. Click an icon in the diagram to view details for the corresponding asset.

Viewing Project History

You can review the activities performed within a project to gain insights into the project, or to perform root cause analysis to uncover issues with a project.

You can review the following activities:

- All activities within the project
- Activities performed on worksheets within the project
- Publication activities for assets within the project

You can click  to view project history or perform the following steps to view the history for a project:

1. In the **Projects** view, select a project.
2. From the **Manage Projects** menu, select **View Project History** .

Sharing a Project

Share a project with other users to collaborate on the data assets in the project. When you share a project with a user, you can also assign a role to the user.

1. In the **Projects** view, select a project.
2. On the Project Details page, you can perform one of the following operations:

   - Click  to share a project. The red badge on the icon indicates the number of collaborators in a project.
   - Click the manage projects icon and then select **Share Project**.

   The **Share Project** dialog box appears.
3. Enter the full name of the user that you want to add as a collaborator on the project. If you enter the login name, the search will not yield any results.

   Enterprise Data Preparation displays a list of names that match the name you enter.
4. Select the user name from the list.
5. In the list, choose to assign a permission to the user to enable the user to view, edit, or become a project co-owner:

   - Assign the **View** permission to enable the user to view the project. The user can view the project and cannot edit, share, publish, or prepare the worksheet.

- Assign the **Edit** permission to enable the user to view and edit the project. The user can edit the properties, view data assets, prepare data assets, and publish data assets.
- Assign the **Co-owner** permission to enable the user to view, edit, and add data assets or work on the worksheets in the project. The user can also edit properties, share the project, prepare, and publish. Co-owners can share projects with other users and give view and edit permissions to other users. Co-owners cannot assign the co-owner permission to other users.

Collaborators cannot change the owner of the project, add another co-owner to the project, or delete the project. The owner of the project can perform all these operations. However, the owner cannot unlock a project. Domain administrators and users with administrator privileges can perform all the operations in projects including unlocking the projects.

6. Click **OK**.

Changing the Project Owner

Change the project ownership from one user to another user. Project owners, administrators, and users with administrator privileges can change the project ownership.

1. In the **Projects** view, select a project.
2. Click the manage projects icon, and then select **Change Owner**.

   The **Change Owner** dialog box appears.
3. Enter the full name of the new owner. If you enter the login name, the search will not yield any results.

   Enterprise Data Preparation displays a list of names that match the name you enter.
4. Select the user name from the list.
5. Click **OK**.

Locked Projects

When you edit shared projects, the projects are locked automatically for editing. Locks on shared projects are automatically released when you sign out of Enterprise Data Preparation.

A project is locked when a user performs any of the following tasks:

- Edits a project.
- Shares a project.
- Prepares a project.
- Publishes a project.
- Changes the project owner.
- Deletes the project.

When other users try to access this project, a project locked icon appears. They cannot perform any of these tasks when a project is locked. To view the name of the user who is working on a project, hover the mouse over the locked icon.

If a user is performing tasks other than preparing and publishing, other users can access the project and view the data assets in the worksheet. However, if a user has a worksheet open in view mode, other users cannot access the worksheet.

Administrators and users with administrator privileges can unlock projects.

Adding a Worksheet

You can add a worksheet to a project.

1. In a project, click **Add a Worksheet**.

   You can view the recommendations panel for suggestions on data assets.
2. Select the data assets and a connection.
3. Click **Add Selected Worksheet**.

Recommendations

Enterprise Data Preparation uses CLAIRE to recommend additional data assets to add to a project based on other data assets added to the project.

You can view recommendations for data assets used in different projects by trusted people in your organization or for data assets that share similar columns. You can set the maximum number of recommendations to display from **App Configuration** > **Advanced Properties**. Enterprise Data Preparation makes the following types of recommendations:

Additional recommendations.

Data assets that are used in similar projects by other users.

Alternate recommendations.

Data assets that share similar columns with assets in the project.

Hover over the **Why?** text in a recommendation card for details on why Enterprise Data Preparation recommends that you add a data asset to the project.

Deleting a Worksheet

You can delete a worksheet from a project that you own. You cannot delete a worksheet if it has dependent worksheets.

1. In the **Projects** view, select a project.
2. In the project page, select a worksheet.
3. Click the **Manage Worksheets** icon, and then select **Delete Worksheet**.

   The **Delete Worksheet** dialog box appears.
4. Click **Yes**.

Deleting a Project

You can delete projects that you own from Enterprise Data Preparation.

1. In the **Projects** view, select a project.
2. Click the manage projects icon and select **Delete**.

   The **Delete Project** dialog box appears.
3. Click **Yes**.

CHAPTER 7

Prepare Data

This chapter includes the following topics:

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Overview

Data preparation is the process of combining, cleansing, transforming, and structuring data from one or more data assets so that it is ready for analysis.

When you add a data asset to a project, Enterprise Data Preparation creates a corresponding worksheet in the project. A worksheet has an interactive data-driven spreadsheet like interface. The worksheet contains the data and a recipe that tracks the changes you make to the data as you prepare it. Enterprise Data Preparation loads sample data into the worksheet depending on your sampling and filtering selection when you open a worksheet.

You can prepare data that have multibyte characters in the worksheet or column names that are configured using Data Preparation Service custom property with the following exceptions:

- Column names that have same as database keywords
- Column names that contains special characters: [\s!@#$%^&*()\-+={}[]|\\;:"'<>,\.\/\?]
- Column names that starts with a digit or underscore

When you prepare data, you use the sample data loaded in the worksheet. You perform all data preparation operations on this sample data. You do not directly change the data in the input source. When you publish the prepared data, Enterprise Data Preparation applies the recipe to the data in the input source and creates a new data asset.

You can prepare data from the following sources:

- Amazon S3
- MapR-FS
- Microsoft Azure Data Lake Storage
- Microsoft Windows Azure Storage Blob
- Microsoft Azure SQL Server
- Microsoft Azure SQL Data Warehouse

Consider the following guidelines for Microsoft SQL Server and Microsoft Azure:

- The blank values are sampled as NULL values.
- You cannot perform operations on the sample data containing blank values.
- The Data Preparation Service shows the blank values as blank data after data preparation. The blank values are displayed as NULL values only after publishing.

Data Preparation Features

Enterprise Data Preparation provides a number of features you can use to prepare data in a worksheet.

Let's explore the data preparation tasks that you can perform in a worksheet:

Filter data

Use the **Filter** icon to filter the data on any column in the worksheet. You can choose to add a filter or modify a filter condition. You can also clear all filters. When you filter your data, you select a column to filter and choose to search for a value in a column or create a custom filter and select your filter conditions. Different filters are available for String, Date, and Number columns depending on the column selected. You can filter using the top menu or you can use the filter panel available at the bottom or the window.

Edit a worksheet and its columns

Use the **Edit** menu to edit the columns in a worksheet. You can choose to delete blank columns or columns with one value, or hide or unhide columns in the worksheet.

You can also delete duplicate rows within a selected column. When delete duplicate rows, you can choose to ignore case as well as leading or trailing whitespace in rows containing Text values. To delete duplicate rows, select a column, and then select **Delete** > **Duplicate rows** from the **Edit** menu.

Note: You can only delete duplicate rows from a worksheet that contains a maximum of 100 columns.

You can right-click the worksheet tab to rename, delete, or copy the worksheet. You can perform operations to change the case of the column data, trim the column data, and split the column data. You can also right-click the columns to perform these operations on column data.

You can hide and unhide columns in a worksheet. You can also easily find and navigate to specific columns using the **Columns** panel, which also shows you the percentage of unique versus duplicate values a column contains. These features are especially useful when you prepare large data sets with dozens of columns.

View suggestions for improving the data

Enterprise Data Preparation provides an overview of the data in the worksheet and of the values in each column. The application also offers suggestions on how you can manipulate or improve the data.

Refresh the data in a worksheet

Use the **Refresh** icon to update the data in the worksheet after you perform your data preparation tasks or import a new data asset . If you refresh a worksheet, the data from the Hive table or storage is refreshed. If

you refresh a joined, merged, or aggregated worksheet, the data is refreshed from the parent worksheets if there is any change in the parent worksheets.

- Use the revert icon to revert the data to its previous state before making the refresh.
- Click **Highlight Changes** to view the changes that are made during the refresh.
- Click **Refresh Summary** to view the description of the changes made as part of the refresh in the form of instructions.

Blend data

Use the **Blend** icon to combine worksheets. You can perform operations such as inner join, left join, and right join on the worksheets. You can merge worksheets into one worksheet, or you can add columns from another sheet into your current sheet based on a common lookup key.

You can also group similar values in a column to into categories to make analysis easier.

Summarize data

Use the **Summarize** icon to aggregate or summarize data in the worksheet. You can aggregate the data on one or more columns and perform mathematical operations like sum or average on aggregated columns.

You can pivot columns to reshape the data into a pivot table format for analysis. You can also unpivot columns in a worksheet into rows containing the column data in key value format.

Apply formulas, rules and window functions to data

Use the **Functions** icon to formulas, rules and window functions to worksheets.

You can apply formulas to columns using the Text, Numeric, Date, and other functions available. You can also create your own formulas.

You can apply rules to data in a worksheet to help you cleanse, transform, or validate the data. You can apply passive rules, which operate on columns in the current worksheet; or active rules, which operate on one more columns in one or more worksheets within the project.

You can use window functions in Enterprise Data Preparation to perform operations on a small subset of a larger data set. A window function operates on a group of rows in a worksheet, and calculates a return value for every input row.

View and manage your data preparation steps in the worksheet recipe

Use the recipe icon to view and manage the steps you take to prepare the data in the worksheet.

You can edit a step in a recipe, or insert a new step. You can also reuse recipe steps. After you modify a recipe, you can use the back in time mode to review the impact of each recipe step on the data in the worksheet.

Data Masking

Enterprise Data Preparation integrates with Informatica Dynamic Data Masking, a data security product, to enable masking of sensitive data in data assets. When you view or perform operations on columns containing masked data, the actual data is fully or partially obfuscated based on the masking rules applied.

Data masking can be applied to data in the following scenarios:

When previewing data.

When you preview a data asset that is in the data lake or in an external database, data is masked according to the masking rules defined on the source data.

When importing data from an external database.

When you import data from an external database, data masking rules are applied to the data in the Hive table created for the asset.

When preparing data.

During data preparation, columns in worksheets are populated with masked data according to the masking policies defined on the source data. When you perform a blending operation such as a join or lookup, data in masked columns cannot be blended with unmasked columns.

When exporting or downloading data to an external database or file.

When you export data from the data lake to an external database, or downloads data to a file, data is written to the database or file in masked format.

When publishing data.

When you publish prepared data, data in columns that were masked during data preview, data sampling, and data preparation is written to the Hive table in masked format.

Sampling Data

You must sample the data for each asset you add to a project before you can begin preparing the data. You perform data preparation operations using the sample data in the worksheet created for the asset.

The sampling preview page in the Enterprise Data Preparation application displays the sample data to load based on the sampling selection settings specified for the asset. You can edit the sampling settings to meet your needs.

Sampling Table Data

You must sample the data in each database table you add to your project as the first step in data preparation. Enterprise Data Preparation presents the data in a worksheet that you use to sample the data.

The sampling preview page displays the sample data based on the current sampling selection settings. By default, the sampling happens when you add an asset to a project. You can change the sampling criteria if required. You can also change the default sampling criteria from **App Configuration** > **Advanced Properties**.

1. On the project page, click **Prepare**.

   The Sampling Preview panel appears along with the worksheet.

2. To edit the current sampling selection settings, click the **edit** icon in the Current Sampling Selection panel.

3. In the **Columns** tab, select the columns that you want to work with. To search for a column, enter the column name in the Search text box. You can search for multiple columns by entering a comma-separated list.

   Note that if columns are added to the source data after you complete sampling the data, and you want to sample the data again, you must select the new columns in the **Columns** tab.

4. Click the **Filters** tab.

   a. Click **New Filter**.

   b. In the **Create filter with** text box, enter the name of the column to use for filtering.

c. From the **select** drop down list, click one of the following options for string columns:

- **Equals**. Select to filter and display rows with the column value equal to this entered value. You can enter multiple values in this field, separated by a comma. If you enter multiple values, rows that match each value are filtered. Only the first value is considered and the other entries are ignored.
- **Contains**. Select to filter and display data containing the entered value. You can enter multiple values in this field, separated by a comma. If you enter multiple values, rows that match each value are filtered.
- **Starts with**. Select to filter and display rows with the column value that start with the entered value. You can enter multiple values in this field, separated by a comma. If you enter multiple values, rows that match each value are filtered.
- **Ends with**. Select to filter and display rows with the column value that end with the entered value. You can enter multiple values in this field, separated by a comma. If you enter multiple values, rows that match each value are filtered.
- **ISNULL and ISNOTNULL**. Select to filter NULL or non-NULL values in number, text, or date columns.

For date or numeric columns, select **Equals**, **Less than**, or **Greater than**.

d. Enter the value to use to filter the column.

e. Click **Save**.

Note: Filters are not available when you try to configure sampling for worksheets created for HDFS files.

5. Click the **Sampling** tab.
6. Select the Sampling type:

- **Random Rows**. Select to view the rows at random from the entire data asset. The number of rows displayed may be less than or equal to the sample size.
- **First Rows**. Select to view consecutive rows starting with the first row in the file.

Note: Random sampling is not available when you try to configure sampling for worksheets created for HDFS files.

7. Enter the number of rows to sample. The Informatica administrator can set the maximum number of rows in the Administrator tool.
8. Click **Save** to save your changes.
9. Click **Load** to load the sample data.

Sampling Delimited Files

You must sample the data in each delimited file you add to your project as the first step in data preparation. Enterprise Data Preparation presents the data in a worksheet that you use to sample the data.

The sampling preview page displays the sample data based on the current sampling selection settings. By default, the sampling happens when you add an asset to a project. You can change the sampling criteria if required.

Note that if columns are added to the source data after you complete sampling the data, and you want to sample the data again, you must select the new columns in the **Columns** tab. Columns that have changed in the source data but are used in a recipe step are included in the sample data by default.

1. On the project page, click **Prepare**.

   The Sampling Preview panel appears along with the worksheets.

2. To edit the current sampling selection settings, click the **edit** icon in the Current Sampling Selection panel.
3. In the **Format** tab, enter the code page, delimiter and text qualifier format details for the flat file.
   a. Start at row. Enter the row number from which you want to start the sampling.
   b. Column Names. Select **Get column names from the specified Start at Row** if the first row contains column names.
4. In the **Columns** tab, select the columns that you want to work with. To search for a column, enter the column name in the Search text box. You can search for multiple columns by entering a comma-separated list.
5. In the **Sampling** tab, enter the number of rows to sample.

   The Informatica administrator can set the maximum number of rows in the Administrator tool.
6. Click **Save** to save your changes.
7. Click **Load** to load the sample data.

Sampling Avro, JSON, and Parquet Files

You can sample files in the Avro, JSON Lines (JSONL), and Parquet formats. Enterprise Data Preparation flattens the data in the file into a flat structure and presents the data in a worksheet that you use to sample the data.

The Sampling Preview panel displays the file structure as a tree. Each node in the tree represents a key that exists in one or more lines within the file.

Enterprise Data Preparation creates columns in the worksheet for the keys you include in the sample data. The values corresponding to the keys within each line in the file are displayed in rows in the worksheet.

The nodes for top-level keys, such as a string, number, or boolean, are selected in the tree by default. To select a nested key, you can either select the key to include all of the keys below it, or expand the key and select each key to include.

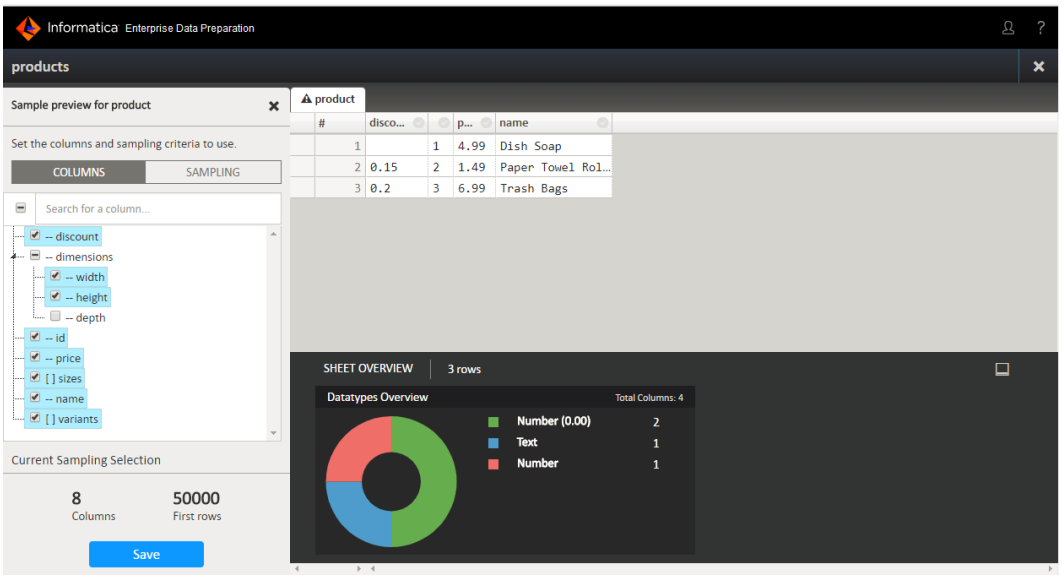
When you select key that is an array, Enterprise Data Preparation explodes the values in the array into rows in the worksheet. Rows are created for each array element in each line in the file. A file that contains numerous arrays might result in a large number of rows added to the worksheet, up to the maximum number of rows specified in the sampling settings.

If a line in the file does not contain a particular key or array, the cell in the corresponding column is empty.

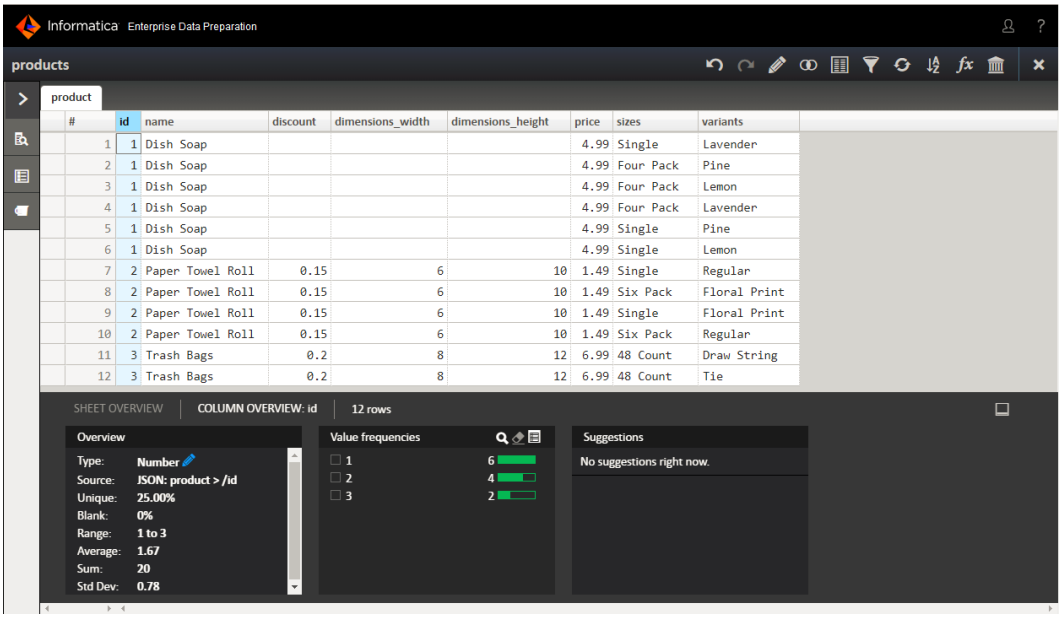
The following image shows the contents of a JSON Lines file. The file contains a key with nested keys and two arrays.


```
{ "id": 1, "name": "Dish Soap", "price": 4.99, "variants": [ "Lemon", "Pine", "Lavender" ], "sizes" : ["Single", "Four Pack"] }
{ "id": 2, "name": "Paper Towel Roll", "price": 1.49, "discount": 0.15, "variants": [ "Regular", "Floral Print" ], "sizes" :
  ["Single", "Six Pack"], "dimensions": { "height": 10, "width": 6, "depth": 6 } }
{ "id": 3, "name": "Trash Bags", "price": 6.99, "discount": 0.2, "variants": [ "Draw String", "Tie" ], "sizes" : ["48 Count"],
  "dimensions": { "height": 12, "width": 8, "depth": 6 } }
```

The following image shows the keys to sample selected in the tree. Note that the width and height keys within the dimensions key are selected.



The following image shows the worksheet created for the JSON file. The width and depth keys within the dimensions key are added as columns to the worksheet. The sizes and variants arrays are exploded into rows.



Because line 1 in the JSON file does not contain the discount or dimensions keys, the cells in the corresponding rows are empty.

Sampling an Avro, JSON, or Parquet File

You must sample the hierarchal data in each file you add to your project as the first step in data preparation.

The sampling preview page displays the sample data based on the current sampling selection settings. By default, the sampling happens when you add an asset to a project. You can change the sampling criteria if required.

Note that if columns are added to the source data after you complete sampling the data, and you want to sample the data again, you must select the new columns in the **Columns** tab. Columns that have changed in the source data but are used in a recipe step are included in the sample data by default.

1. On the project page, click **Prepare**.

   The Sampling Preview panel displays the file key hierarchy in a tree structure.
2. To edit the current sampling selection settings, click the **edit** icon in the Current Sampling Selection panel.
3. In the **Columns** tab, select or deselect each node to add the corresponding key as a column in the worksheet.

   The nodes for primitive keys, such as strings and numbers, are selected and displayed as columns in the worksheet by default.

   To select an array or a key that contains other primitive keys, you can either select the node to include all of the keys below, or expand the node and then select each key to include. Note that if a key within a top level key contains a control character, whitespace, or any punctuation other than an underscore, you cannot select the key.
4. In the **Sampling** tab, enter the number of rows to sample.

   The Informatica administrator can set the maximum number of rows in the Administrator tool.
5. Click **Save** to save your changes.
6. Click **Load** to load the sample data.

JSON Data Preparation Errors

You might encounter an error when you attempt to load sample data from a JSONL file. The most common causes of errors that might occur include the following:

- The file is not in JSON Lines format.
- The file and the file schema do not match.
- The file is corrupted.
- A column name in the file exceeds 128 characters.

Data preview of a JSON file that contains a column name that includes double quotes fails. For example, the following column causes data preview to fail:

```
{"id": 2,"\"first_name\"": "Bonnie","\"last_name\"":
"Cruz","Salary": 8489}
```

Limitations and Constraints

When there is a change in the source data, note the following constraints before you try to edit the sampling settings.

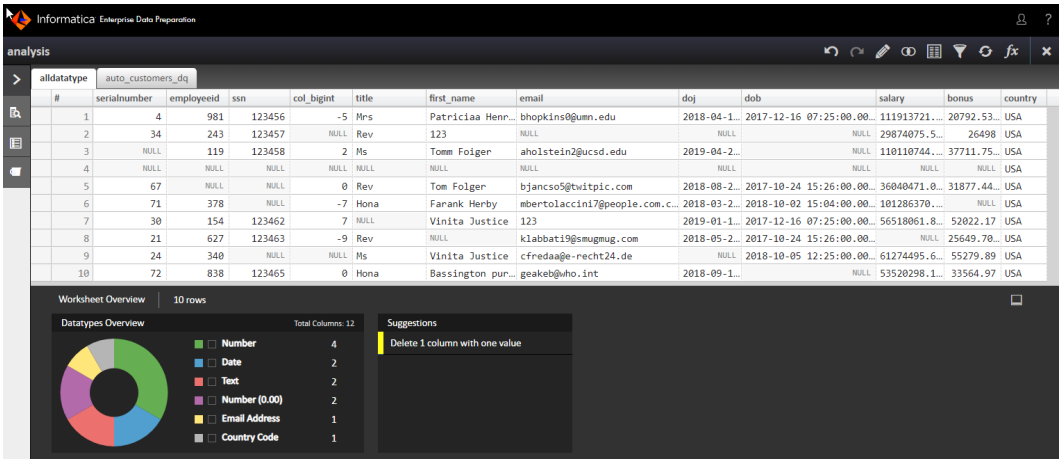
- Missing columns used in the recipe. You can't edit the sample if one or more columns used in the recipe are missing in the source. Restore the missing columns, or remove the columns from the sample.

- Missing columns used in sampling filter criteria. If one or more columns used to filter the sample are missing in the source, the filters will be automatically removed when you edit the sample. Restore the missing columns, or remove the columns from the sample.
- Missing columns selected for sampling. If you selected all columns during sampling, a warning will appear if some of the columns are missing in the source. Restore the missing columns, or remove the columns from the sample.
- Data type of a column has changed. If the data type of one or more columns used to filter the sample changes in the source, the filters are automatically removed when you edit the sample. Revert the data types for these columns before editing, or select different filters.

Working with Worksheets

When you use a worksheet to prepare your data, Enterprise Data Preparation provides an overview of the data in the worksheet and of the values in each column. The application also offers suggestions on how you can manipulate or improve the data.

When you open a worksheet, Enterprise Data Preparation provides you with an overview of the data within the worksheet, as shown in the following image:

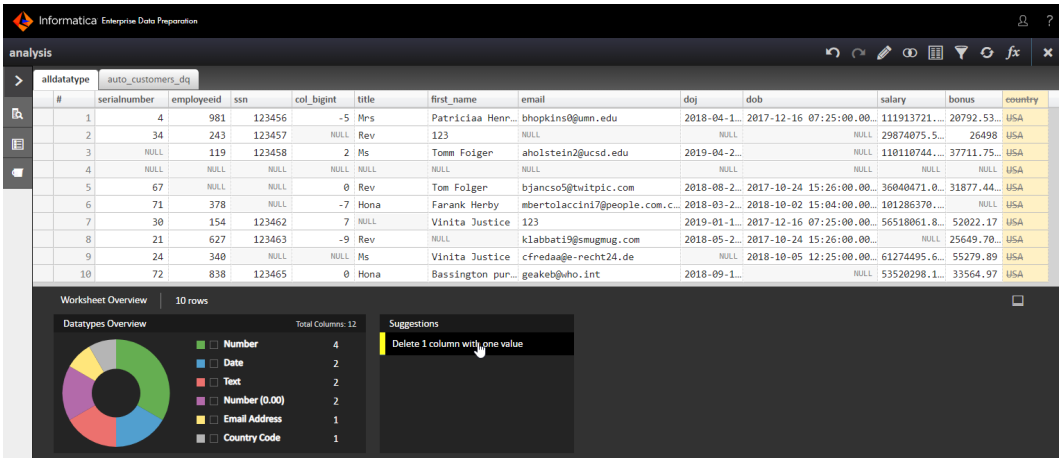


A grey-colored `NULL` string in a column indicates a NULL value. A red-colored `NULL` string indicates an error value. Hover your mouse over a red-colored `NULL` string to see the corresponding error. The application processes NULL values and error values as NULL values when you publish the asset to the data lake.

The **Datatypes Overview** panel shows you how many columns in the worksheet are of each type or data domain. You can select one or more types to filter in the worksheet.

The **Suggestions** panel offers suggestions on how you might improve the worksheet. When you hover over a suggestion, the application updates the worksheet with a preview of the impact of accepting the suggestion.

The following image shows the impact of accepting the suggestion to delete a column that contains only one distinct value has on the worksheet, and may not add any value when you publish the asset to the data lake:



When you select a column in the worksheet, the application provides an analysis of the data and makes suggestions on how you can improve or manipulate the data.

The **Overview** panel provides a summary of the column data, including column type and the source of the data. When you prepare a worksheet, Enterprise Data Preparation infers types or data domains for string values in each column based on the data types or data domains assigned to data in similar assets. For example, if all of the values in a column contain digits, Enterprise Data Preparation infers that the column is a Number type. If values in the column contain digits in the format of U.S. postal codes, Enterprise Data Preparation might infer that the column values belong to the US 9 Digit Zip Code data domain.

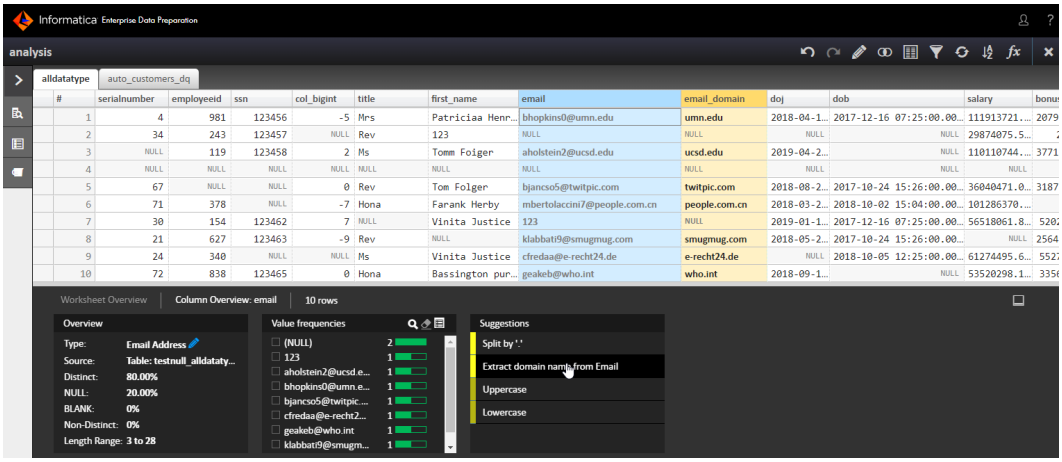
You can choose to accept the inferred type or data domain for a column, or you can revert the column back to the source data type. You can also convert some types to another type, such as Date to Text. For more information, see "Reverting an Inferred Data Type" on page 52.

The Source property indicates the source of the data displayed in the column. For more information, see "Viewing Column Data Sources" on page 53.

The **Value frequencies** panel displays the occurrence of each distinct value in the column. You can select one or more values to filter in the worksheet, including NULL values. You can also set a value in the column to NULL.

Suggestions panel offers suggestions on how you might manipulate the column data. When you hover over a suggestion, the application shows you a preview of the impact of accepting the suggestion.

The following image shows a preview of the columns that selecting the "Extract domain name from Email" suggestion adds to the worksheet:



If a project contains multiple worksheets, you can reorder the worksheets in the data preparation page by moving a worksheet to a different position. For example, you might want to move the second worksheet displayed in the page to the third position. To move a worksheet, select the worksheet tab, and then select Move Worksheet. You can then click the icon indicating the position you want to move the worksheet to.

Upgrading Worksheets for NULL Value Handling

Effective in 10.4.0, the application handles NULL values following the ANSI-92 SQL standard or Informatica Data Engineering processing behavior during data preparation and publication. After the service upgrade, you must upgrade all worksheets to maintain consistency. The upgrade process resamples the worksheet from the data source and refreshes the recipe with the new behavior. You must reschedule the activities to use the refreshed recipe.

To upgrade all worksheets in a project, complete the following steps:

1. Log in to Enterprise Data Preparation.
2. Click **Prepare** and open a project.
   A dialog box appears enlisting all the worksheets in that project.
3. To upgrade all worksheets, click **OK**.

Tip: You can click **Close** to leave the project while the upgrade is in progress. The application continues to upgrade all worksheets within the project in the background.

Note: If you choose not to upgrade worksheets, the scheduled activities will remain as is. In order to maintain consistency with the ANSI-92 SQL standard or Informatica Data Engineering processing behavior, you must upgrade all worksheets created in the previous releases.

NULL value handling behavior prior to version 10.4.0

Previously, Enterprise Data Preparation interpreted NULL values in a worksheet column as blank values. For example, a NULL value in an arithmetic equation was interpreted as zero:

```
NULL + 5 = 5
```

Similarly, a NULL value in a function was interpreted as a blank value:

```
ABS(NULL) = 0
```

NULL value handling behavior effective in 10.4.0

In accordance with the ANSI-92 SQL standard or Informatica Data Engineering processing behavior, an operation containing a NULL value returns NULL. For example, an arithmetic equation containing a NULL value returns NULL:

```
NULL + 5 = NULL
```

Similarly, a function containing a NULL value also returns NULL:

```
ABS(NULL) = NULL
```

Known Issues

If you face the following issues, contact Informatica Global Customer Support:

- When you use a pivot operation on the date, text, or number column with COUNT BLANK in a worksheet and later upgrade the worksheet for NULL handling, the data preparation and data publication display inconsistent results.
- When you use Cluster and Categorize or Categorize operations to replace blank values from 10.2.2 and NULL values from 10.4.0 to non-NULL or non-blank values in a worksheet and later upgrade the worksheet for NULL handling, the data preparation and data publication display inconsistent results.

Converting a Column Data Type

You can convert column data to a different type.

You can convert the following data types:

- Number to Text
- Date to Text
- Text to Number
- Text to Date

The conversion options available depend on the column data type.

The application adds a new column containing the converted data to the worksheet. You can specify

1. Select the column containing the data to convert.
2. Right-click the column, and then select the operation to perform:
   - Select **Convert to Text** to convert values in a Date or Number column to Text format.
   - Select **Convert to Number** to convert values in a Text column to numerical values.
     Enter a name for the column that contains the converted data, and then select the separator to use for thousand or decimal values.
   - Select **Convert to Date** to convert values in a Text column to Date format.
     Enter a name for the column that contains the converted data, and then select the Date format to apply to the data.

You can also specify a custom format. Custom formats are not case sensitive. The following table describes supported custom Date formats:

Time period	Format
Year	- YYYY: four digit year - YY: last two digits of year
Month	- MONTH: full name of month - MON: abbreviated name of month (Jan-Dec) - MM: numeric designation of the month (01-12)
Day	- DD: day of month (1-31) - DY: abbreviated name of day (Sun-Sat) - DAY: full name of day
Hour	- HH24: hour of day (0-23) - HH12: hour of day (1-12) - HH: hour of day (1-12)
Minute	MI
Second	SS
Millisecond	MS
A.M./P.M.	AM, PM

You can use the following delimiters with custom Date formats:

comma, dash, forward slash, space

You can also use a colon to separate formats in a timestamp, such as in the following example:

HH24:MI:SS

Reverting an Inferred Data Type

You can revert inferred types and data domains in a worksheet back to the source type. You might want to revert an inferred type or data domain back to the source type if you want to use the column data in a formula.

String values that appear to be text, numbers or dates might be inferred as being of the Text, Number, or Date type. You can convert the inferred type to another type. For example, you might want to convert a string inferred as the Date type to the Text type.

String column values that appear to match a data domain might be inferred as being of that data domain type. You can revert the inferred data domain type to the source type.

When you create a derived worksheet through an aggregate, join, or union operation, the types and data domains inferred for columns in the source worksheet are applied to the corresponding columns in the derived worksheet. If you change a data type in a source worksheet, you must refresh all of the corresponding derived worksheets.

You cannot revert the type or data domain assigned to a column if the column is included in a step in a recipe.

After you revert or convert a column type, you can revert the column back to the inferred type if needed.

1. On the project page, click **Prepare**.
2. Select a string column in a worksheet.

   The inferred type or data domain for the column appears in the Type attribute in the Overview panel.
3. Hover over the **Change type** icon, and then select the operation to perform:
   - Select **Revert column data to <type>** to revert a column inferred as belonging to a data domain to the source type.
     The column reverts to the source type.
   - Select **Convert column data to <type>** to convert an inferred type to another type.
     The column converts to the selected type, and the action is added as a step to the worksheet recipe.
   - Select **Re-infer column type** to revert the column back to the inferred type.
     The column reverts back to the inferred type, and the action is added as a step to the worksheet recipe.

Reverting All Inferred Data Types

You can revert all inferred types and data domains applied to source columns in a worksheet back to their original types.

The operation does not revert inferred types and data domains applied to new columns or to columns used in data preparation operations. Note that if you revert all inferred types and data domains applied to source columns, you cannot undo the operation results.

After you revert or convert a column type, you can revert the column back to the inferred type if needed.

1. On the project page, click **Prepare**.
2. Right-click the worksheet tab, and then select **Revert all inferred data types**.

   All source columns in the worksheet revert to the original types.

Viewing Column Data Sources

You can view the source of the data for a selected column in the Source property in the **Overview** panel. You might want to view the source of the data in a column to help you troubleshoot an issue.

Columns added by importing source data into a worksheet.

Displays the source type, the source name, and the name of the column from which the column data is derived.

Columns added through operations such as join, lookup, or aggregate.

Displays the names of the worksheet and column from which the column data is derived.

Columns added by a union operation.

Displays the names of each of the worksheets and columns from which the concatenated column data is derived, separated by a comma.

Columns added by operations such as split, concatenate, or extract.

Displays the step in the recipe that generated the column data. Open the Recipe panel and mouse over the Source property to highlight the step in the recipe.

Columns added by applying a formula or a rule to a column.

Displays the step in the recipe that generated the column data. Because the formula or rule modifies the source data, Enterprise Data Preparation cannot display the actual source. Open the Recipe panel and mouse over the Source property to highlight the step in the recipe.

Recovering Worksheets

Enterprise Data Preparation prompts you to recover data in a worksheet in the case of a failure or data corruption during data preparation.

The Data Recovery dialog opens with a list of worksheets to recover. Click **Recover** to recover the data in all of the worksheets listed.

Using Recipes

A recipe defines the steps Enterprise Data Preparation follows to combine, cleanse, transform, and structure the data in a worksheet. Each operation you perform in a worksheet during data preparation is added as a step to the recipe.

You can edit or delete a step in a recipe. You can also apply a filter to the columns used in a step.

You can insert a new step below an existing step in a recipe. You can insert a step below any step except the last step in the recipe. If filters are applied on the step above the inserted step, Enterprise Data Preparation applies the filters to the inserted step. You cannot perform some operations when you insert a step in a recipe, including aggregate, join, or union operations.

To insert a step, select **Insert Step** from the menu in the existing step. The application adds a placeholder step below the selected step. Select the column you want to perform an operation on, and then select the operation to perform.

You can add or edit comment in a recipe step. Use comments to improve collaboration with other users, or to provide details to comply with auditing requirements.

Reviewing and Troubleshooting Recipe Steps

You can use the back in time mode to review the impact of each recipe step in the worksheet. The back in time mode is useful for enabling collaborators on a project to see what has been done thus far, or if you are troubleshooting issues discovered in a worksheet.

To view a step's impact, select the step in the recipe, and then select **Review steps back in time** from the menu. The worksheet updates based on the impact of the selected step.

You can edit or delete a step in back in time mode. Enterprise Data Preparation recalculates the subsequent steps in the recipe based on your changes. You can also apply a temporary filter to a step to preview the impact of adding a filter, although you must create and apply the filter in the regular editing mode.

Reusing Recipes

You can reuse recipe steps created in a worksheet, including steps that contain complex formulas or rule definitions. You can reuse recipe steps within the same worksheet or in a different worksheet, including a

worksheet in another project. You can copy and reuse selected steps from a recipe, or you can reuse the entire recipe.

When you copy a recipe step, Enterprise Data Preparation includes all input columns used in the step, including the column name and type. When you insert a copied step into the recipe in the target worksheet, the application prompts you to map each column in the step to a column of the same type in the target worksheet. The target worksheet must contain columns of the same type as the columns included in each copied recipe step.

If a column with the same name and type exists in the worksheet, Enterprise Data Preparation populates the column name. The application notes the number of columns modified in or added to the worksheet as a result of inserting the copied recipe steps.

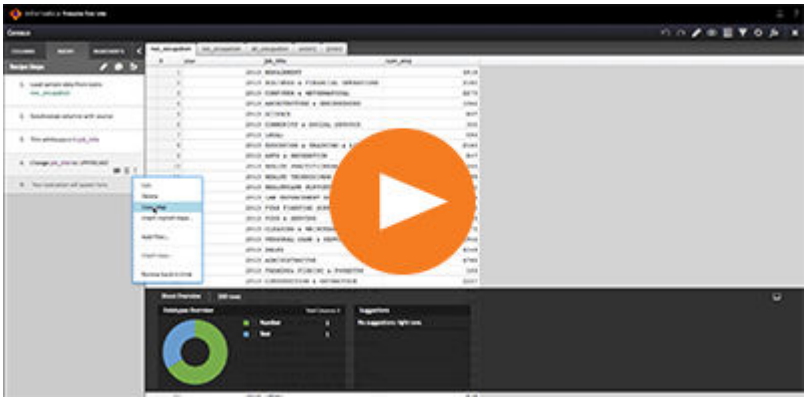
Filters applied in the source worksheet to columns used in copied steps are also copied with the steps. You can choose to apply or ignore the filters when you insert the copied steps into the worksheet.

Input parameters used in formulas or rules are included with the copied recipe steps. You can modify the inputs inserted with a formula or rule after the copied steps are saved in the worksheet.

You can insert copied steps below any step in a recipe, except above any step that is followed by a Synchronize Columns step. The copied steps remain available for insertion into a recipe until you copy any other content in Enterprise Data Preparation or your session ends.

You can reuse recipe steps in a worksheet in a different project. If you copy a lookup step into a worksheet in another project, you must select a worksheet in the project that contains a lookup key and added columns with the same names and types as those in the worksheet used in the lookup operation. If a matching worksheet does not exist in the project, you might need to add the data asset used in the lookup operation as a worksheet to the project.

Click the image below to view a video about copying recipe steps:



Copying a Single Recipe Step

You can copy and reuse a single step from a recipe. Filters applied to columns in the step are copied by default.

1. On the project page, click **Prepare**.

   The Sampling Preview panel appears.
2. Select the worksheet containing the recipe step you want to reuse.
3. Click the **Recipe** icon.
4. Select the recipe step you want to copy.
5. Select **Actions** > **Copy Step**.

6. Open the worksheet in which you want to reuse the step.
7. Select the recipe step below which you want to add the copied step.
8. Select **Actions** > **Insert Copied Steps**.
9. To use filters included with the copied step, click **Apply filters from <source sheet>**.
10. Map the column in the step to a corresponding column of the same type in the worksheet.
11. Click **Done**.

    Enterprise Data Preparation updates the worksheet based on the step, and displays the number of columns modified in or added to the worksheet as a result of inserting the copied recipe steps.

Copying Multiple Recipe Steps

You can copy and reuse selected steps from a recipe, or you can reuse the entire recipe.

You must select contiguous steps to copy. For example, if you want to copy the second and fourth steps in the recipe, you must select the second, third, and fourth steps. Filters applied to columns in a step are copied by default.

1. On the project page, click **Prepare**.

   The Sampling Preview panel appears.
2. Select the worksheet containing the recipe step you want to reuse.
3. Click the **Recipe** icon.
4. Select the steps you want to copy.
   - To copy all of steps in the recipe, click the **Copy Recipe Steps** icon and select **Copy All Steps**.
   - To copy multiple contiguous steps in the recipe, use **Shift + Click** to select the steps, and then click the **Copy Recipe Steps** icon and select **Copy Selected Steps**.
5. Open the worksheet in which you want to reuse the steps.
6. Select the recipe step below which you want to insert the copied steps.
7. Select **Actions** > **Insert Copied Steps**.
8. To use filters included with the copied steps, click **Apply filters from <source sheet>**.
9. Map each column in the step to a column of the same type in the worksheet.
10. Click **Done**.

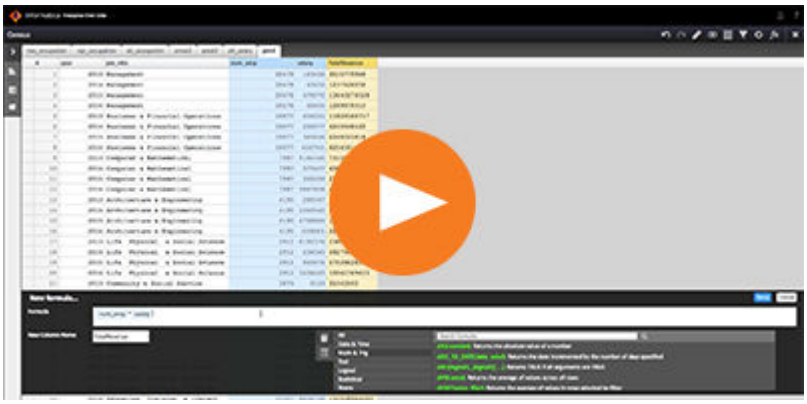
    Enterprise Data Preparation updates the worksheet based on the step, and displays the number of columns modified in or added to the worksheet as a result of inserting the copied recipe steps.

Using Formulas

You can use formulas to perform calculations and return values in a worksheet. Enterprise Data Preparation adds a column containing the formula results to the worksheet.

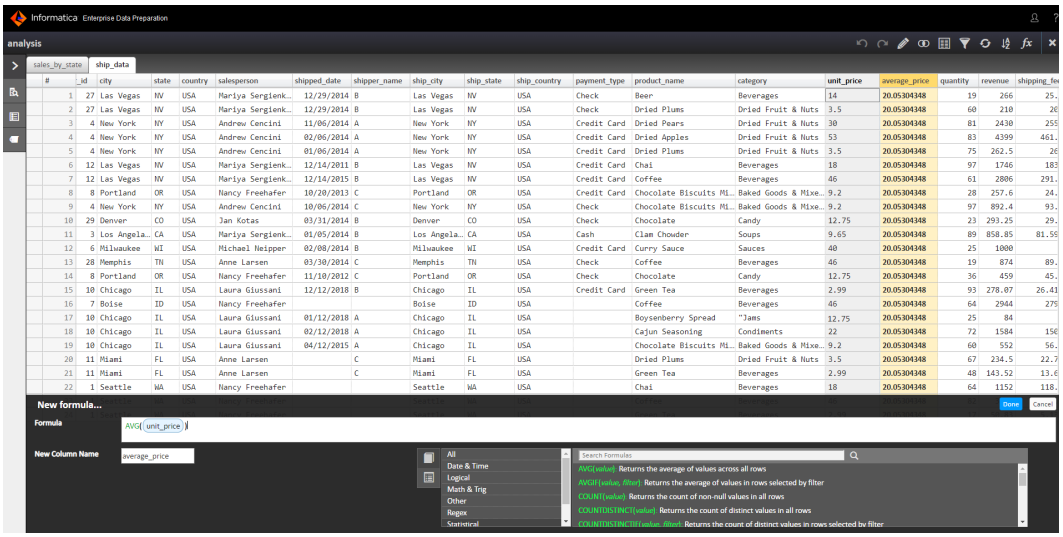
The application provides a variety of functions that you can use in formulas. To help you find the function you need, functions are organized into groups, such as Date and Time functions.

Click the image below to view a video about using formulas:

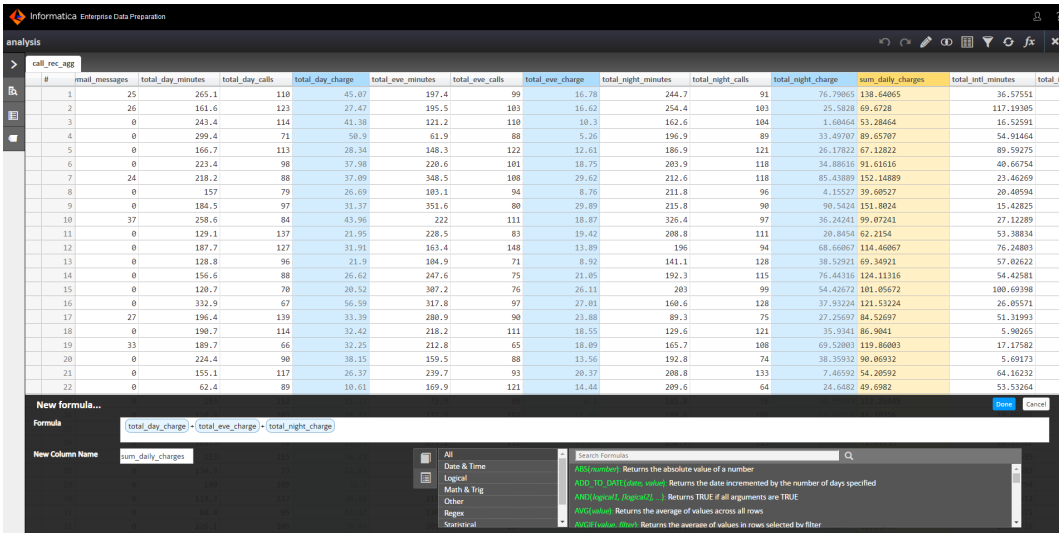


Example

The following example uses the AVG aggregate function to calculate the average of the values in the unit_price column. The results are displayed in the average column, as shown in the following image:



You can also create your own formulas. The following example adds the total day, evening and night call charges and provides the results in the sum_daily_charges column:



Applying a Formula

Select one or more functions to use in a formula, or create your own formula.

1. On the project page, click **Prepare**.
2. Select the worksheet to which you want to apply the formula.
3. Click the **Functions** icon, and then select **New formula**.
4. Enter the name of the column containing the formula results to add to the worksheet.
5. Select a function to use in the formula from the list, or type a custom formula in the **Formula** field.

   To add a column to the formula, start typing the name of the column, and then select the column name from the list.
6. Click **Done** to apply the formula.

Limitations and Constraints

Enterprise Data Preparation uses lenient parsing of input values in the DATE and DATETIME formulas. If you specify a value for a DATE or DATETIME formula that contains a value that is out of range, the application returns a value within the preceding or following unit of time.

For example, if you specify DATE(2018, 13, 30), the application returns 01/30/2019, because 13 is outside of the valid range for the month argument.

In addition, you cannot use nested aggregate functions in a worksheet. For example, the following nested aggregate function is not valid:

SUM(AVG(quantity) + AVG(unit_price))

Using Window Functions

You can apply window functions in Enterprise Data Preparation to groups of rows within a worksheet.

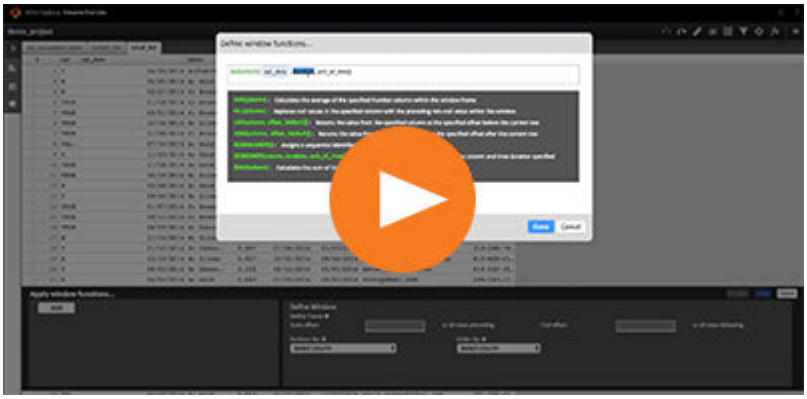
The group of rows on which a function operates is called a window. A window function calculates a return value for every input row within a window.

You can use window functions to perform the following tasks:

- Retrieve data from a previous row or a subsequent row.
- Calculate a sum or an average based on a group of rows.
- Assign a sequential row number to each row in a group of rows.
- Replace null values in rows with the preceding non-null value within a group of rows.
- Generate session identifiers you can use to group rows based on a specific time period, such as web site visits recorded in a log file.

You can apply multiple window functions to a worksheet. For example, you might apply a function to calculate the sum of values for each row preceding or following the current row within a window, and apply another function to calculate the average of the same values. Enterprise Data Preparation adds a column containing the results of each function you apply to the worksheet.

Click the image below to view a video about applying window functions:



Window Configuration

When you use a window function, you configure the windowing properties associated with the function. Windowing properties define the frame, partitioning, and ordering boundaries associated with a particular input row.

Windowing properties define the following:

- A frame that defines the boundaries associated with a particular input row. You can define a frame if you apply a SUM or AVG function to a worksheet.
- Partition by columns containing the categorical values to use to group rows into windows.
- Order by columns that define the order of rows in each window.

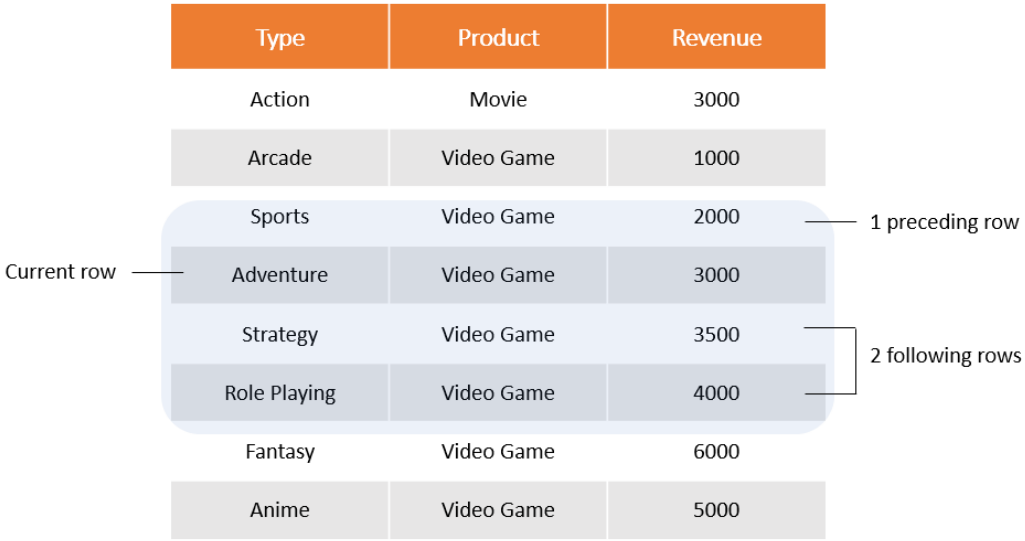
Frame

The frame definition determines which rows are included in the calculation for the current input row, based on the relative position of the rows to the current row.

You can configure frames when you use the SUM and AVG functions. The application ignores frame definitions when you use all other functions.

The start offset defines the number of rows before or after the current row at which to begin the calculation. The end offset describes the number of rows after the current input row at which to end the calculation. For example, a start offset of -1 and an end offset of 2 describes a frame including the row before the current row, the current input row, and the two rows after the current row.

The following image shows a frame with a start offset of -1 and an end offset of 2:



You can also specify a frame that does not include the current input row. For example, a start offset of 2 and an end offset of 4 describes a frame that includes three rows, from the second to the fourth row after the current row.

Note: The start offset must be less than or equal to the end offset.

Offsets of **All Rows Preceding** and **All Rows Following** represent the first row of the partition and the last row of the partition. For example, if the start offset is All Rows Preceding and the end offset is -1, the frame includes one row before the current row, and all rows before it.

The following image shows a frame with a start offset of 0, indicating that the start offset is the current row, and an end offset of All Rows Following:



Partition By and Order By Columns

You use partition by columns to group rows into windows, and use order columns to define the order or sequence of rows within each window.

Partition by columns

Partition by columns define window boundaries, based on categorical values within the columns. A window function operates across the rows that fall into the same window as the current row.

Partition by columns are optional for all functions. If you do not specify partition by columns, the function treats the entire data set as a window. You must select columns containing categorical values as partition by columns.

You can select multiple partition by columns to apply to a window function. The application creates a window based on each unique combination of categorical values in the selected columns.

Order by columns

Order by columns determine the order of the rows within a window.

Order by columns are required for all functions except AVG and SUM. For the AVG and SUM functions, specifying order by columns is optional when you select **All Rows Preceding** and **All Rows Following** as the offsets in the frame definition.

You can select multiple order by columns to apply to a window function. The application orders rows based on precedence of the order by columns selected.

When you apply an order by column to a function, you choose to arrange the data in ascending or descending order.

Example

In this example, you want to calculate the revenue for coffee and tea products.

The following image shows a table listing the product types, the corresponding product categories, and the revenue from each product:

Type	Product	Revenue
Coffee	Espresso	600
Tea	Black	650
Coffee	Cappuccino	300
Coffee	Americano	500
Tea	Oolong	250
Coffee	Macchiato	450
Tea	Green	300
Tea	White	550

Select the Type column, which contains categorical values, as the partition by column. You select the Revenue column as the order by column, and choose to order the data by descending revenue.

The following image shows the data grouped into two windows based on the values in the Type column. Within each window, the revenue data is displayed in descending order:

Type	Product	Revenue
Coffee	Espresso	1000
Coffee	American	500
Coffee	Macchiato	450
Coffee	Cappuccino	300
Tea	Black	650
Tea	White	550
Tea	Green	300
Tea	Oolong	200

Supported Window Functions

AVG

The AVG function calculates the average of the values preceding or following the current row within the specified window.

You can apply the function to Number columns in a worksheet. The function adds a column containing the calculation results to the worksheet.

You must specify the Start Offset and End Offset properties in a window frame definition to define the rows preceding or following the current row to include in the calculation. The function iterates through each row within the window, performing the calculation based on the offset values specified.

For example, if you set Start Offset to 2 and End Offset to 4, the function calculates the average of the values in the second, third, and fourth rows after the current row.

To set the current row as the start offset, enter 0 as the value for the Start Offset property. To set a row before the current row as the start offset, enter a negative number.

For example, if you set Start Offset to -2 and End Offset to 4, the function calculates the average of the values in the current row, the two rows before the current row, and the four rows after the current row.

Partition by columns are optional. Specifying order by columns is optional when you select **All Rows Preceding** and **All Rows Following** as the offsets in the frame definition; otherwise, you must specify order by columns.

Syntax

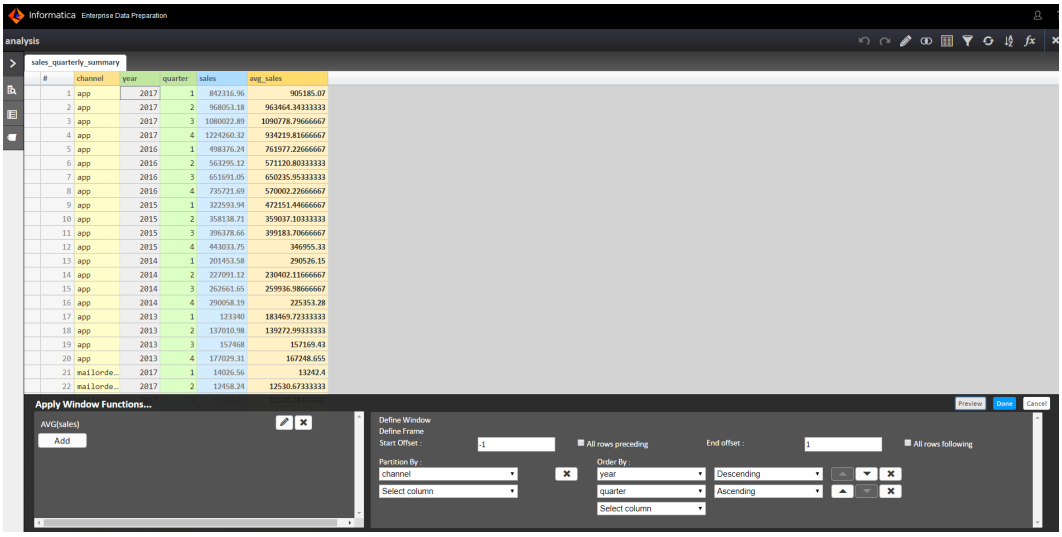
```
AVG (Column)
```

The following table describes the function arguments:

Argument	Required/Optional	Description
Column	Required	The name of the column containing the values to include in the calculation. The column must be of the Number type.

Example

The following example calculates the average sales per channel for the current, previous, and following quarters. To include the rows for the previous and following quarters, set the offset values to -1 and 1.



Enter the following function arguments:

- Column: sales

Enter the following window properties:

- Start Offset: -1
- End Offset: 1

- Partition By column: sales
- Order By columns: year, quarter

FILL

The FILL function replaces null values in rows in the specified column with the preceding non-null value within a window, based on the partition by and order by columns specified. You must specify at least one order by column.

Syntax

```
FILL (Column)
```

The following table describes the function arguments:

Argument	Required/Optional	Description
Column	Required	The name of the column name within which to replace null values.

LAG

The LAG function returns the value from the specified column at the specified offset before the current row.

Use the LAG function to compare the values in the current row with the values in a preceding row. You must specify at least one order by column.

Use the offset parameter to specify the number of rows before the current row for which to return the value. For example, set the offset parameter to 3 to return the value of the row that is three rows before the current row.

Syntax

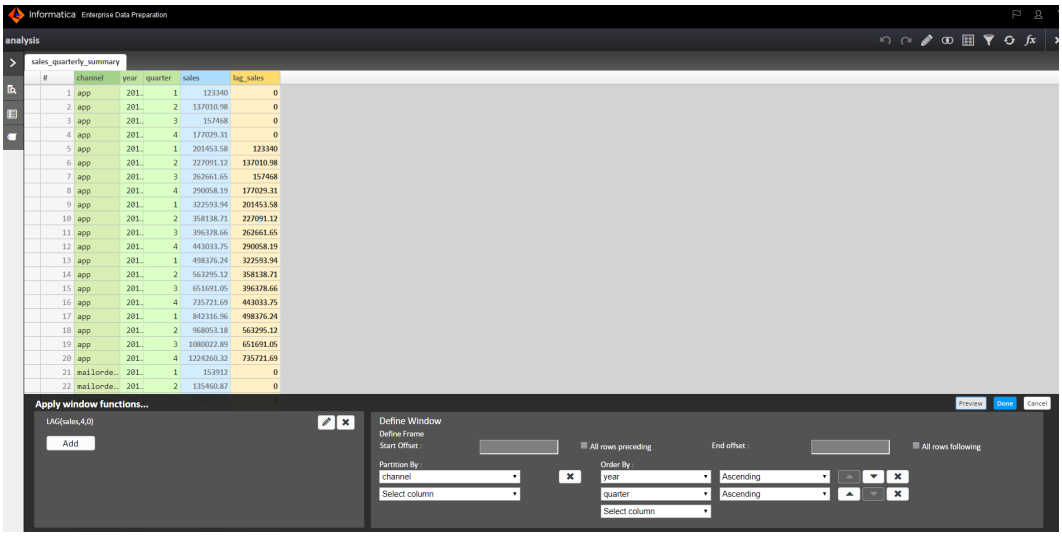
```
LAG (Column, Offset, [Default])
```

The following table describes the function arguments:

Argument	Required/ Optional	Description
Column	Required	The name of the column name containing the value to retrieve.
Offset	Required	The number of rows preceding the current row from which to retrieve the data. The value must be a positive number.
Default	Optional	The default value to return if a row does not exist at the specified offset value. If you apply the function to a Date column, enter the parameter value in the in the following formats: 'mm/dd/yyyy' or 'mm/dd/yyyy hh24:mi:ss' If you apply the value to a Text column, the value must be enclosed in quotes. For example: 'Annual' If you do not specify a value, the default value is null. Remove this argument from the function if it is not needed.

Example

The following example compares sales per product channel for each quarter with sales for the same quarter in the previous year. Enter 0 as the default value to return for rows that are not within the specified offset value.



Enter the following function arguments:

- Column: sales
- Offset: 4
- Default: 0

Enter the following window properties:

- Partition By column: sales
- Order By columns: year, quarter

LEAD

The LEAD function returns the value from the specified column at the specified offset after the current row.

Use the LEAD function to compare the values in the current row with the values in a following row. You must specify at least one order by column.

Use the offset parameter to specify the number of rows after the current row for which to return the value. For example, set the offset parameter to 3 to return the value of the row that is three rows after the current row.

Syntax

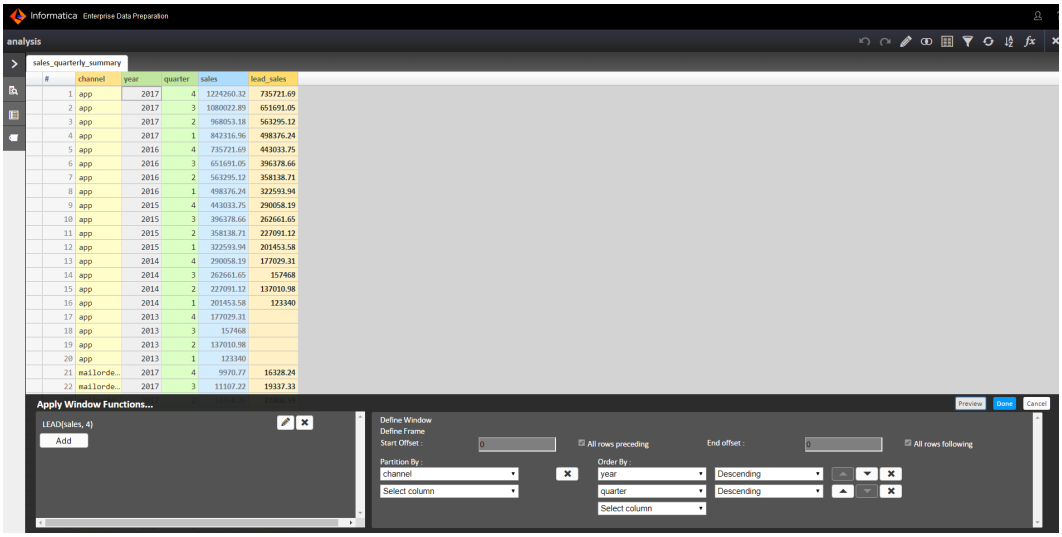
```
LEAD (Column, Offset, [Default])
```

The following table describes the function arguments:

Argument	Required/ Optional	Description
Column	Required	The name of the column name containing the value to retrieve.
Offset	Required	The number of rows preceding the current row from which to retrieve the data. The value must be a positive number.
Default	Optional	The default value to return if a row does not exist at the specified offset value. If you apply the function to a Date column, enter the parameter value in the in the following formats: 'mm/dd/yyyy' or 'mm/dd/yyyy hh24:mi:ss' If you apply the value to a Text column, the value must be enclosed in quotes. For example: 'Annual' If you do not specify a value, the default value is null. Remove this argument from the function if it is not needed.

Example

The following example compares sales per product channel for each quarter with sales for the same quarter in the following year. Remove the default argument from the function to return null values for rows that are not within the specified offset value.



Enter the following function arguments:

- Column: sales
- Offset: 4
- Default: remove this argument from the function

Enter the following window properties:

- Partition By column: sales
- Order By columns: year, quarter

ROWNUMBER

The ROWNUMBER function assigns a sequential row number, starting at 1, to each row within a window.

The function adds a column containing the row identifiers to the worksheet.

When the function reaches the end of a window, it assigns identifiers to the rows in the next window, starting at 1.

Syntax

```
ROWNUMBER ()
```

The function does not take any arguments.

SESSIONIZE

The SESSIONIZE function generates session identifiers based on the Date column and the period of time specified. You can use the generated session identifiers to group rows based on a specific time period, such as web site visits recorded in a log file.

You can apply the function to Date columns in a worksheet. The function adds a column containing the session identifiers to the worksheet.

When you configure the function properties, you must select the name of the Date column you use in the function as the first order by column. You must also configure the order by column to list rows in ascending order.

Syntax

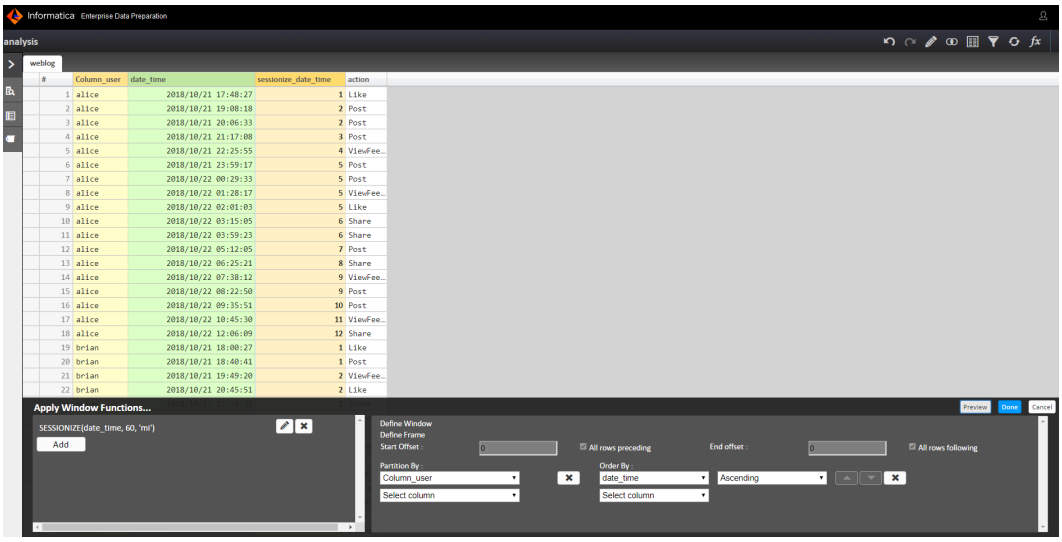
```
SESSIONIZE (Column name, Duration, Unit of time)
```

The following table describes the function arguments:

Argument	Required/Optional	Description
Column	Required	The column to which to apply the function. The column must be of the Date type. You must select the same column as the value of the Order By property.
Duration	Required	The time period by which to group rows included in a session. The value must be greater than 0. You can specify fractional values, such as 1.5.
Unit of time	Required	The unit of time to apply to the time interval value. Valid values are: - 'ms': milliseconds - 'ss': seconds - 'mi': minutes - 'hh': hours - 'dd': days Enclose values in quotes.

Example

The following example groups user visits to a particular web site recorded in a log file based on the date and time noted in the date_visited column. The function generates a new session identifier when the date_visited column value is greater than the previous value plus the specified time period.



Enter the following function arguments:

- Column: date_time
- Time Interval: 60
- Unit of time: 'mi'

Enter the following window properties. Note that you must set the value specified in the Column argument as the value for the first Order By column. You must also set the Order By column to list results in ascending order:

- Partition By column: column_user
- Order By columns: date_time

SUM

The SUM function calculates the sum of values preceding or following the current row within the specified window.

You can apply the function to Number columns in a worksheet. The function adds a column containing the calculation results to the worksheet.

You must specify the Start Offset and End Offset properties in the window frame definition to define the rows preceding or following the current row to include in the calculation. The function iterates through each row within the window, performing the calculation based on the offset values specified.

For example, if you set Start Offset to 2 and End Offset to 4, the function calculates the sum of the values in the second, third, and fourth rows after the current row.

To set the current row as the start offset, enter 0 as the value for the Start Offset property. To set a row before the current row as the start offset, enter a negative number.

For example, if you set Start Offset to -2 and End Offset to 4, the function calculates the sum of the values in the current row, the two rows before the current row, and the four rows after the current row.

Partition by columns are optional. Specifying order by columns is optional when you select **All Rows Preceding** and **All Rows Following** as the offsets in the frame definition; otherwise, you must specify order by columns.

Syntax

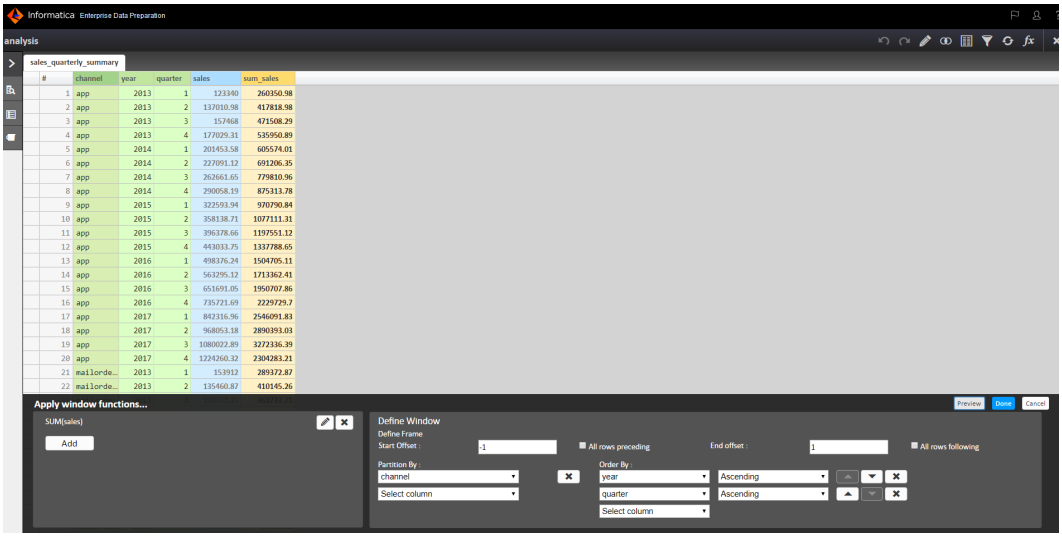
```
SUM (Column)
```

The following table describes the function arguments:

Argument	Required/Optional	Description
Column	Required	The name of the column containing the values to include in the calculation. The column must be of the Number type.

Example

The following example calculates the total sales per channel for the current, previous, and following quarters. To include the rows for the previous and following quarters, set the offset values to -1 and 1.



Enter the following function arguments:

- Column: sales

Enter the following window properties:

- Start Offset: -1
- End Offset: 1
- Partition By column: sales
- Order By columns: year, quarter

Applying Window Functions

You can apply one or more window functions to a worksheet.

For each function, optionally select one or more partition by columns to create windows from groups of rows. Select one or more order by columns to specify the order or sequence of rows within each window.

If you apply an AVG function or a SUM function, you must specify the rows preceding and following the current row to include in the frame within which to perform the calculation. The application ignores frame definitions for all other window functions.

If you select **All Rows Preceding** and **All Rows Following** in the frame definition for an AVG function or a SUM function, you do not need to select an order by column.

1. On the project page, click **Prepare**.
2. Select the worksheet to which you want to apply the functions.
3. Click the **Functions** icon, and then select **Apply window functions**.
4. Click **Add**.
5. Click or type the name of the function you want to apply.
6. Click each function argument to enter a value. To specify a column value, start typing the name of the column, and then select the column name from the list.
7. Click **Done**.

   Enter the window properties in the Define Window panel.
8. If you apply an AVG function or a SUM function, enter the number of rows before or after the current row at which to begin the calculation from the **Start Offset** menu.

   To begin the calculation at a row before the current row, enter a negative number.

   To include all rows before the current row in the calculation, select **All Rows Preceding**.
9. If you apply an AVG function or a SUM function, select the number of rows following the current row to include in the calculation from the **End Offset** menu.

   To include all rows following the current row in the calculation, select **All Rows Following**.
10. Select a column to use as a partition key from the **Partition By** menu.

    The window function operates across the rows that fall into the same group as the current row.
11. Select a column to use to specify how rows in the group or partition are ordered in the function results from the **Order By** menu.

    If you do not select an order by column, the rows have no particular order.
12. Select whether to order rows in the output column in ascending or descending order.
13. Click **Preview** to preview the function results in the worksheet.
14. Click **Done** to apply the function.

Using Rules

You can apply rules during data preparation to help you cleanse, transform, or validate data. Rules provide you with consistent, complex logic for transforming data that can be reused across projects.

Enterprise Data Preparation provides hundreds of rules that you can apply to data. Rules can also be created by developers using the Administrator tool, or by analysts using the Analyst tool. To use rules, your Informatica administrator must deploy rules-related services and content in your Informatica installation. Your Informatica administrator must also give you privileges to access rules.

You can apply two types of rules: passive rules and active rules.

Passive Rules

A passive rule operates on columns in the current worksheet. The application adds columns containing the rule output to the current worksheet.

Active Rules

An active rule uses all rows within a data set as input. You can select multiple worksheets containing data to use as inputs to the rule. The application adds a worksheet containing the rule output to the project.

An active rule contains one or more input groups. For each input group, you select a worksheet that contains the data that you want to use in the rule. You can select the same worksheet for multiple input groups, or you can select a different worksheet for each group.

You map inputs in the input group to columns in the worksheet that contain the data you want to pass to the rule. You can also specify a constant value as an input, instead of a column name.

An active rule might also include parameters that are passed to the rule at runtime. Each parameter has a default value that you can modify.

An active rule contains one or more output groups. Each output group contains the names of columns containing the rule results that are added to the generated worksheet. You can select a single output group to use in an active rule.

Using Passive Rules

A passive rule operates on columns in the current worksheet. The application adds columns containing the rule output to the current worksheet.

When you apply a rule, Enterprise Data Preparation adds the rule as a step to the worksheet recipe. You can edit the rule in the recipe, or remove it from the recipe.

1. On the project page, click **Prepare**.
2. Select the worksheet to which you want to apply the rule.
3. Click the **Functions** icon, and then select **Apply rule**.
4. Select a rule from the menu.

   To find the rule you want to use, start typing the name of the rule.
5. Start typing the name of a column to which you want to apply the rule.
6. Select the name of the column to add to the worksheet containing the rule output.
7. Click **Apply**.

Using Active Rules

An active rule uses all of the rows within a data set as input. You can select multiple worksheets to use as input sources. Enterprise Data Preparation adds a worksheet containing the output columns chosen from the output group to the project.

When you apply a rule, the application adds the rule as a step to the worksheet recipe. You can edit the step created for the rule in the recipe, or you can remove the step from the recipe.

You can configure and run one active rule at a time.

1. On the project page, click **Prepare**.
2. Click the **Functions** icon, and then select **Apply rule**.
3. Select an active rule from the list.

To find the rule you want to use, start typing the name of the rule.

4. Select a worksheet to use as an input source.

   The input group displays the columns in the worksheet that contain values you can use as inputs to the rule.

5. Provide the inputs to the rule. You can select the names of columns in the worksheet, or enter constant values.

   - To use a column in the worksheet as an input for a field, select the column name from the menu. Select at least one column in each input group.
   - To use a constant value as an input for a field, select **Enter a Constant Value** from the menu, and then enter the value. For more information about valid values, see "Valid Rule Inputs" on page 72.

6. If the rule contains parameters, the data type and default value of each parameter applied in the rule definition appear. To change a parameter value, click the **edit** icon next to the parameter.
7. Expand an output group.

   The output group displays the columns that you can include in the generated worksheet containing the rule results. If the rule contains multiple output groups, select only one output group.

8. Select the output column associated with each input. You must select an output column for each input you include in the rule.
9. Click **Run**.

   Enterprise Data Preparation runs the rule and adds a worksheet containing the rule output to the project.

Valid Rule Inputs

You can use the name of a column in the worksheet as a rule input in a worksheet. You can also use strings, dates and numbers as constant value inputs.

The following table describes the valid formats for values you can use as rule inputs:

Value Type	Valid Format
Number	Up to 19 numeric characters and a single decimal point. For example, 3.15 is valid; 3.1.5 is not valid.
String	No restrictions.

Blending Data

You can create blended data sets by combining, importing, or merging data in worksheets within a project. Enterprise Data Preparation provides you with several methods for blending data.

You can use the join operation to combine data from two different worksheets based on a common column containing identical distinct values. Similarly, you can perform a lookup to import selected columns from another worksheet into your current worksheet based on a common column in both worksheets that contains identical distinct values.

You can use the union operation to merge data from one worksheet into another worksheet. When you perform a union, you match the columns in both worksheets that contain similar data.

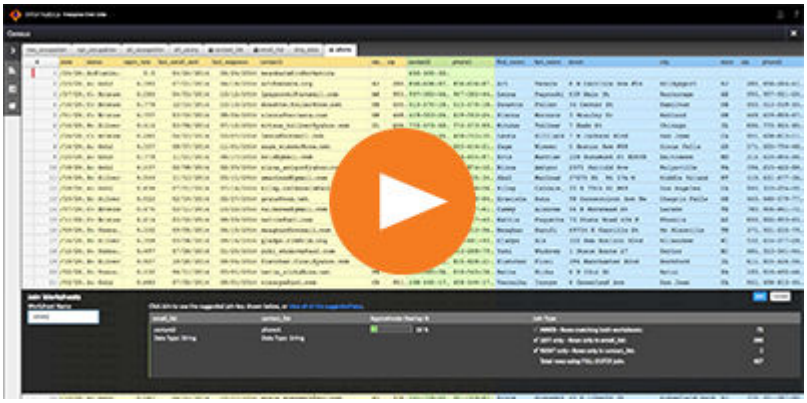
You can categorize or group similar values in a column to make analysis easier. For example, you might want to group values in a column containing the color of various item according to the corresponding primary color, such as red, blue or yellow.

Joining Worksheets

You can use the join operation to combine data from two different worksheets within a project. The operation adds a new worksheet containing the combined data to the project.

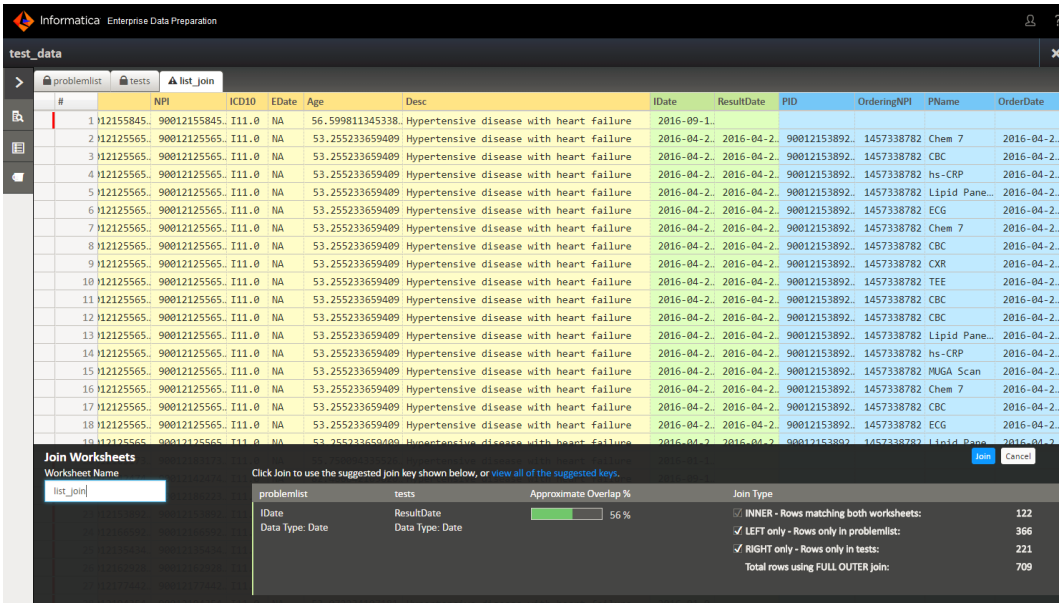
When you perform a join, you specify columns to use in each worksheet that contain the same distinct values. Both worksheets must contain at least one column with identical distinct values. For example, if both worksheets contain a column with a customer identifier, you can use these columns to combine columns containing data associated with each customer identifier in a single worksheet.

Click the image below to view a video about the join operation:



Example

Enterprise Data Preparation suggests columns to use as join keys. The application adds a worksheet containing the combined data as a preview, as shown in the following image:

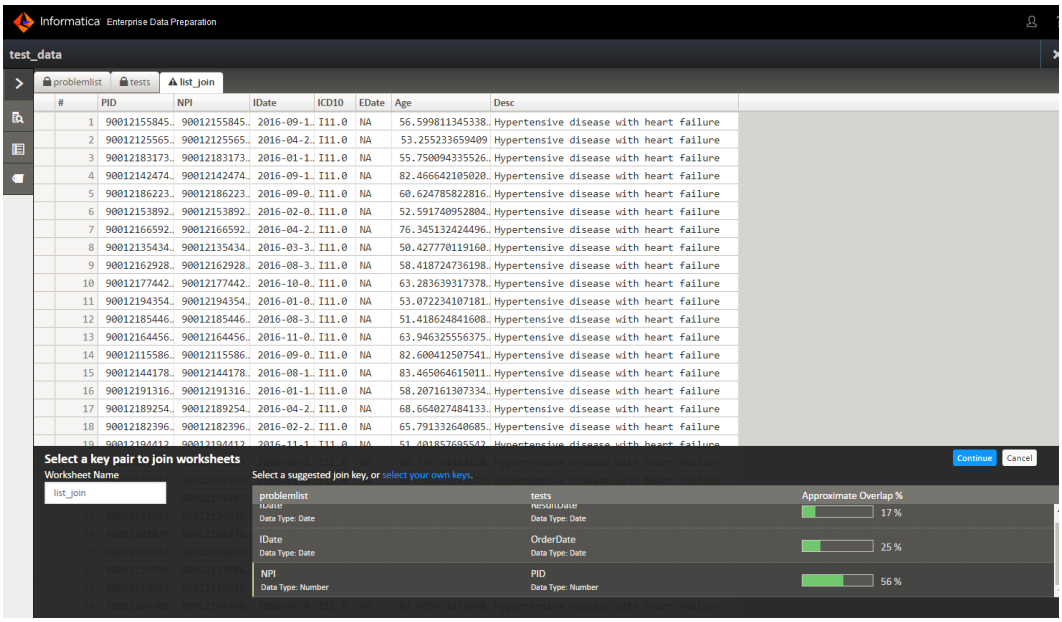


Enterprise Data Preparation performs a FULL OUTER JOIN by default, which combines all rows containing matching data in the selected columns in both worksheets. You can change the selections in the Join Type

column to determine which join type best meets your needs. You can also see the number of rows each join type adds to the worksheet.

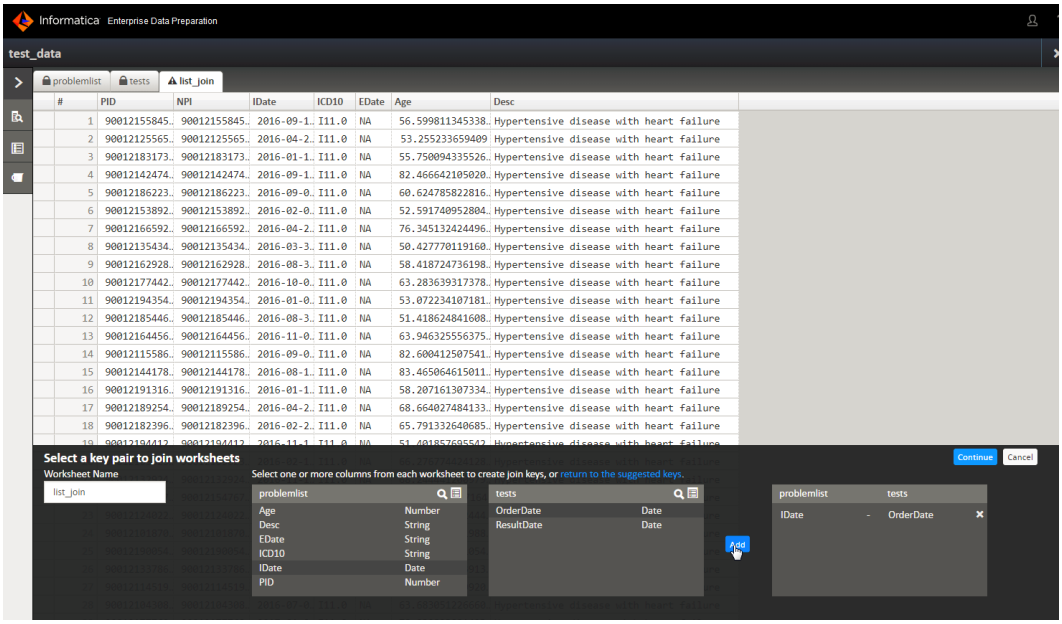
You can click **Join** to perform the join using the selected key pair, or you can view all of the suggested key pairs.

The following image shows all of the suggested key pairs:



If you know that the worksheets contain other columns that also contain matching data, you can choose your own join key pairs from the list of columns in both worksheets. Select a column in each worksheet, and then click **Add**. You must select columns of the same type.

The following image shows how you can select your own join keys:



Joining Data in Worksheets

A project must contain at least two worksheets to perform a join operation. You can modify the suggested columns to use as join keys.

1. On the project page, click **Prepare**.
2. Click the **Blending** icon, and then select **Join worksheets**.
3. In the **Join Worksheets** dialog box, select the worksheet that you want to join with the current worksheet.
4. Select the worksheet in the project containing the data you want to join with the current worksheet.

   The application adds a worksheet containing the joined columns to the project. Matching columns that occur in both worksheets are displayed in green. Columns that occur in the current worksheet only are shown in yellow. Columns that occur only in the worksheet selected for the join are shown in blue.
5. Enter a name for the worksheet.
6. Review the suggested key pair. You can view the occurrence of identical distinct values in both columns in the Approximate Overlap % column.

   Decide which keys to use to join the worksheets:

   - Use the suggested keys.
   - Click **View all of the suggested keys** to choose from the complete list of selected join keys. Select a row containing the keys to use, and then click **Continue**.
   - Click **Select your own keys** to choose one or more join key pairs from all possible keys in the worksheets. Click a key in each worksheet, and then click **Add**. You must select columns of the same type to use as join keys. Click **Continue** when you finish selecting the columns.
7. Select the type of join to perform in the Join Type column. Enterprise Data Preparation updates the worksheet based on your selection.
   - Select **INNER** to join the matching rows in both the worksheets.
   - Select **LEFT** to join all of the rows from the first worksheet and the rows with matching values from the second worksheet.
   - Select **RIGHT** to join all of the rows from the second worksheet and the with matching values from the first worksheet.
   - Select **RIGHT** and **LEFT** to perform a FULL OUTER JOIN of all rows containing matching distinct values in the selected columns in both worksheets.
8. Click **Join** to join the worksheets using the join keys you select.

   Enterprise Data Preparation adds the join worksheet to the project.

Editing Joins

On a joined worksheet, you can edit the join and choose different join keys. You cannot edit joins if any non-key columns used in the recipe are missing from the source sheets. You cannot edit joins created in an earlier version of Enterprise Data Preparation.

1. Select the joined worksheet.
2. Select the join step in the recipe.
3. Click the **Edit** icon to edit the join.
4. Review the suggested selected key. You can view the approximate overlap percentage and results.
5. Click **Change Key** to select another join key.

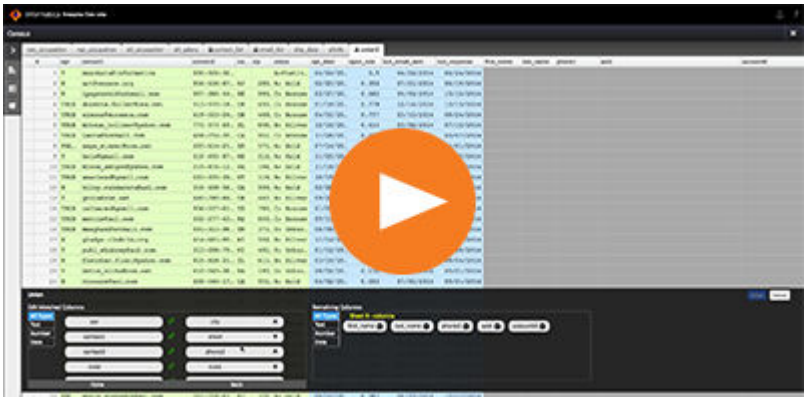
   Note: If the join key is not present, you can select a new join key.
6. From the Results displayed, select or deselect to modify the type of join: inner join, left join, or right join.

7. Click **Done**. To undo the edits to the join, click **Cancel**.

Merging Worksheets

You can use the union operation to merge data from another worksheet into the current worksheet. The operation adds a new worksheet containing the merged data to the project.

Click the image below to view a video about the union operation:



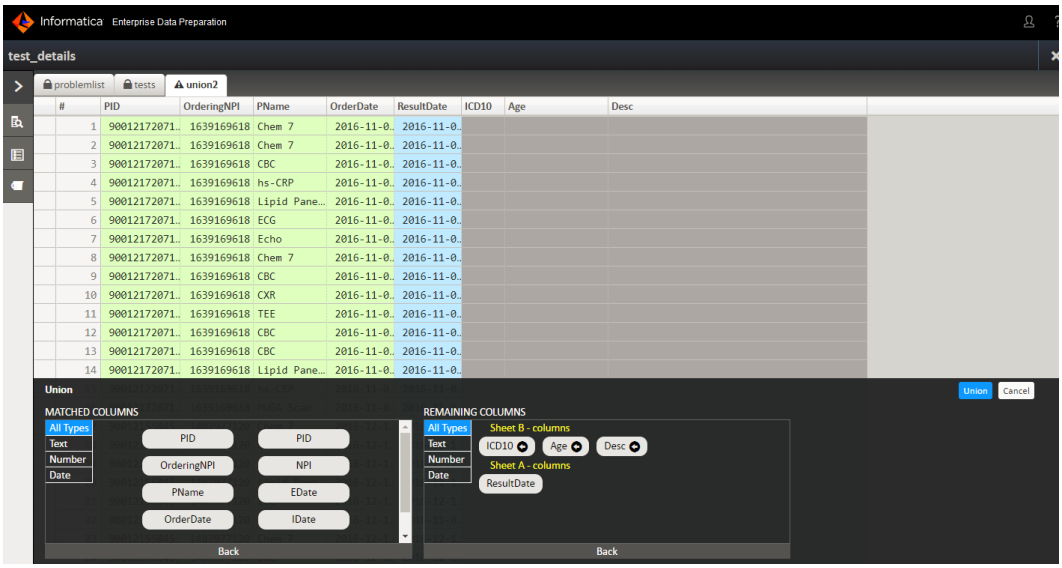
When you perform a union, you match columns in both worksheets that contain similar data. Enterprise Data Preparation suggests matching columns of the same type to include in the union. The application merges the data in a matched column in the other worksheet into the corresponding matched column in the current worksheet.

You can modify the suggested matching columns by removing a column from a matched pair. You can also select columns in the other worksheet to match with columns in the current worksheet. The columns you select must be of the same type.

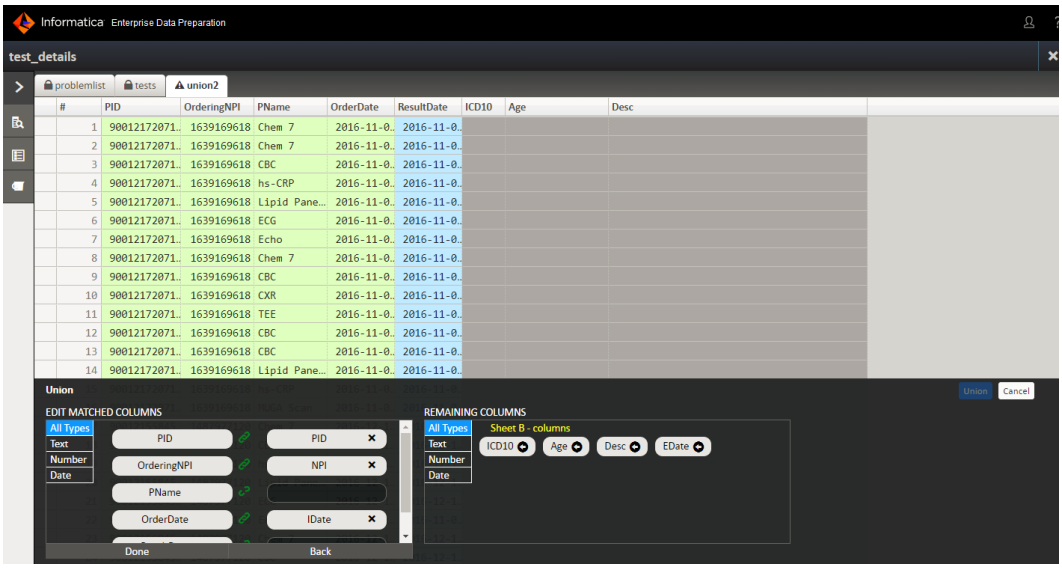
Example

To perform a union, you select the worksheet you want to merge with the current worksheet. Enterprise Data Preparation updates the worksheet with a preview of the union operation results. Matched columns are displayed in green. Columns in the tests worksheet that the application does not match with columns in the problemlist worksheet are shown in blue.

Click **View** in each panel to review the matched column pairs, as shown in the following image:

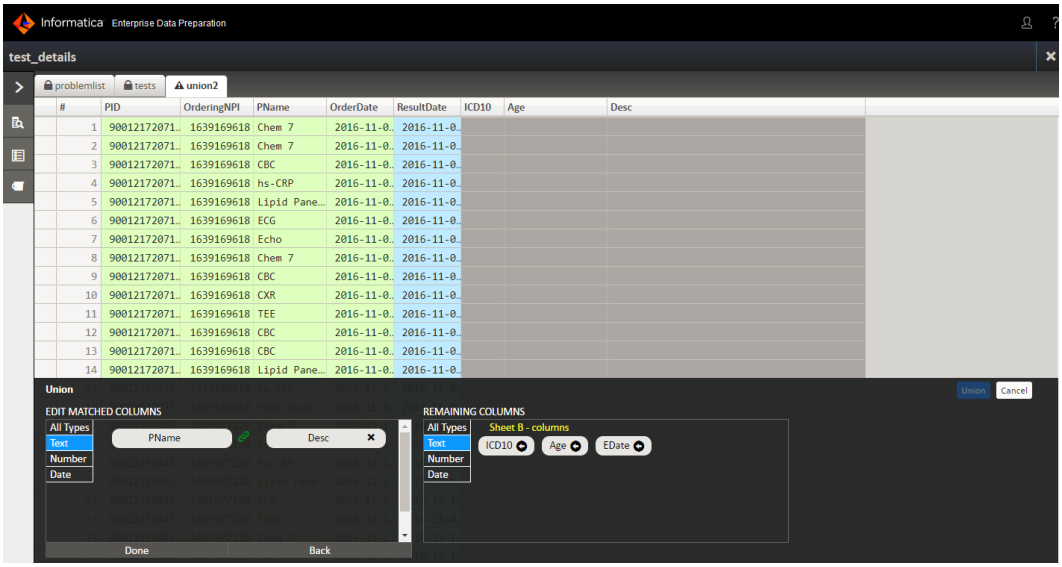


You realize that the data in the PName column in the tests worksheet does not match the EDate column in the problemlist worksheet. Click the **Change** button in the **Matched Columns** panel, and then click EDate to remove it from the panel, as shown in the following image:

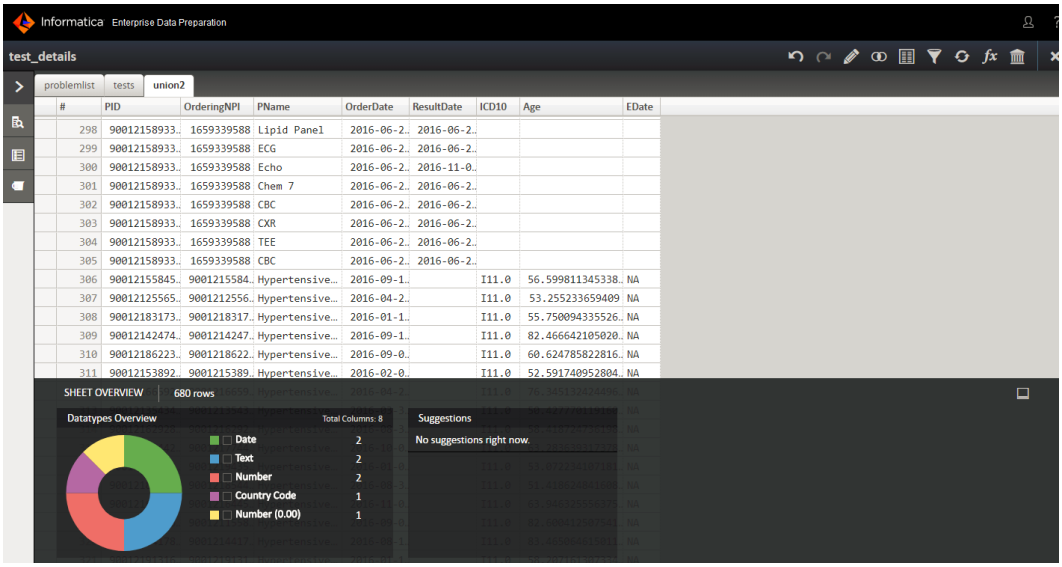


Select **Text** in each panel to display only values of the same type. You know that the data in the Desc column in the problemlist worksheet is a good match with the PName column. Click the empty field next to PName,

and then click Desc in the **Remaining Columns** panel to select it as the column to match with the PName column, as shown in the following image:



When you are satisfied that the matched column pairs are correct, click **Done** in the **Matched Columns** panel, and then click **Union** to complete the union. The application adds the union worksheet to the project, as shown in the following image:



Merging Columns in Worksheets

A project must contain at least two worksheets to perform a union operation. You can modify the suggested matched columns. Columns matched together must be of the same type.

1. Open the worksheet you want to union with another worksheet.
2. Click the **Blend** icon, and then select **Union worksheets**.
3. Select the worksheet in the project containing the data you want to merge into the current worksheet.

The application adds a worksheet containing the merged data to the project. Matching columns that occur in both worksheets are displayed in green. Columns that occur in the current worksheet only are shown in blue. Columns that occur only in the worksheet selected for the union are shown in yellow.

4. The **Matched Columns** panel contains the matching column pairs found in both worksheets. Columns are matched based on type. The **Remaining Columns** panel contains columns for which a matching column cannot be found.

   Click **View** in each panel to view the columns. Click **Back** when finished.

5. Click **Change** in the **Matched Columns** panel to modify the matched column pairs.

   a. If a matched column pair is not correct in the **Matched Columns** panel, click the column to the right to remove it from the panel.

   b. To match a column in the **Remaining Columns** panel with a column in the **Matched Columns** panel, click the entry field for the corresponding column in the **Matched Columns** panel, then click a column of the same type in the **Remaining Columns** panel.

6. Click **Done** when you finish making changes to the matching columns.

7. Click **Union**.

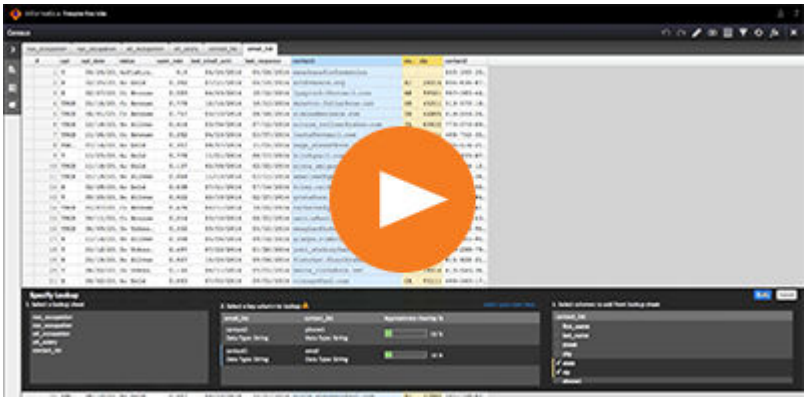
   Enterprise Data Preparation adds the union worksheet to the project.

Using Lookup

You can use the lookup operation to import selected columns from another worksheet into the current worksheet.

When you perform a lookup, you select a column in the current worksheet and a column in another worksheet that contain similar distinct values to use as lookup keys. For example, if both worksheets contain a column with a customer e-mail address, you can select these columns to use as lookup keys. You then select columns in the other worksheet that contain data associated with each customer email address, and import the columns into the current worksheet.

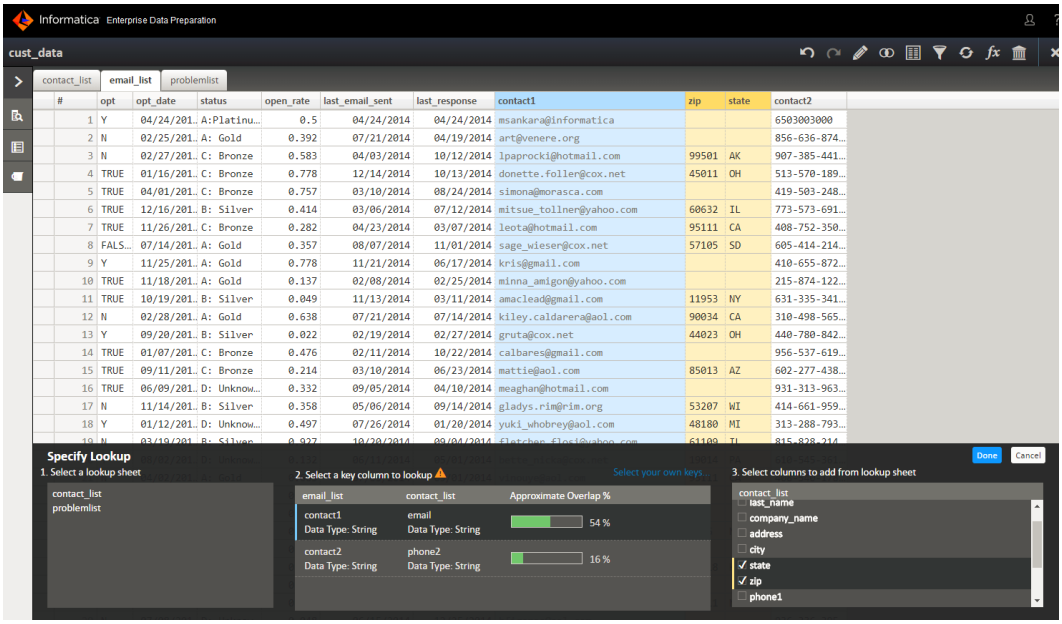
Click the image below to view a video about the lookup operation:



Example

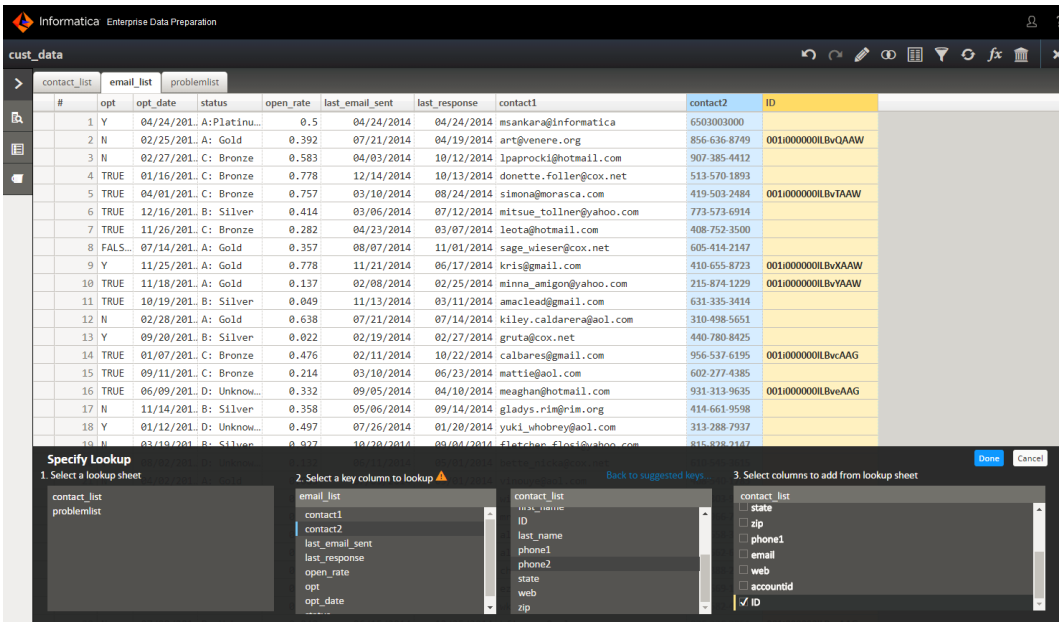
After you select the lookup worksheet, which is the worksheet containing the data to import, you select the columns to use as lookup keys in both the current worksheet and in the lookup worksheet. You then select columns in the lookup worksheet that you want to import into the current worksheet.

Enterprise Data Preparation suggests columns that contain a high frequency of identical distinct values to use as lookup keys, as indicated in the Approximate Overlap % property. The application adds the imported columns to the current worksheet as a preview, as shown in the following image:



You can also select other columns to use as lookup keys. You must select columns of the same type.

The following image shows the selected lookup keys:



Performing a Lookup on Worksheets

A project must contain at least two worksheets to perform a lookup operation. You can modify the suggested columns to use as lookup keys. Both worksheets must exist in the same project.

You cannot perform a lookup on derived columns, worksheets derived from other, or sheets with filter options applied.

1. On the project page, click **Prepare**.
2. Click the **Blending** icon, and then select **Lookup**.
3. Select the worksheet containing the data you want to import from the list of worksheets.
4. The application displays the columns with the highest amount of overlap as suggested keys as indicated in the Approximate Overlap % column. Select the columns to use as lookup keys:
   - Select a suggested key pair.
   - Click **Select your own keys** to choose a column in each worksheet to use as a lookup key. You must select columns of the same type.
5. Select the columns to add to the current worksheet.
6. Click **Done**.

General Limitations for Lookup

The following limitations apply to the lookup operation.

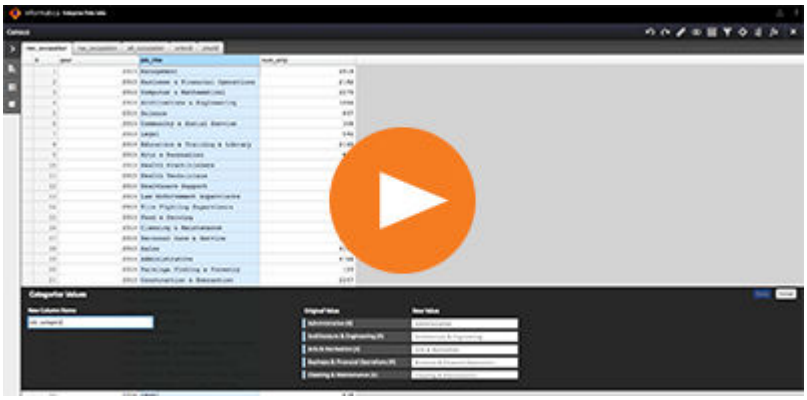
- Duplicate keys. You cannot perform a lookup on a worksheet that contains duplicate keys.
- Derived worksheets. You can perform a lookup only on existing worksheets within a project. You cannot perform lookup on worksheets derived from other worksheets using operations such as union, aggregate, or join. However, you can perform a lookup on copied worksheets.
- Filtered worksheets. Publication is disabled for lookup worksheets that use column filters. A lookup worksheet is the worksheet created by a lookup operation.
- Source filters. Publication is disabled for lookup worksheets that use source filters.
- New columns. You can perform a lookup only on existing columns. You cannot perform a lookup on new columns added to a worksheet, such as columns derived after performing operations on existing columns.
- Existing columns. You cannot perform a lookup on columns on which any operation has been performed.
- Data type mismatch. You cannot perform a lookup on two columns with different data types such as decimal and double data types.
- Duplicate rows removed. You cannot perform a lookup on a worksheet from which duplicate rows were removed.

Categorizing Column Data

You can categorize similar values in a column to make analysis easier.

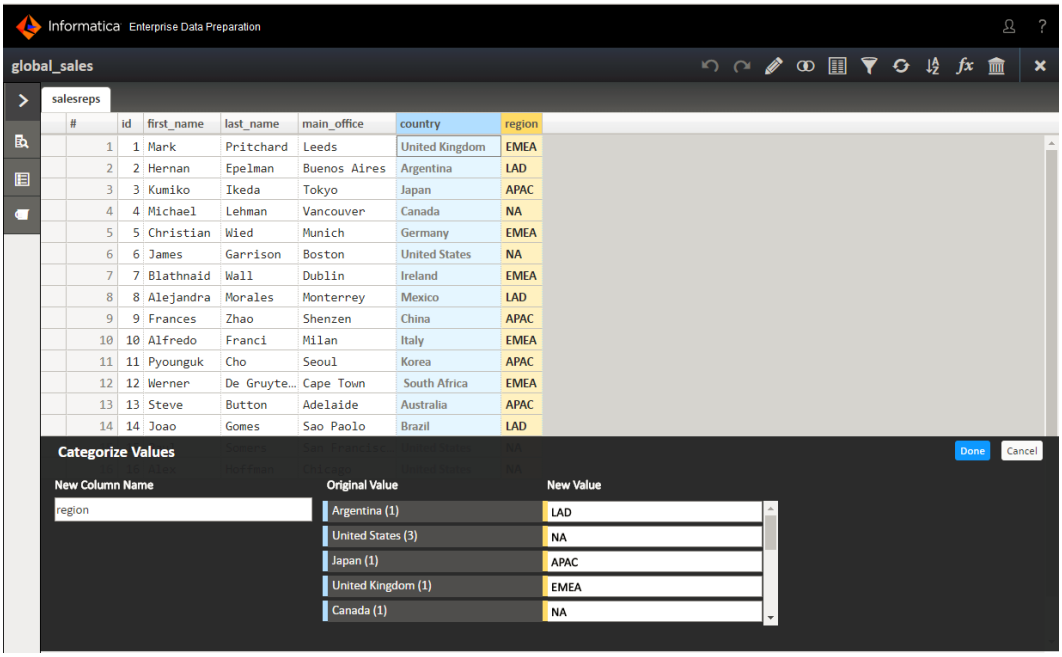
To categorize data, you map a category value to each value in a column. The category values are displayed in a new column you add to the worksheet.

Click the image below to view a video about categorizing column data:



The following example shows each country in a worksheet categorized by sales region. The corresponding category value for each country is shown in a new column named region that is added to the worksheet.

The following image shows the categorized data:



The number next to each value in the Original Value column indicates the occurrence of the value in the asset. An occurrence value of 0 indicates that the value is no longer available in the asset.

1. On the project page, click **Prepare**.

   The Sampling Preview panel appears along with the worksheet.

2. Select the column in the worksheet containing the data that you want to categorize.
3. Click the **Blending** icon and select **Categorize**.
4. Enter the name of the new column that contains the category values.
5. For each value in the Original Value column, enter the corresponding category value in the New Value column.

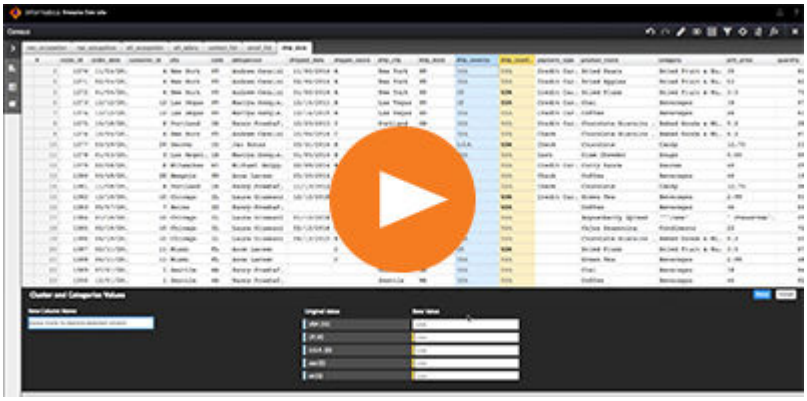
   If you do not enter a new column name, each value you enter in the New Value column overwrites the values in the Original Value column. Note that you can enter NULL as a value.

6. Click **Done**.

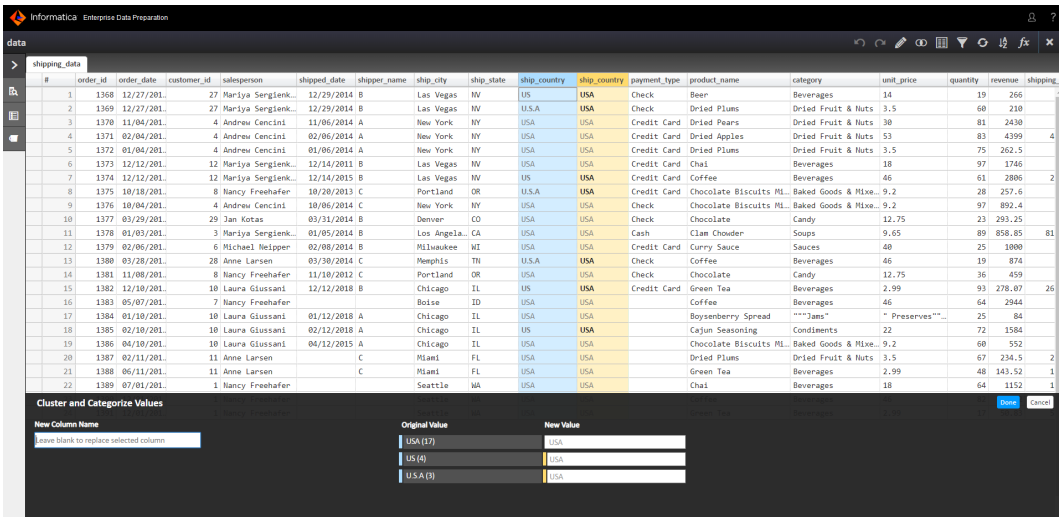
Clustering and Categorizing Column Data

You can cluster similar Text values in a column, and then categorize the values using values suggested by Enterprise Data Preparation. The application uses a phonetic algorithm to identify similar values, and then suggests that you replace the less frequently occurring values with the most frequently occurring value.

Click the image below to view a video about clustering similar values in a column:



For example, if the string "USA" occurs 17 times in a column, and the strings "U.S.A" and "US" each occur fewer times, the application suggests that you replace the less frequently occurring values with "USA", as shown in the following example:



The features is also useful for replacing misspelled values. For example, the application might suggest that you replace "northdakota" or "North Dakkotta' with the more frequently occurring "North Dakota".

Note that updates are made to the sample data only. Values in rows not included in the sample data are published as is.

1. On the project page, click **Prepare**.

   The Sampling Preview panel appears along with the worksheet.

2. Select the column in the worksheet containing the data that you want to categorize.
3. Click the **Blending** icon and select **Cluster and Categorize**.

4. Enter the name of the new column that contains the category values.

   If you do not enter a column name, the application updates the values in the selected column.

5. For each value in the Original Value column, the application displays suggested values in the New Value column, with the most commonly occurring similar value shown first.

   You can modify a suggested value to suit your needs. Note that you can enter NULL as a value.

6. Click **Done**.

Summarizing Data

When you prepare your data, you can summarize the data to make analysis easier. Enterprise Data Preparation offers you several options for summarizing data.

You can aggregate data in a worksheet by grouping selected columns together, and then performing calculations on columns containing number values to summarize the data.

You can reorganize and summarize data in selected columns in a worksheet using a pivot table. For example, you might want to summarize home sales data to analyze the average price of single family homes sold in each city by month.

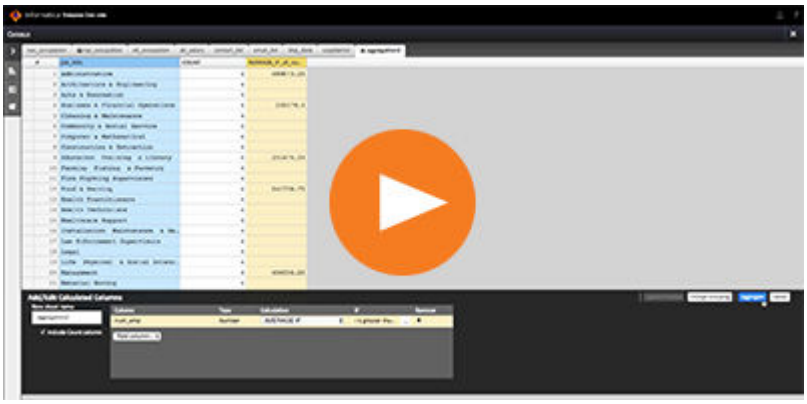
You can also transform columns in a worksheet into rows containing the column data in key value format. For example, if a worksheet includes columns containing price and supplier data for various products, you might want to summarize the data using products as the key, and price and supplier data as the values.

Aggregating Data

You can aggregate data in a worksheet by grouping selected columns together, and then performing calculations on columns containing number values to summarize the data.

The aggregate operation adds a worksheet with a copy of the data in the original worksheet. The **Suggested columns** panel displays columns that Enterprise Data Preparation suggests you group together. The application updates the worksheet with the columns you select.

Click the image below to view a video about the aggregate operation:

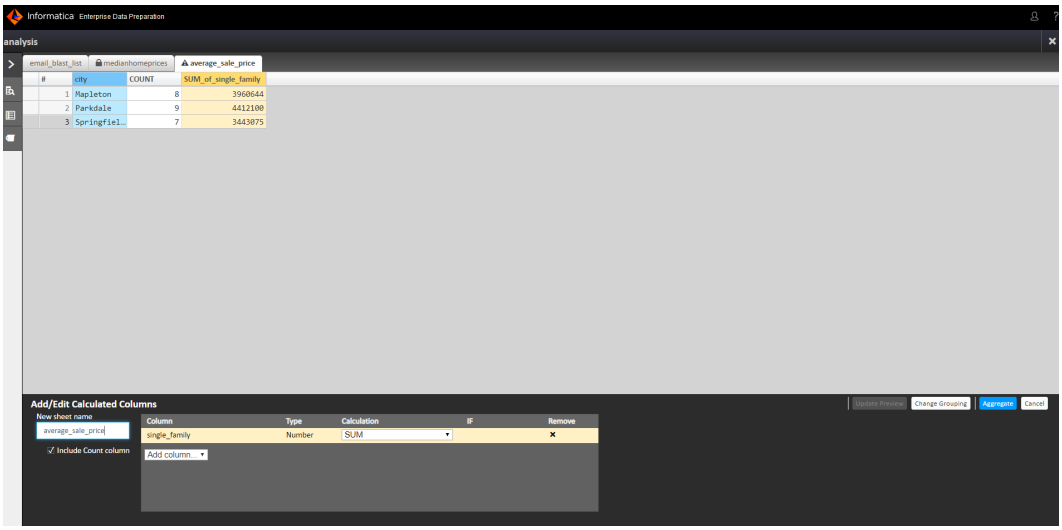


Example

The following image shows an aggregation worksheet created in a project. Enterprise Data Preparation suggests that you include the city and month columns in the grouping.



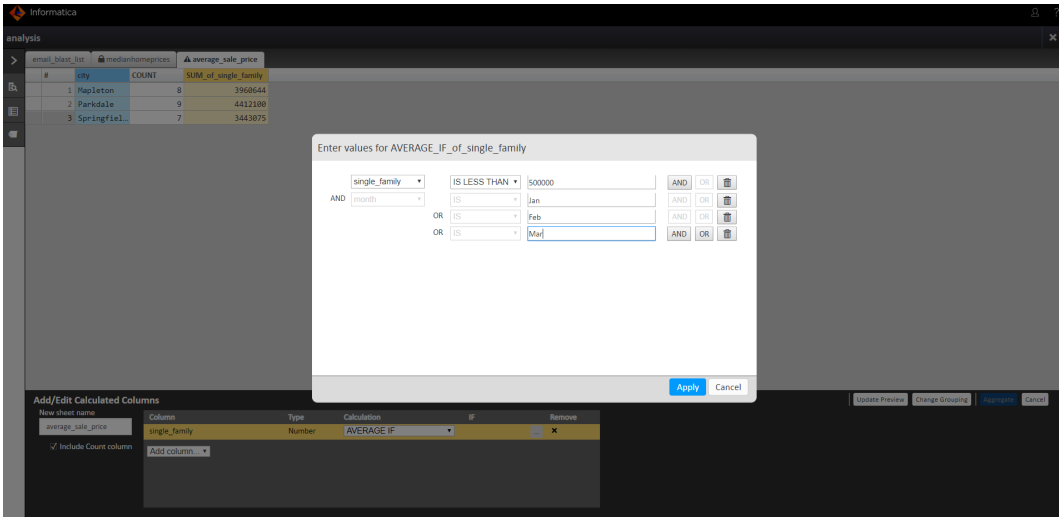
Enterprise Data Preparation displays the rows in the selected columns grouped by distinct values. The application adds a Count column indicating the occurrence of each value combination in the source data to the worksheet, as shown in the following image:.



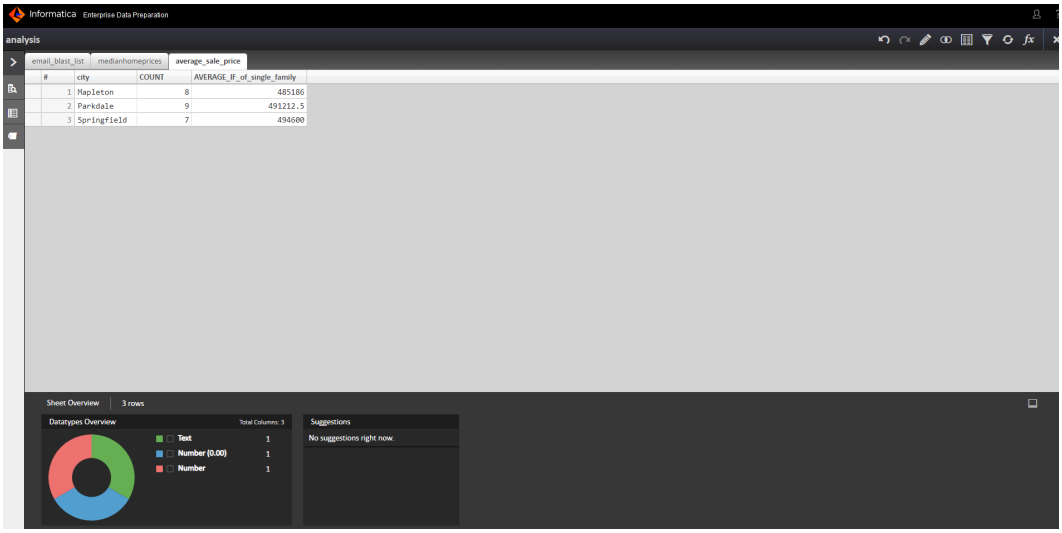
Select a column you want to include in the aggregation, and then select the calculation to apply to the column data. If you select an IF calculation, you can set conditions to restrict the calculation to a range of values. Use AND and OR logic to apply more than one condition.

- Use AND with all operators to include more than one column in a condition.
- Use OR with the IS, IS NOT and IS BETWEEN operators to include more than one value within a column in a condition.

The following example calculates the average price of single family homes sold for less than $500,000 in each city for first three months of the year. Use AND to include the single_family and month columns in the calculation, and then use OR to include specific values in the month column, as shown in the following image:



Enterprise Data Preparation aggregates the values in the calculated columns, as shown in the following image:



Aggregating Columns in a Worksheet

1. Open the worksheet containing the columns you want to aggregate.
2. Click the **Summarize** icon, and then select **Aggregate**.

   The application adds a worksheet with a copy of the data to the project.
3. The Suggested columns field displays columns that Enterprise Data Preparation suggests you group together. Click each column you want to include in the grouping.

   Click **Select your own columns** to select other columns to include in the grouping.
4. Click **Continue**. Enterprise Data Preparation displays the rows in the selected columns grouped by distinct value pairs.

5. Enter a name for the worksheet.
6. The application adds a Count column indicating the occurrence of each value combination in the source data to the worksheet. Clear the Count checkbox if you do not want to include the column in the aggregation.
7. Select the columns on which to perform an aggregate calculation from the **Add Column** menu, and then select the calculation to perform.

   You can perform the following aggregate calculations:

   - SUM: Calculates the sum of the aggregated columns.
   - SUM IF: Calculates the sum of the aggregated columns based on the conditions specified.
   - AVERAGE: Calculates the average value of the aggregated columns.
   - AVERAGE IF: Calculates the average value of the aggregated columns based on the conditions specified.
   - COUNT IF: Calculates the number of value occurrences in the aggregated columns based on the conditions specified.
   - COUNT NULL: Calculates the number of NULL occurrences for aggregated Date, Number, and Text columns.
   - COUNT BLANK: Calculates the number of blank cell occurrences for aggregated Text columns.
   - MIN: Calculates the minimum values in the aggregated columns.
   - MIN IF: Calculates the minimum values in the aggregated columns based on the conditions specified.
   - MAX: Calculates the maximum values in the aggregated columns.
   - MAX IF: Calculates the maximum values in the aggregated columns based on the conditions specified.
   - STANDARD DEVIATION: Calculates the spread of values across the aggregated columns.
   - STANDARD DEVIATION IF: Calculates the spread of values across the aggregated columns based on the conditions specified.
   - VARIANCE: Calculates the average of the squared differences from the mean across the aggregated columns.
   - VARIANCE IF: Calculates the average of the squared differences from the mean across the aggregated columns based on the conditions specified.
8. You can set IF conditions on certain calculations to limit the calculation to a range of values.
   a. Select a calculation containing IF in the calculation title.
   b. Enter the variables to use in the IF statement.
   c. Click **AND** or **OR** to add another condition.
   d. Click **Apply**.
9. Click **Update Preview** to update the worksheet with the calculation results.

   If you want to change the columns included in the operation, click **Change Grouping**.
10. Click **Aggregate**.

    Enterprise Data Preparation adds the worksheet to the project.

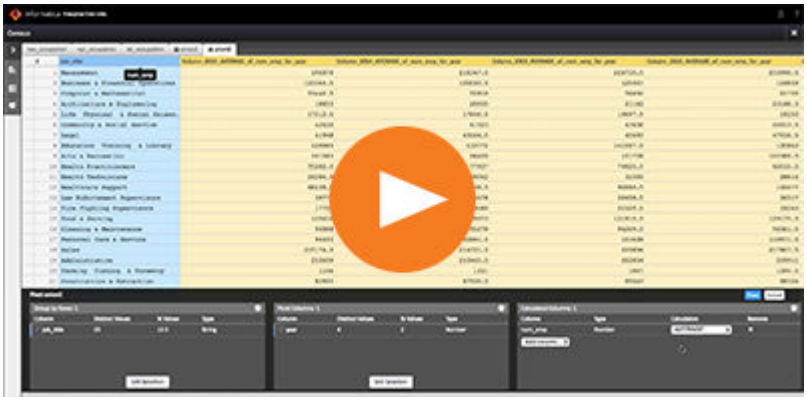
Pivoting Data

You can use the pivot operation to reshape the data in selected columns in a worksheet into a summarized format for analysis.

The pivot operation generates a pivot table in a new worksheet added to the project. To perform a pivot operation, you select one or more columns in the worksheet containing the values you want to group into rows in the pivot table. You also select one or more columns containing distinct values to pivot into new columns added to the pivot table.

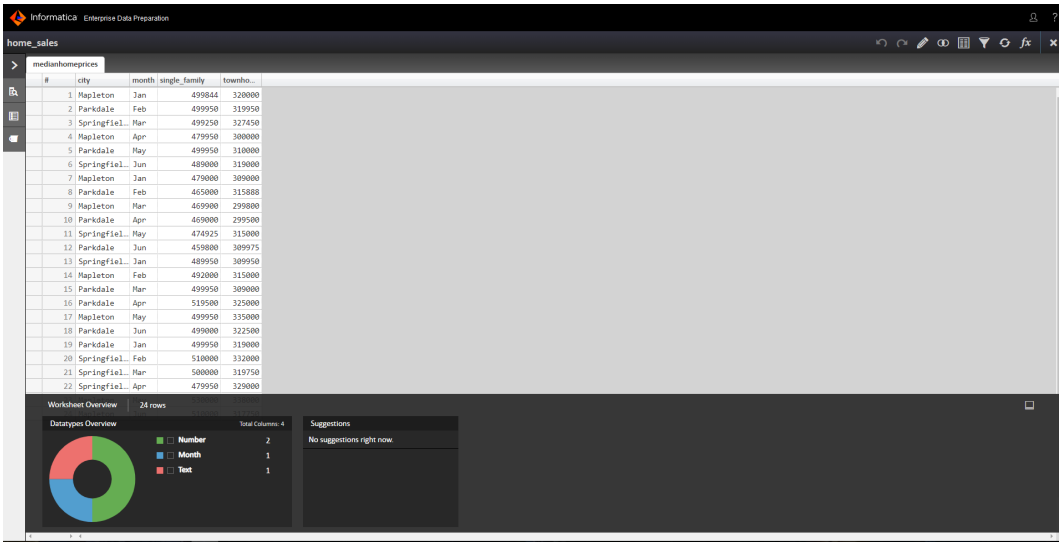
To summarize the data, you select the columns on which to perform a calculation, such as SUM or AVERAGE. Enterprise Data Preparation displays the calculation results for each row in the corresponding column.

Click the image below to view a video about the pivot operation:



Example

The following example generates a pivot table summarizing the average price of single family homes sold in each city for the first six months of the year. The worksheet contains sale price data for single family homes and town homes sold in each city by month, as shown in the following image:

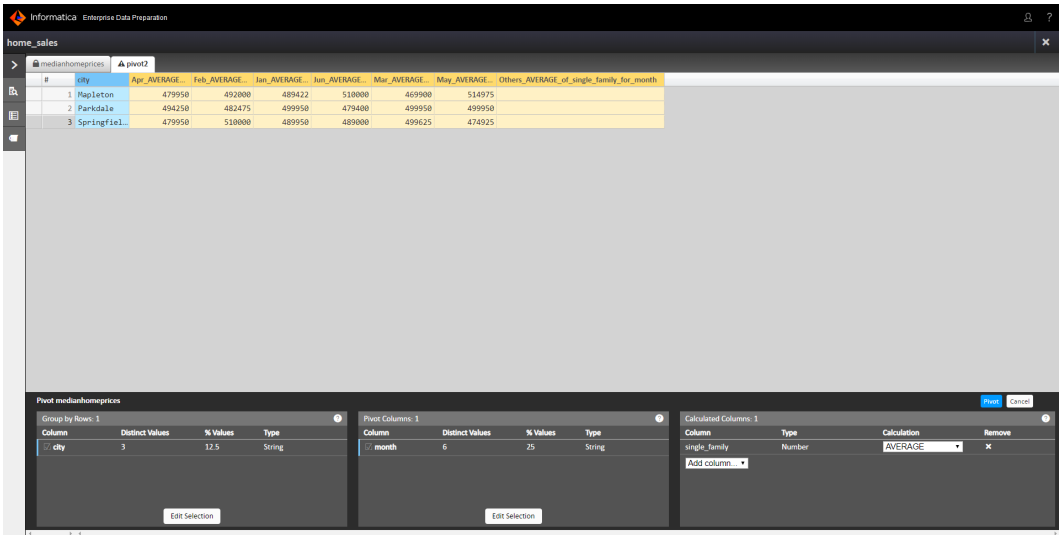


To pivot the data:

- Select the city column as the group by row.
- Select the month column as the pivot column.

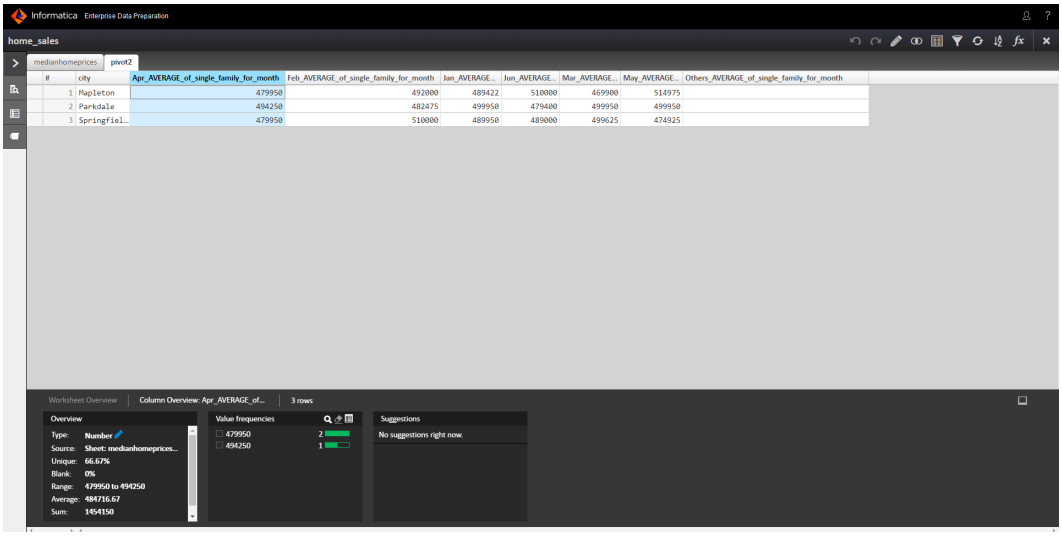
- Select single_family as the calculated column, and then select AVERAGE as the calculation to perform on the column values.

The following image shows the selected columns to pivot:



Note that the Group by Rows panel and the Pivot Columns panel display the number of distinct values in each column, and the percentage of total values they comprise. You can use this information to determine the best columns to select.

The following image displays the resulting pivot table with the summarized data:



The name of each output column is derived from the pivot column value and the calculated data. Special characters in pivot column values are replaced with an underscore in the output column name. Examples of special characters include @,#,*,-, and %. To avoid underscores in column names, remove any special characters from pivot columns before performing a pivot operation.

The pivot operation is performed on the sampled data. To accommodate distinct values beyond the number of rows to load specified in the sampling criteria for the source worksheet, the operation adds a placeholder column named other_<calculationName> to the pivot worksheet. The placeholder column enables the additional values to be added to the Hive table when the asset is published.

Pivoting Columns in a Worksheet

1. Open the worksheet containing the columns you want to pivot.
2. Click the **Summarize** icon, and then select **Pivot**.

   The application adds a new worksheet containing a pivot table to the project.
3. Select the columns containing the keys to group into rows in the **Group By Rows** panel.
4. Click **Apply Selection**.

   The application groups the selected columns into rows in the pivot table.
5. Select the columns to pivot into new columns in the **Pivot Columns** panel.

   The operation creates a column in the pivot table for each distinct value in the selected columns. By default, the operation adds a maximum of 500 columns to the table. To limit the number of columns created in the pivot table to a manageable level, select columns that have lower distinct value counts. You can view the distinct value count for each column in the DISTINCT_VALUES column in the panel.
6. Click **Apply Selection**.

   The application adds a pivot column for each value in the selected columns to the pivot table.
7. Select each column on which to perform an aggregate calculation, and then select the calculation to perform in the **Calculated Columns** panel.

   You can perform the following calculations:

   - SUM: Calculates the sum of the calculated columns.
   - AVERAGE: Calculates the average value of the calculated columns.
   - COUNT: Calculates the number of value occurrences for the calculated columns.
   - COUNT NULL: Calculates the number of NULL occurrences for calculated Date, Number, and Text columns.
   - COUNT BLANK: Calculates the number of blank cell occurrences for calculated Text columns.
   - MIN: Calculates the minimum values in the calculated columns.
   - MAX: Calculates the maximum values in the calculated columns.
   - STANDARD DEVIATION: Calculates the spread of values across the calculated columns.
   - VARIANCE: Calculates the average of the squared differences from the mean across the calculated columns.

   The calculated values are displayed in the pivot columns in the worksheet. Note that columns used as group by or pivot columns cannot be selected as calculation columns.
8. Click **Pivot**.

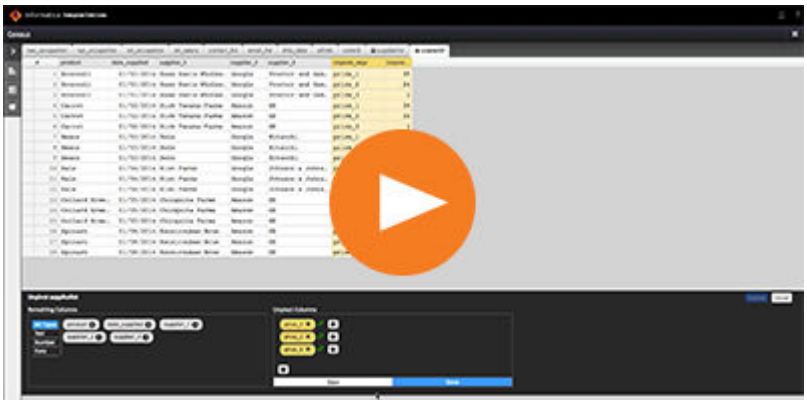
   Enterprise Data Preparation generates the pivot table.

Unpivoting Data

You can use the unpivot operation to transform columns in a worksheet into rows containing the column data in key value format. The unpivot operation is useful when you want to visualize and organize data based on keys and corresponding values.

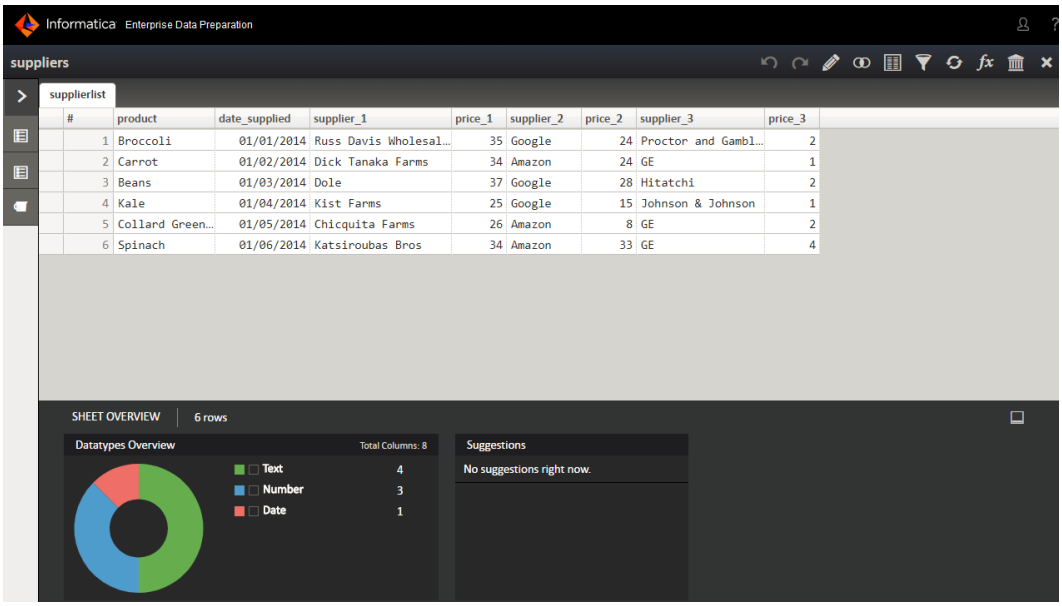
The unpivot operation adds a worksheet containing key and value columns to the project. The key column contains the name of each column included in the operation. The value column contains the corresponding column values.

Click the image below to view a video about the union operation:



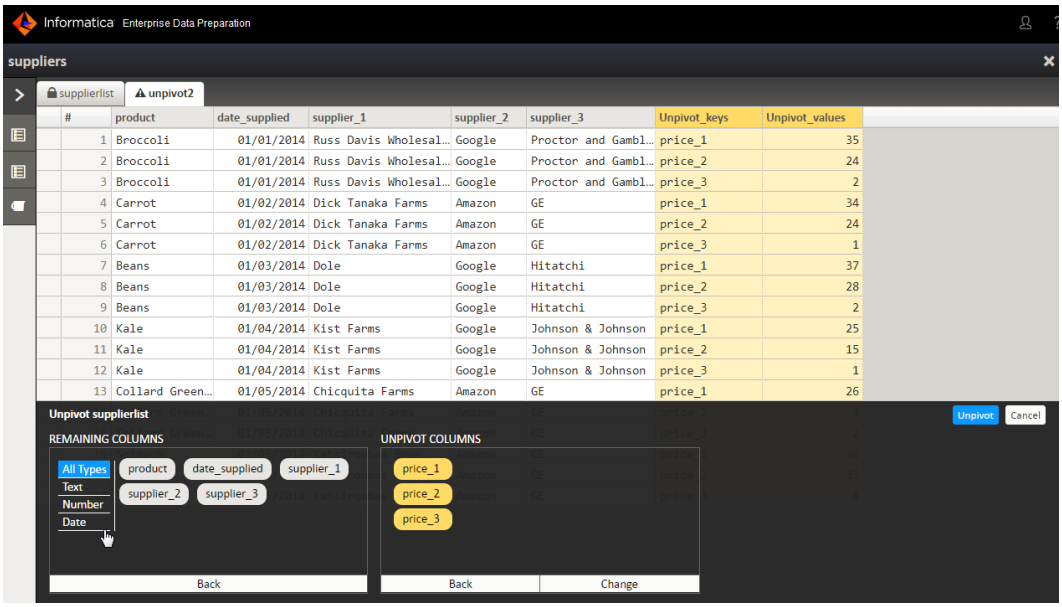
Example

The following example shows how related columns can be transformed into key value format. The following image shows a worksheet containing supplier and price data for various products:



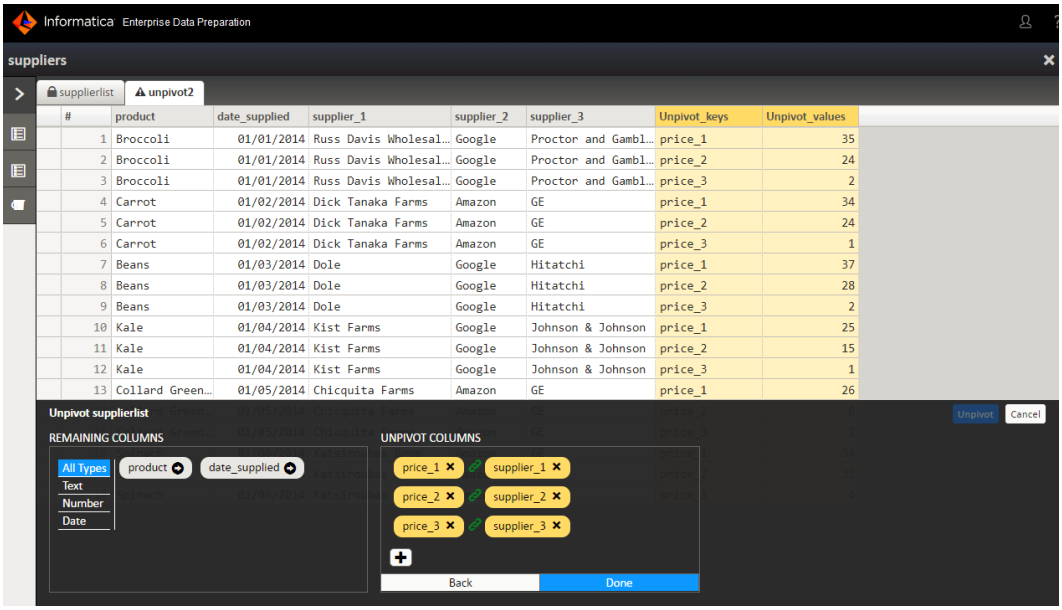
Enterprise Data Preparation suggests that all number columns be included in the unpivot operation by default, and adds the columns in the **Unpivot Columns** panel. Columns not suggested for inclusion are displayed in the **Remaining Columns** panel. You can remove columns you do not want to include in the operation from the **Unpivot Columns** panel, or add columns in the **Remaining Columns** panel to the operation. All columns that you include in the operation must be of the same type.

The following image shows the worksheet with the selected columns to unpivot:

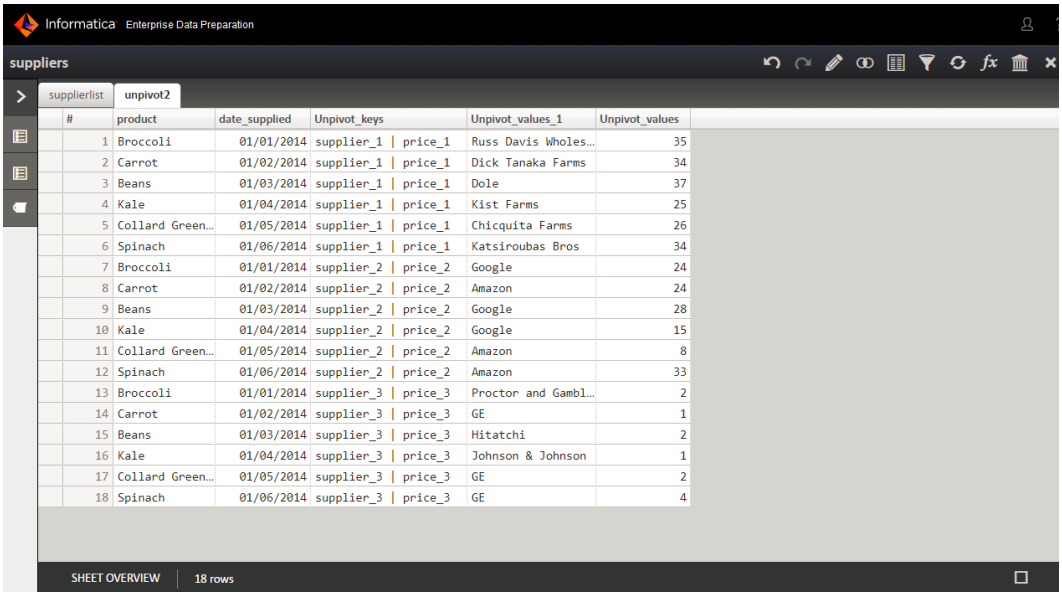


Because the price and supplier columns are related, you can link the columns in the **Unpivot Columns** panel. Note that if you link one column in the **Unpivot Columns** panel with a related column, you must link all of the columns in the panel with related columns.

The following image shows the linked unpivot columns:



When you apply the unpivot operation, the price and supplier column names are concatenated into keys. The corresponding column values appear in each row, as shown in the following image:



#	product	date_supplied	Unpivot_keys	Unpivot_values_1	Unpivot_values
1	Broccoli	01/01/2014	supplier_1 | price_1	Russ Davis Wholes...	35
2	Carrot	01/02/2014	supplier_1 | price_1	Dick Tanaka Farms	34
3	Beans	01/03/2014	supplier_1 | price_1	Dole	37
4	Kale	01/04/2014	supplier_1 | price_1	Kist Farms	25
5	Collard Green...	01/05/2014	supplier_1 | price_1	Chicquita Farms	26
6	Spinach	01/06/2014	supplier_1 | price_1	Katsiroubas Bros	34
7	Broccoli	01/01/2014	supplier_2 | price_2	Google	24
8	Carrot	01/02/2014	supplier_2 | price_2	Amazon	24
9	Beans	01/03/2014	supplier_2 | price_2	Google	28
10	Kale	01/04/2014	supplier_2 | price_2	Google	15
11	Collard Green...	01/05/2014	supplier_2 | price_2	Amazon	8
12	Spinach	01/06/2014	supplier_2 | price_2	Amazon	33
13	Broccoli	01/01/2014	supplier_3 | price_3	Proctor and Gambl...	2
14	Carrot	01/02/2014	supplier_3 | price_3	GE	1
15	Beans	01/03/2014	supplier_3 | price_3	Hitatchi	2
16	Kale	01/04/2014	supplier_3 | price_3	Johnson & Johnson	1
17	Collard Green...	01/05/2014	supplier_3 | price_3	GE	2
18	Spinach	01/06/2014	supplier_3 | price_3	GE	4

Unpivoting Columns in a Worksheet

1. Open the worksheet containing the columns to unpivot.
2. Click the **Summarize** icon, and then select **Unpivot**.

   The application adds a worksheet containing the suggested key and value columns to the project. The **Unpivot Columns** panel contains the suggested columns to unpivot. The **Remaining Columns** panel contains the columns not included in the operation.
3. Click **Change** in the **Unpivot Columns** panel to add or remove columns.

   Select each column to add in the **Remaining Columns** panel. Each column you add must be of the same type as the columns displayed in the **Unpivot Columns** panel.

   To link two related columns, click the **+** symbol next to a column in the **Unpivot Columns** panel, and then click the related column in the **Remaining Columns** panel.
4. Click **Done** when you finish selecting the columns to use in the operation.
5. Click **Unpivot**.

   Enterprise Data Preparation transforms the unpivoted columns into key value format in the worksheet.

One Hot Encoding

You can use the one hot encoding operation to determine the existence of a string value in a selected column within each row in a worksheet. You might use the one hot encoding operation to convert categorical values in a worksheet to numeric values required by machine learning algorithms.

The operation creates output columns in the worksheet for the 500 most frequently occurring distinct values in the selected column. The output column for each value contains either a 1 indicating that the value exists in the row, or a 0 indicating that the value does not exist.

The existence of additional distinct values not included in the 500 most frequently occurring distinct values is indicated in an output column named Is_otherValue_<column_name>. A value of 1 in the Is_otherValue_<column_name> column indicates that additional distinct values exist in the selected column.

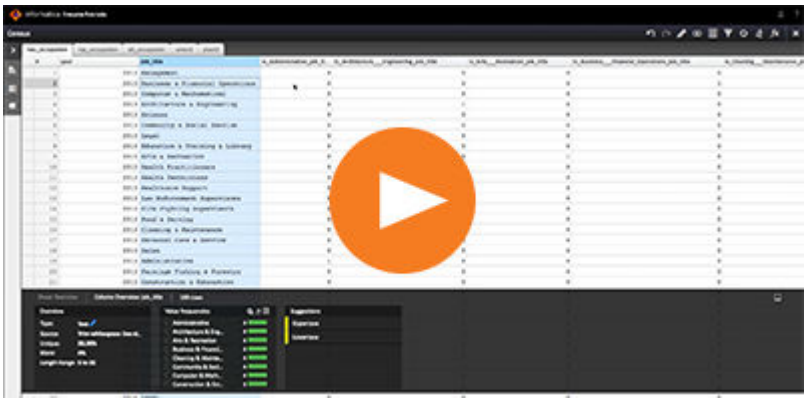
Output columns for NULL values are named Is_NULL_<columnName>. Output columns for empty values are named Is_BLANK_<columnName>.

If you apply one hot encoding to a column with values that contain special characters, the special characters are replaced with an underscore in the output column names. Examples of special characters include @,#,*,-, and %. To avoid underscores in column names, remove any special characters from the columns.

You can copy a recipe step that includes a one hot encoding operation into a worksheet. Map the column in the copied step to a column in the worksheet that contains the same data as the column in the source worksheet. When you insert the copied step into a worksheet, the output columns created in the source worksheet are copied into the worksheet.

When you refresh the data in a worksheet that uses one hot encoding, the output columns for values removed from or added to the source worksheet are persisted in the worksheet.

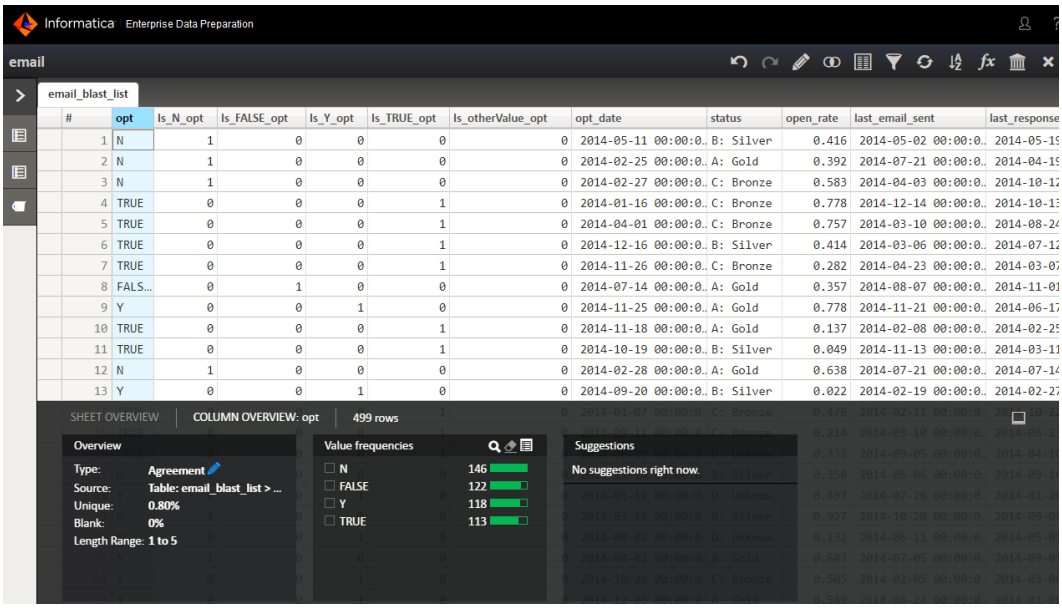
Click the image below to view a video about the one hot encoding operation:



Example

The following example shows the one hot encoding operation applied to the opt column in the worksheet. The existence of each value in the selected column is indicated in output columns added to the worksheet. Each output column is named Is_<valueName>_<columnName>.

The following image shows the one hot encoding operation applied to the worksheet:



Using One Hot Encoding

1. Open the worksheet in which you want to apply one hot encoding.
2. Select the column on which you want perform the one hot encoding operation.
3. Click the **Edit** icon, and then select **Apply one hot encoding**.

   Enterprise Data Preparation adds an output column for each distinct value in the selected column to the worksheet.

CHAPTER 8

Publish Data

This chapter includes the following topics:

- Data Publication Overview, 96
- Publishing Prepared Data to a Hive Table in the Data Lake, 97
- Publishing Prepared Data to an HDFS Location in the Data Lake, 98
- Publishing Prepared Data to Amazon S3 and Azure Data Lake Storage, 99
- Publishing Prepared Data to Microsoft Azure SQL Server or Data Warehouse, 99
- Saving a Recipe as a Mapping, 100
- Downloading a Data Asset, 101
- Operationalize Mappings, 101

Data Publication Overview

Data publication is the process of making prepared data available.

When you publish prepared data, Enterprise Data Preparation applies the recipe to the data in the input source. Enterprise Data Preparation writes the transformed input source to the data lake. You can use a third-party business intelligence or advanced analytic tool to run reports to further analyze the published data. Other analysts can add the published data to their projects and create new data assets.

You can publish prepared data to a Hive table in the data lake, or to an HDFS location in the data lake. To publish to an HDFS location, your catalog administrator must create and configure an HDFS resource that represents the location in Enterprise Data Catalog.

You can publish prepared data to a target that has multibyte characters in the asset name, path, or schema with the exception of known limitations for multibyte file name or path in Hive tables.

You can also publish prepared data to Amazon S3 buckets and to Azure Data Lake Storage Gen2 locations.

When you publish data, you can select the file type to write the data to in the data lake. For example, if choose to publish data as a comma separated value file, the application writes the data to the data lake as a .csv file.

You can schedule publication of data assets. Scheduling publication activities enables you to publish updated data assets on a recurring basis. For more information about scheduling publication activities, see Chapter 10, “Schedule Export, Import and Publish Activities” on page 105.

You can also choose to save a recipe as a mapping that developers can modify and operationalize using Informatica Developer (the Developer tool). You can save the mapping to the Model repository associated

with the Enterprise Data Preparation Service, or you can save the mapping to an .xml file and import it into a different Model repository.

Publishing Prepared Data to a Hive Table in the Data Lake

After you prepare the data in a worksheet, you can publish the data to a Hive table in the data lake.

1. In the **Projects** view, select the worksheet that you want to publish.
2. Click the **Publish** icon.

   The **Publish** dialog appears.
3. Select the Hive resource that represents the Hive target in the data lake to which to publish the data.
4. Select the Hive schema to which to publish the data.
5. Enter a name for the Hive table to which to publish the data.
6. Select **Hive** as the file type to publish the data to a Hive table.
7. Select a publish option:
   - Select **Create** to create a new table in the data lake.
   - Select **Overwrite** to overwrite the existing table in the data lake.
   - Select **Append** to append the data to an existing Hive table with the same structure. You can only select this option if you publish to a Hive table. The data you publish and the table must have the same structure.
8. Click **Next**.
9. If sampling filters are set on columns in the worksheet, the **Apply sampling filter on publication** option is selected by default. Clear this option if you do not want to apply the filters shown in the dialog when publishing the data asset.
10. Select **Enable Full Profiling** if you want to apply full profiling.

    Note that the checkbox is not selected if profiling is disabled on the Hive resource. Contact your Enterprise Data Catalog to enable profiling on the resource.
11. Click **Next**.
12. Specify when you want to publish the worksheet.
    - Select **Publish Now** to publish the asset now.
    - Select **Schedule Publication** to schedule the publish activity.
      If you decide to schedule the publish activity, specify what to do if the asset already exists in the data lake.
      - Select **Append** to append the data to an existing table. Make sure the column names are same, the data types of the columns are same, precision and scale of columns is same, and sequence of columns is same for the importing table and the existing table. The import will fail if any of these do not match.
      - Select **Overwrite** to overwrite the existing table. It drops the table and re-creates the table with the data from the importing table.

      For more information about scheduling a publish activity by creating a schedule or using an existing schedule, see “Scheduling Overview” on page 105.

13. Click **OK**.

    Check the status of the publish activity on My Activities page.

Publishing Prepared Data to an HDFS Location in the Data Lake

After you prepare data in a worksheet, you can publish the prepared data to an HDFS location in the data lake.

To publish to an HDFS location, your catalog administrator must create and configure an HDFS resource that represents the location in Enterprise Data Catalog.

1. In the **Projects** view, select the worksheet that you want to publish.
2. Click the **Publish** icon.

   The **Publish** dialog appears.
3. Select the resource that represents the HDFS location in the data lake to which to publish the data in the Resources panel.

   The application displays the folders within the location.
4. Select the target folder in the location.
5. Verify the file name to which to write the data.
6. Select the file type to the file type to write the data to in the data lake.

   For example, select **CSV** to publish the file as a comma separated value file.
7. Select a publish option:
   - Select **Create** to create a new table in the data lake.
   - Select **Overwrite** to overwrite the existing table in the data lake.
8. Click **Next**.
9. Select the action to take:
   - Select **Create** to create a new asset in the data lake.
   - Select **Overwrite** to overwrite the existing asset in the data lake.
10. Click **Next**.
11. If sampling filters are set on columns in the worksheet, the **Apply sampling filter on publication** option is selected by default. Clear this option if you do not want to apply the filters shown in the dialog when publishing the data asset.
12. Select **Enable Full Profiling** if you want to apply full profiling.

    Note that the checkbox is not enabled if profiling is disabled on the HDFS resource. Contact your Enterprise Data Catalog administrator to enable profiling on the resource.
13. The publication process creates multiple files in HDFS when you publish a .csv, Avro, or Parquet file. If you want to publish all of the multiple files when you publish the asset, select **Use Folder As Source**.
14. Click **Next**.
15. Specify when you want to publish the worksheet.
    - Select **Publish Now** to publish the asset now.

- Select **Schedule Publication** to schedule the publish activity.
  If you decide to schedule the publish activity, specify what to do if the asset already exists in the data lake.
  - Select **Append** to append the data to an existing asset.
  - Select **Overwrite** to overwrite the existing asset.

  For more information about scheduling a publish activity by creating a schedule or using an existing schedule, see "Scheduling Overview" on page 105.

16. Click **OK**.

    Check the status of the publish activity on My Activities page.

Publishing Prepared Data to Amazon S3 and Azure Data Lake Storage

You can publish prepared data to Amazon S3 buckets and to Azure Data Lake Storage Gen2 locations.

For more information about publishing prepared data to Amazon S3, see Configuring Enterprise Data Preparation 10.2.2 and 10.4.0 on Amazon Elastic MapReduce with Autoscaling.

Publishing Prepared Data to Microsoft Azure SQL Server or Data Warehouse

After you prepare the data in a worksheet, you can publish the data to a Microsoft Azure SQL Server or Data Warehouse in the data lakehouse.

1. In the **Projects** view, select the worksheet that you want to publish.
2. Click the **Publish** icon.

   The **Publish** dialog appears.
3. Select the resource and connection that represent the target in the data lakehouse for publishing data.
4. Select the target schema for publishing data.
5. Enter a name for the table.
6. Enter `TABLE` as the file type.
7. Select a publish action:
   - Select **Create** to create a new table in the data lakehouse.
   - Select **Overwrite** to overwrite the existing table in the data lakehouse.
   - Select **Append** to append the data to an existing table with the same structure. You can select this option only if you publish to a table. The data you publish and the table must have the same structure.
8. Click **Next**.

9. If sampling filters are set on worksheet columns, the **Apply sampling filter on publication** option is selected by default. Clear this option if you do not want to apply the filters shown in the dialog when publishing the data asset.
10. Click **Next**.
11. Specify when you want to publish the worksheet:
    - Select **Publish Now** to publish the asset now.
    - Select **Schedule Publication** to schedule the publish activity.
      If you decide to schedule the publish activity, specify what to do if the asset already exists in the data lakehouse.
      - Select **Append** to append the data to an existing table. Make sure that the column names, data types, precision, scale, and the column sequence match in both importing and exporting tables. The publish task fails in case of a column metadata mismatch.
      - Select **Overwrite** to overwrite the existing table. Overwriting drops the table and re-creates it with the data from the importing table.

      For more information about scheduling a publish activity by creating a schedule or using an existing schedule, see "Scheduling Overview" on page 105.
12. Click **OK**.

    Check the status of the publish activity on My Activities page.

    Note: The data publication displays NULL values if prepared data contains blank values.

Saving a Recipe as a Mapping

You can save a recipe as a mapping that developers can modify and operationalize using the Developer tool.

You can save the mapping to the Model repository associated with the Enterprise Data Preparation Service. You must specify the Hive schema and the Hive table defined in the mapping output. The Data Integration Service writes the data to the specified schema and table when it runs the mapping.

You can also choose to save the mapping as an .xml file. Developers can import the file to another Model repository.

1. In the **Projects** view, select the worksheet containing the recipe that you want to save as a mapping.
2. Click the **Manage Worksheets** icon, and then select **Save As Mapping**.

   The **Save As Mapping** dialog box appears.
3. Choose how you want to save the mapping.
   - Click **Model Repository** to save the mapping to the Model repository associated with the Enterprise Data Preparation Service.
     Specify the schema and the table to which the mapping writes data. The application adds the table and schema names to the target definition in the mapping.
   - Click **Local File** to save the mapping as an .xml file.
4. If sampling filters are set on columns in the worksheet, the **Apply sampling filter on publication** option is selected by default. Clear this option if you do not want to apply the filters shown in the dialog when saving the data asset.
5. Click **Save**.

Downloading a Data Asset

Download a data asset or a publication as a comma-separated value (.csv) file or a Tableau Data Extract (.tde) file. You can use the file with a third-party application to run reports on the data.

1. Open the project containing the asset or publication you want to download.
2. In the Worksheets panel, click the name of a data asset in the **Input Sources** column, or of a publication in the **Publications** column.
3. On the Overview page, select the **Manage Data Assets** icon, and then click **Download CSV** or **Download TDE**.

   Enterprise Data Preparation downloads the data asset or publication as a CSV file or TDE file.

Operationalize Mappings

You can work with your Informatica administrator to operationalize your data preparation steps so that your prepared data is uploaded to the data lake on a regular basis.

When you publish your prepared data to the data lake. Enterprise Data Preparation generates a mapping from the worksheet recipe. The administrator can deploy the mapping and run it in the Hadoop environment to load the prepared data back into the data lake on a regular basis.

CHAPTER 9

Visualize and Assess Published Data

This chapter includes the following topics:

- Overview, 102
- Prerequisites, 103
- Visualizing Published Data, 103
- Sharing the Notebook, 104

Overview

You can assess a published asset visually to make sure that the data is appropriate for analysis from content and quality perspectives. You can then choose to fix the recipe and republish the worksheet.

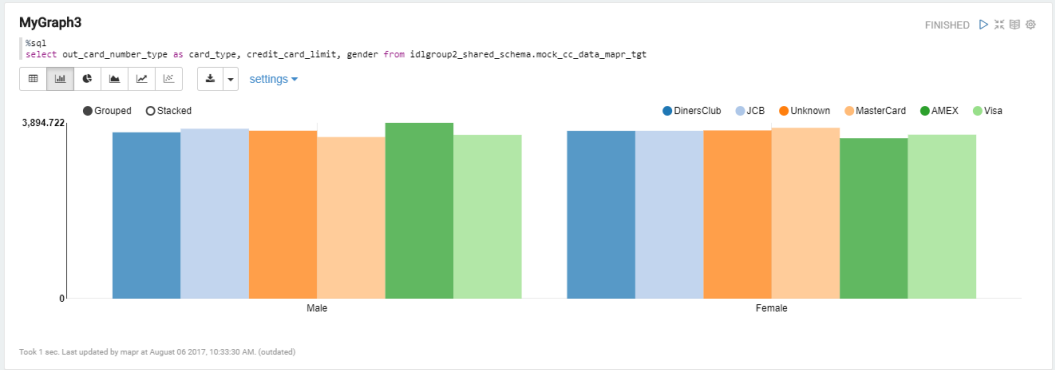
Enterprise Data Preparation uses Apache Zeppelin to view the worksheets in the form of a notebook that contains graphs and charts. You can assess and validate data by performing the following activities:

- View summary statistics. You can view summary statistics for the newly created data assets and assess the completeness, conformity, consistency, integrity, timeliness, and uniqueness of the data. You can view summary statistics at table level and at column level. You can also view statistics for multiple columns.
- View relationships between different columns. You can view the relationships between different columns in the form of pie charts, graphs, and histograms.
- Validate the recipes. You can view the data in the form of graphics to identify any anomalies and correct them.
- Create multiple charts and graphs. You can run new queries on the existing columns to create new charts and graphs to validate the data.

When you click the visualize icon for a published worksheet in a project, Enterprise Data Preparation creates a Zeppelin notebook for the data asset. You can continue to work on the notebook and share it with other Enterprise Data Preparation users.

You can view the Zeppelin notebook from the project page or from the data assets page. You cannot make changes to the notebook when you view it from the Visualization tab of the data assets page.

The following image shows a visualized graph in a visualization notebook:



Prerequisites

You must be able to access Zeppelin to prepare and view notebooks.

- Make sure that Zeppelin is configured by the Enterprise Data Preparation administrator on Spark. For more information about Zeppelin configuration, see Apache Zeppelin documentation.
- Make sure the administrator has configured users to the Zeppelin configuration.
- Make sure you collect the Zeppelin login credentials from the Administrator and use them to log in to the notebook. You can log in to Zeppelin from Enterprise Data Preparation.

Visualizing Published Data

After preparing data and publishing the worksheet, you can visualize the data before you operationalize the recipes created during publishing.

1. In the **Projects** view, select the worksheet that you want to visualize.
2. Click the visualize icon corresponding to the worksheet.

   Note: The Visualize icon will be enabled only if Zeppelin is configured in Informatica Administrator and the successfully published assets are allowed to be visualized.
3. Enter the username and password to access the Visualization page. You can collect these details from the Administrator.
4. Run queries on the notebook page for the worksheet to validate the data. You can run a query on an entire table, single column, or multiple columns in a table. You can also create queries establishing relationship between different columns in the worksheet.

You can delete the Zeppelin notebook if you don't need it for analysis and validation. On the Visualization page, click the menu and select **Delete**. Your assessment notebook associated with the data asset will also be deleted along with this Zeppelin notebook.

Sharing the Notebook

You can share the notebooks with other users of the lake. Other users can access the assessment notebook from the Visualization tab of the published data asset. Unless you share the notebook, only the collaborators of the project can view the notebook.

To share the notebook with all users who have permissions to access the data asset, click **Share**. Make sure you click **Share** every time you update the notebook by adding a graph or running a new query.

Click **Unshare** to avoid letting other users access the notebook. Only the collaborators of the project will be able to view and edit the worksheet, depending on their roles. If the visualization notebook is not shared from the project, a warning message appears indicating that you need to create and share the notebook for others to access it.

CHAPTER 10

Schedule Export, Import and Publish Activities

This chapter includes the following topics:

- Scheduling Overview, 105
- Creating a Schedule, 106
- My Activities View, 106
- Managing Scheduled Activities, 107
- Managing Schedules, 107

Scheduling Overview

When you import, export, or publish a data asset, you can choose to complete the activity at that time, or you can choose to schedule the activity. Scheduling an activity enables you to import, export or publish updated data assets on a recurring basis.

If you decide to schedule an activity, you can create a new schedule, or you can select an existing schedule. Other users can use schedules that you create, and you can use schedules created by other users. If you use a schedule owned by another user, changes the owner makes to the schedule might impact your scheduled activity.

When you schedule an activity, you can specify what to do if an asset with the same name as the asset you are importing, exporting, or publishing already exists. You can choose to overwrite the asset, or you can append the new data to the asset.

The Scheduler Service must be running to create or use a schedule. Contact your Informatica administrator if you encounter an error indicating that the Scheduler Service is not available.

Creating a Schedule

You can create a new schedule when you import, export or publish a data asset. You can also create a new schedule from the Manage My Schedules page.

Contact your Informatica administrator if you do not have the privileges required to create a schedule.

1. To create a new schedule:
   - Select **Create New Schedule** in the import, export or publish wizard.
   - On the Enterprise Data Preparation home page, select **My Activities**. Click the **Manage My Schedules** tab, and then click **New Schedule**.
2. Provide the schedule details.

   The following table describes the schedule detail properties:

Property	Description
Schedule Name	The required schedule name. The name can contain alphanumeric characters and any of the following special characters: _, ., -. or $.
Description	An optional description that might help other users decide whether to use your schedule.
Starts	The date and time the schedule takes effect.
Time Zone	The time zone the schedule is based on.
Frequency	The frequency at which to run the activity. The options displayed are based on the frequency value you select. If you select Weekly, you can select multiple days on which to run the activity.
Repeat Options	The time period for which to repeat the activity.

3. Click **Save**.

   The schedule is added to the repository used by the Scheduler Service and can be selected by other users.

My Activities View

The **My Activities** view displays all activities that you have completed. Activities can include uploading and publishing data assets.

Let's explore what you can do in the **My Activities** view:

- Click the name of a data asset to view its details. The data asset opens in the **Open Asset** view.
- Upload a data asset. Click **Upload Data** to upload a data asset to the data lake.
- View your data assets on which an activity have been performed. The **My Activities** view displays all the data assets on which you performed any of the following activities - copying, deleting, downloading, exporting, importing, publishing, and uploading.

- Filter your list of data assets. Click the **filter** icon to filter data assets by type of activity, time frame and status.
  - Select **Type** to filter by all, copy, delete, download, export, import, publish, and upload types of activities.
  - Select **Updated On** to filter by all, last 24 hours, last week, or last month. You can also select a custom date range.
  - Select **Status** to filter by all, completed, completed (with warnings), in progress, or failed.
- Sort your list of data assets. Click the **sort** icon to sort data assets.
  - Select **Type** to sort by all, copy, delete, download, export, import, publish, and upload types of activities.
  - Select **Updated On** to sort by the time when they were last updated.
  - Select **Name** to sort the data asset names alphabetically.
- Search for a data asset. Use the search box to search for a data asset on which any of these activities have been performed.

You can manage the schedules used to import, export, or publish data assets on a recurring basis from the **My Activities** view. For more information about scheduling import, export, or publish activities, see Chapter 8, "Publish Data" on page 96.

- Click **My Scheduled Activities** page to delete or suspend a scheduled activity, or to select a new schedule to use for an activity.
- Click **Manage Schedules** to edit, delete or suspend schedules that you own. You can also create a new schedule from the page.

Managing Scheduled Activities

Use the My Scheduled Activities page to delete or suspend a scheduled activity, or to select a new schedule to use for an activity.

1. On the Enterprise Data Preparation home page, click **My Activities**, and then click the **My Scheduled Activities** tab.
2. Highlight the scheduled activity you want to modify.
   - Click the **Edit** icon to select a new schedule to use.
   - Click the **Delete** icon to delete all occurrences of the scheduled activity.
   - Click the **Suspend** icon to suspend all occurrences of the scheduled activity. You can resume the scheduled activity at a later time.

Managing Schedules

Use the Manage My Schedules page to modify, delete or suspend schedules that you own.

Modifications that you make to a schedule impact all activities that use the schedule. When you delete or suspend a schedule, the schedule is unavailable to all activities owned by other users, and the activities fail to run. Data Integration Service jobs and other service jobs that use the schedule also fail to run.

You can only manage schedules that you own. Users with administrator privileges can manage any schedule.

1. On the Enterprise Data Preparation home page, click **My Activities**, and then click the **Manage My Schedules** tab.
2. Select the schedule you want to modify in the **All Schedules** panel.

   The details for the selected schedule load in the page.
3. Before you modify the schedule, compare the values shown for the Total Activities and Only My Activities properties to determine if activities owned by other users use the schedule.
4. Select the action you want to take from the **Actions** menu.
   - Select **Edit Schedule** to modify the schedule settings.
   - Select **Delete Schedule** to delete the schedule and all activities that use the schedule from the domain.
   - Select **Suspend Schedule** to suspend the schedule and all activities that use the schedule.

APPENDIX A

Glossary

asset

An information object that is described in the catalog. Assets can include items such as a database table, report, folder, user account, or business glossary definition.

catalog

An indexed inventory of the assets in an enterprise. The assets can come from different types of enterprise systems. The catalog contains metadata about each asset, including profile statistics, asset ratings, data domains, and data relationships.

data asset

Data that you work with as a unit. A data asset is one type of asset described in the catalog. Data assets can include items such as a flat file, table, or view. A data asset can include data stored in or outside the data lake. You can add data assets that are stored in the data lake as Hive tables to a project.

data lake

A centralized repository of large volumes of structured and unstructured data. A data lake can contain different types of data, including raw data, refined data, master data, transactional data, log file data, and machine data.

In most cases, a data lake is a Hadoop cluster used for data engineering initiatives. When different types of data are stored in one repository, data analysts can more easily combine and transform the data to create new insights.

data preparation

The process of combining, cleansing, transforming, and structuring data from one or more data assets so that it is ready for analysis.

In Enterprise Data Preparation, you use worksheets in a project to create data preparation recipes.

data publication

The process of making prepared data available in the data lake.

When you publish prepared data, Enterprise Data Preparation applies the recipe to the data in the input source. Enterprise Data Preparation writes the transformed input source to a Hive table in the data lake. You can use a third-party business intelligence or advanced analytic tool to run reports to further analyze the published data. Other analysts can add the published data to their projects and create new data assets.

data upload

The process of adding data to the data lake from a local drive. When you upload data, you create a new data asset that you can add to a project and prepare for analysis.

You can upload delimited text files to the data lake. Enterprise Data Preparation writes the uploaded data to a Hive table in the data lake.

input source

The data source for a worksheet in a project. An input source can be a Hive table in the data lake.

project

A container used to organize the worksheets and input sources that you use to prepare data.

You can add data assets that are stored in the data lake as Hive tables to a project. By definition, projects are private and can only be viewed by the user who creates the project. You can share projects with other analysts to collaborate on activities related to the project.

recipe

The list of input sources and the steps taken to prepare data in a worksheet.

When you publish prepared data, Enterprise Data Preparation applies the recipe to the data in the input source. Enterprise Data Preparation converts the recipe into an Informatica mapping and stores the mapping in the Model repository.

worksheet

The component within a project where you prepare data. When you add a data asset to a project, Enterprise Data Preparation creates a corresponding worksheet in the project.

A worksheet has an interactive data-driven spreadsheet interface. The worksheet contains the data that you prepare and a recipe that tracks the changes you make to the data as you prepare it. Depending on the size of the input source, Enterprise Data Preparation loads sample data or all data into the worksheet. When you prepare data, you use the data loaded in the worksheet. You do not directly change the data in the input source. When you publish the prepared data, Enterprise Data Preparation applies the recipe to the data in the input source and creates a new data asset.

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